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(54) **RADIO FREQUENCY (RF) SENSING
SESSION BASED ON CLOCK DRIFT
INFORMATION**

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(71) Applicant: **QUALCOMM Incorporated**, San
Diego, CA (US)

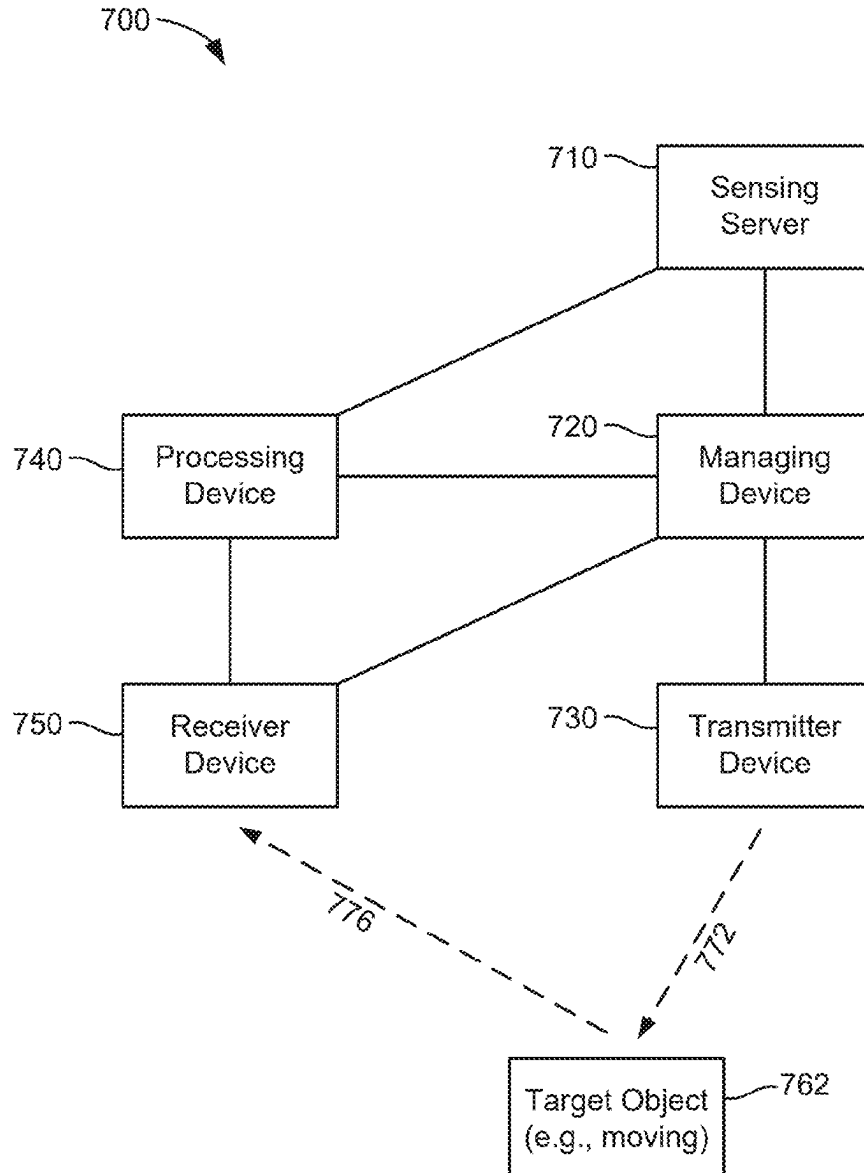
(72) Inventors: **Marwen ZORGUI**, San Diego, CA
(US); **Srinivas YERRAMALLI**, San
Diego, CA (US); **Xiaoxia ZHANG**, San
Diego, CA (US)

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(57) **ABSTRACT**

In an aspect, a sensing node device may transmit, to a managing device, clock drift information corresponding to clock drift of the sensing node device. The sensing node device may receive, from the managing device, a sensing signal configuration for a sensing session based on the clock drift information. The sensing node device may transmit or receive one or more sensing signals during a sensing window of the sensing session based on the sensing signal configuration.



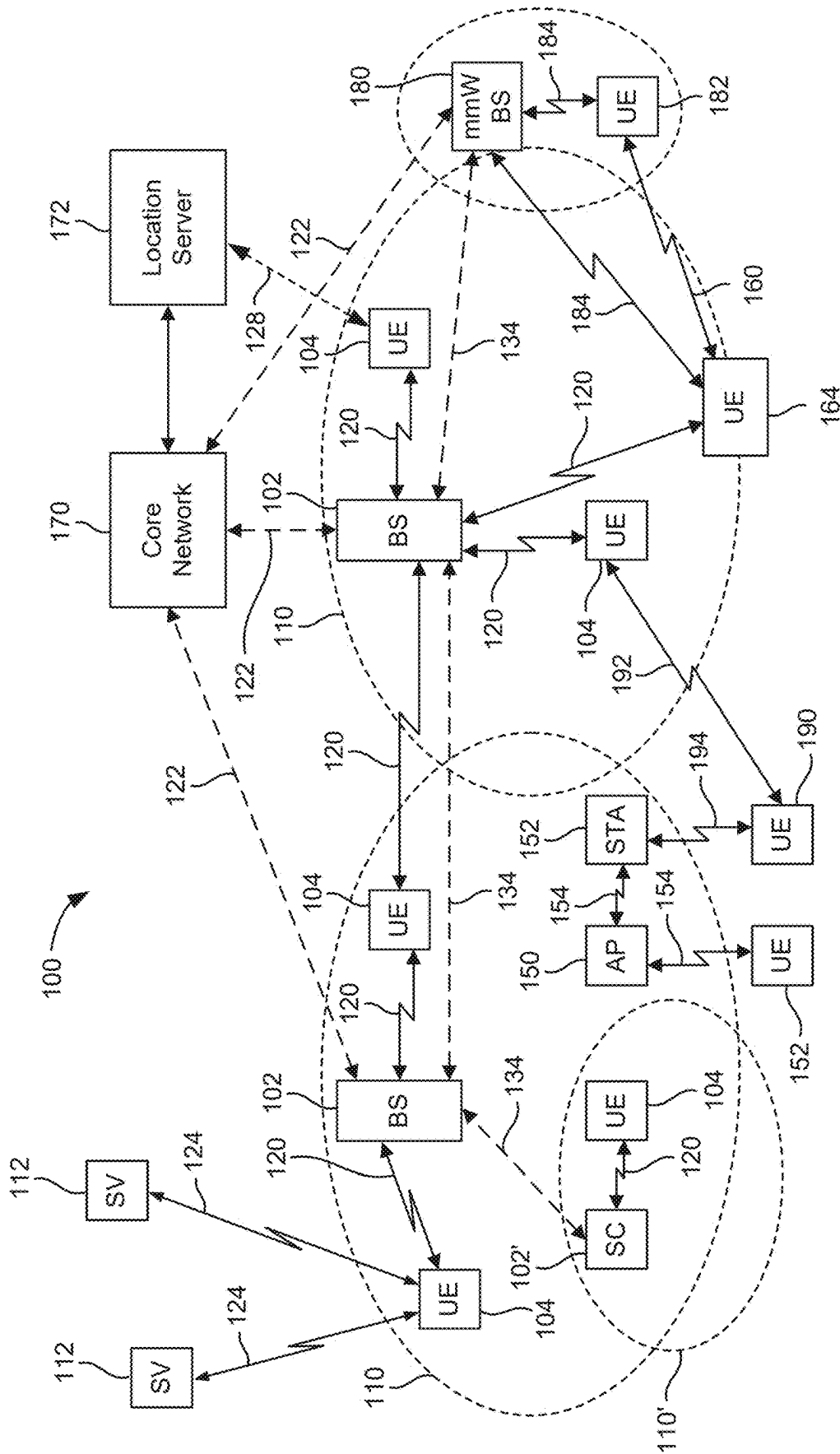


FIG. 1

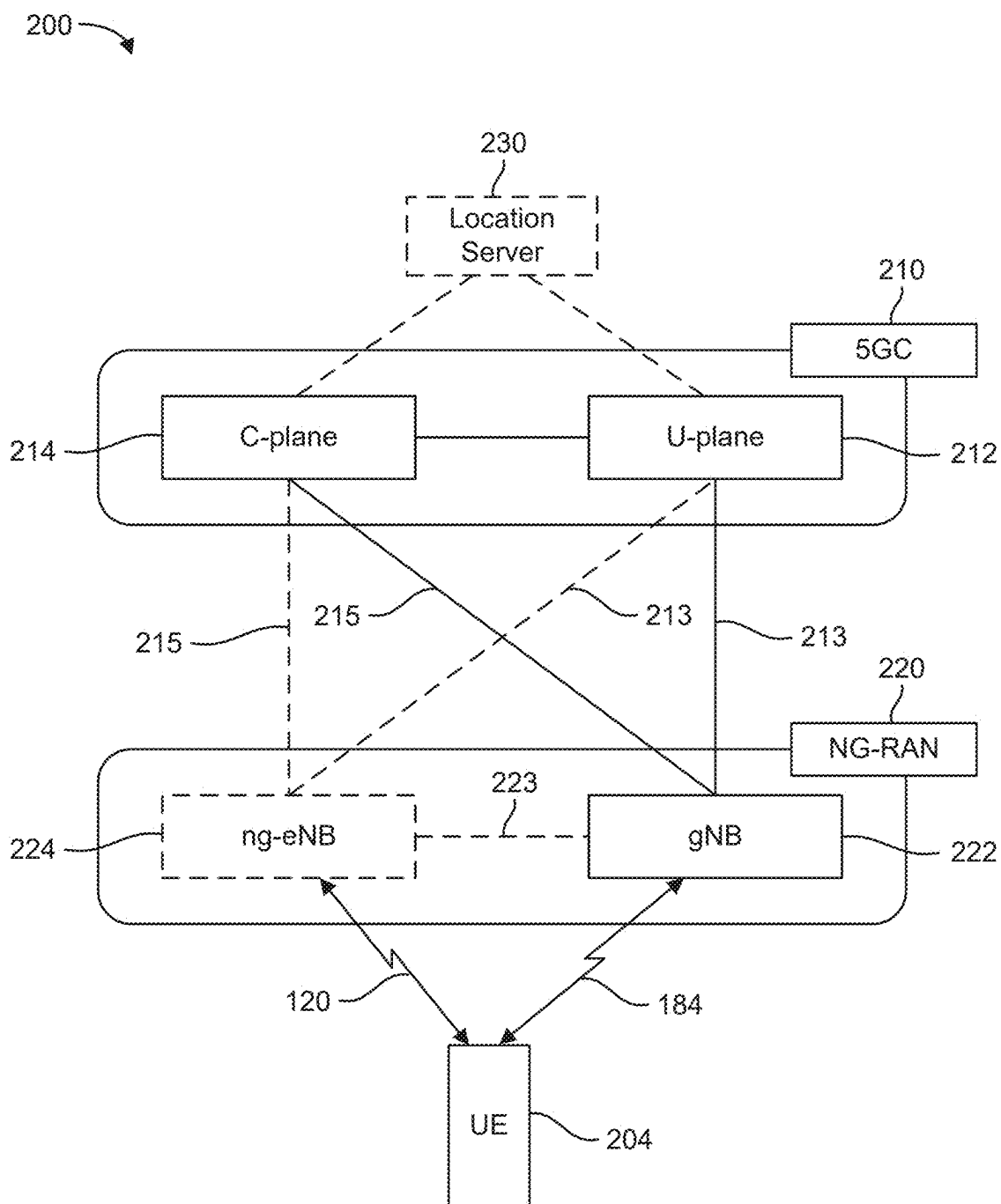


FIG. 2A

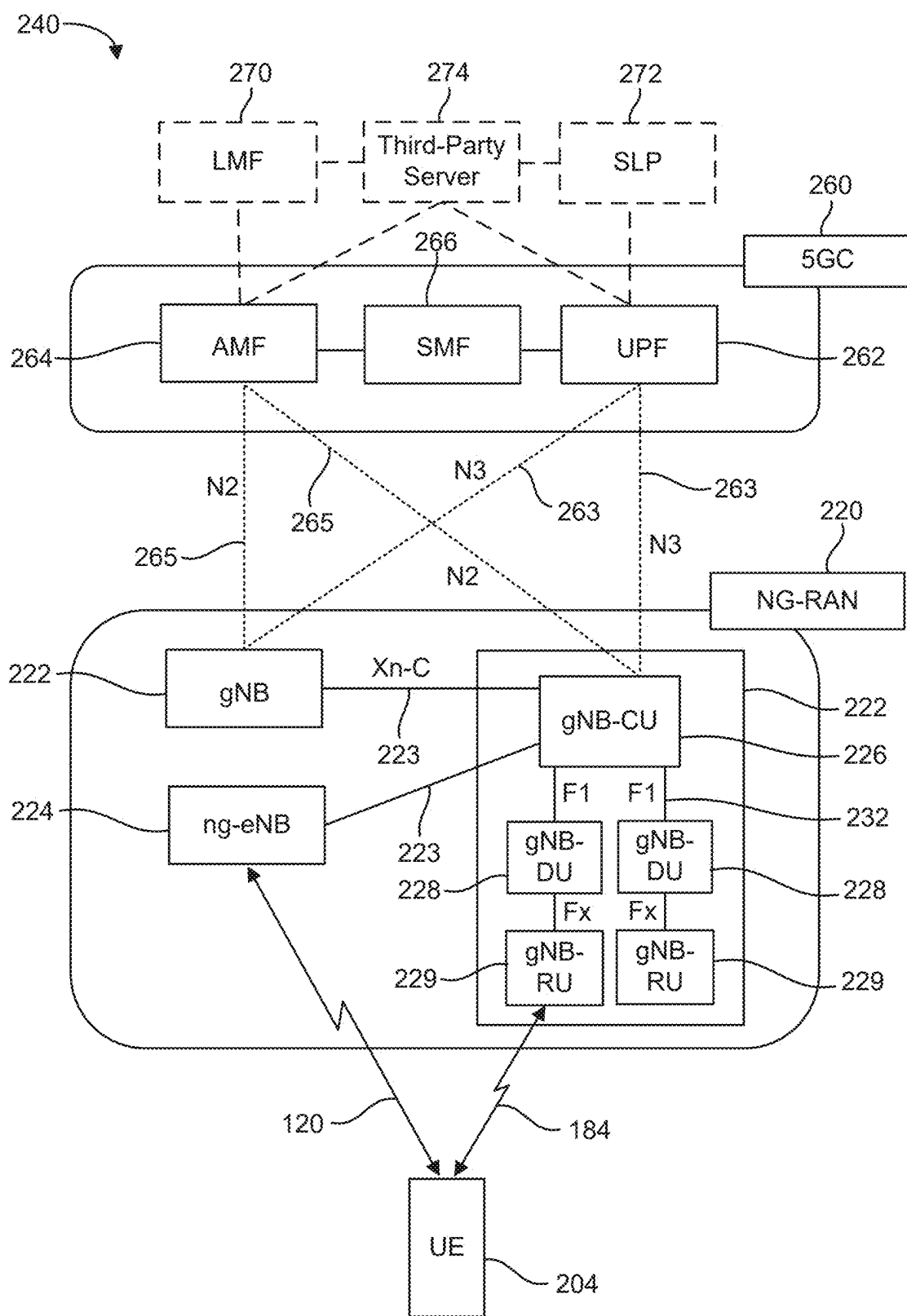
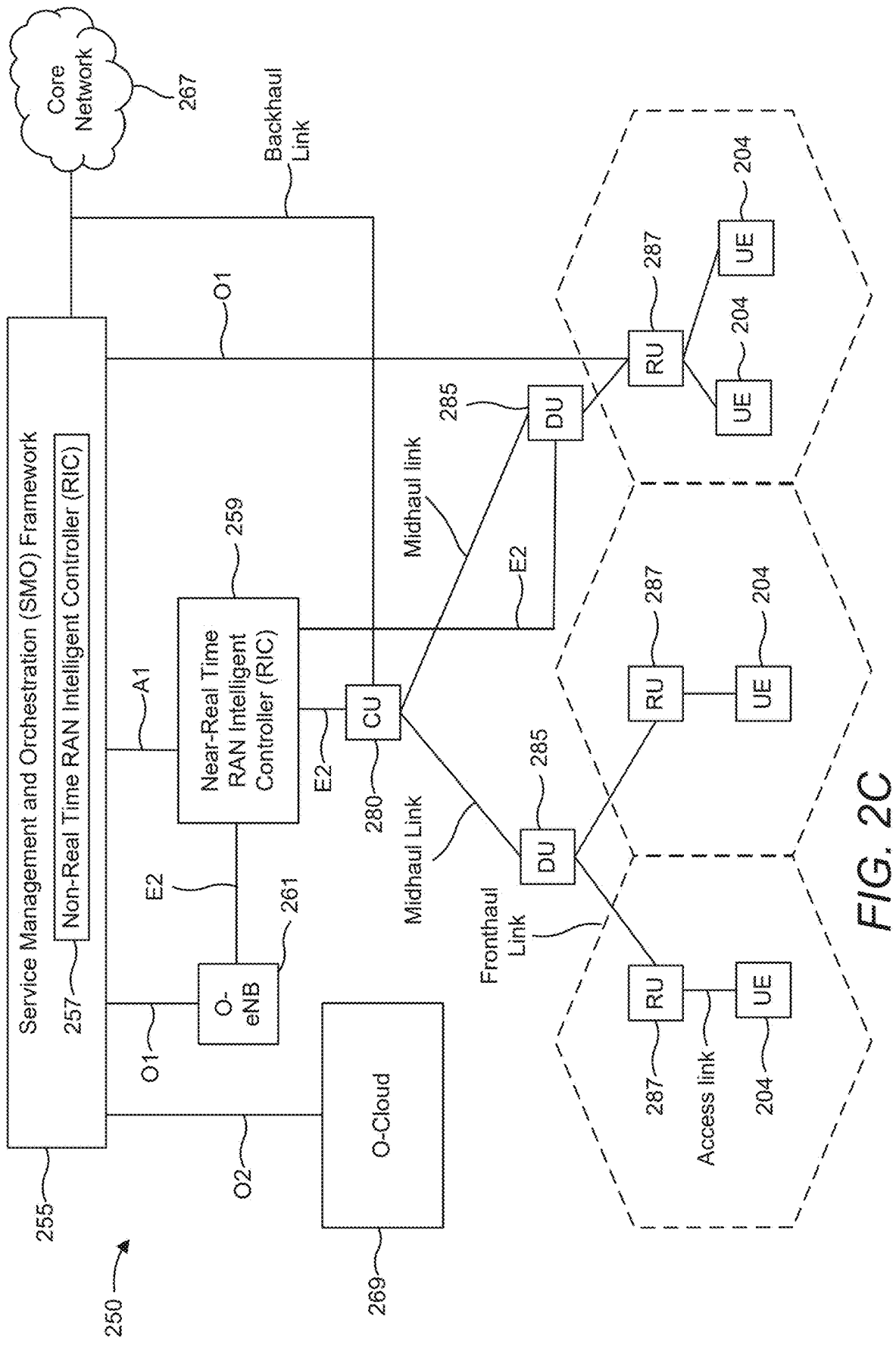


FIG. 2B



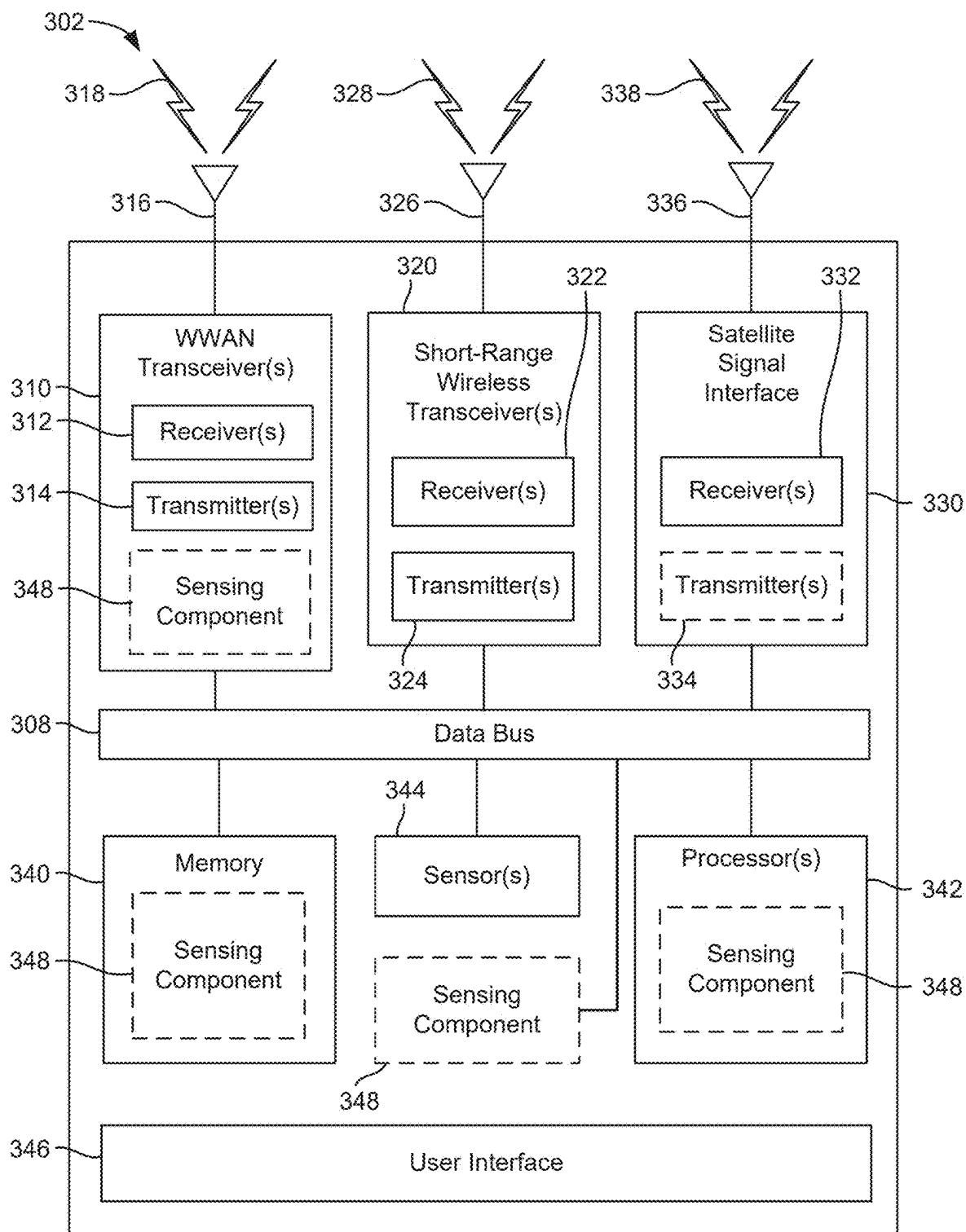


FIG. 3A

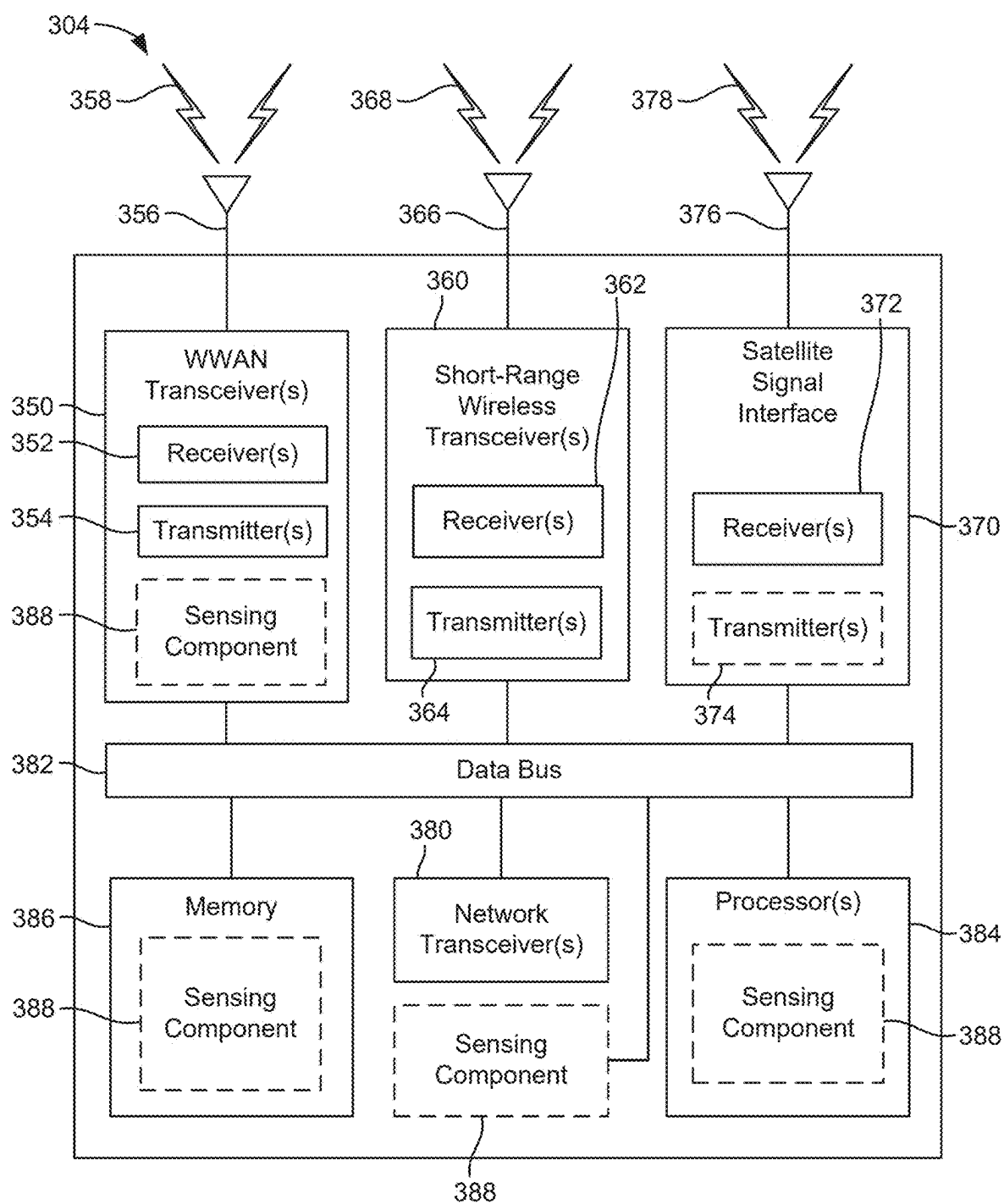
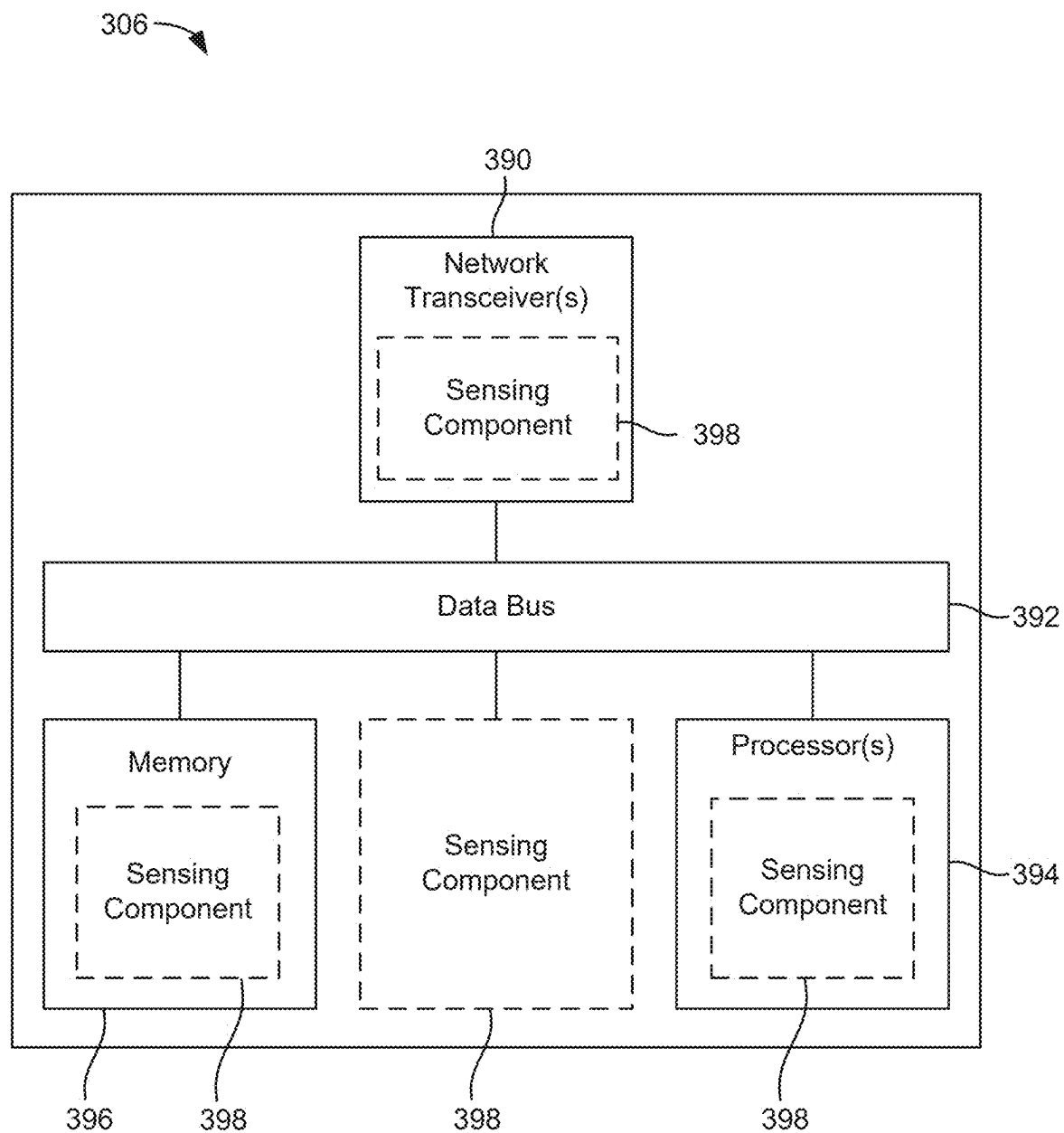
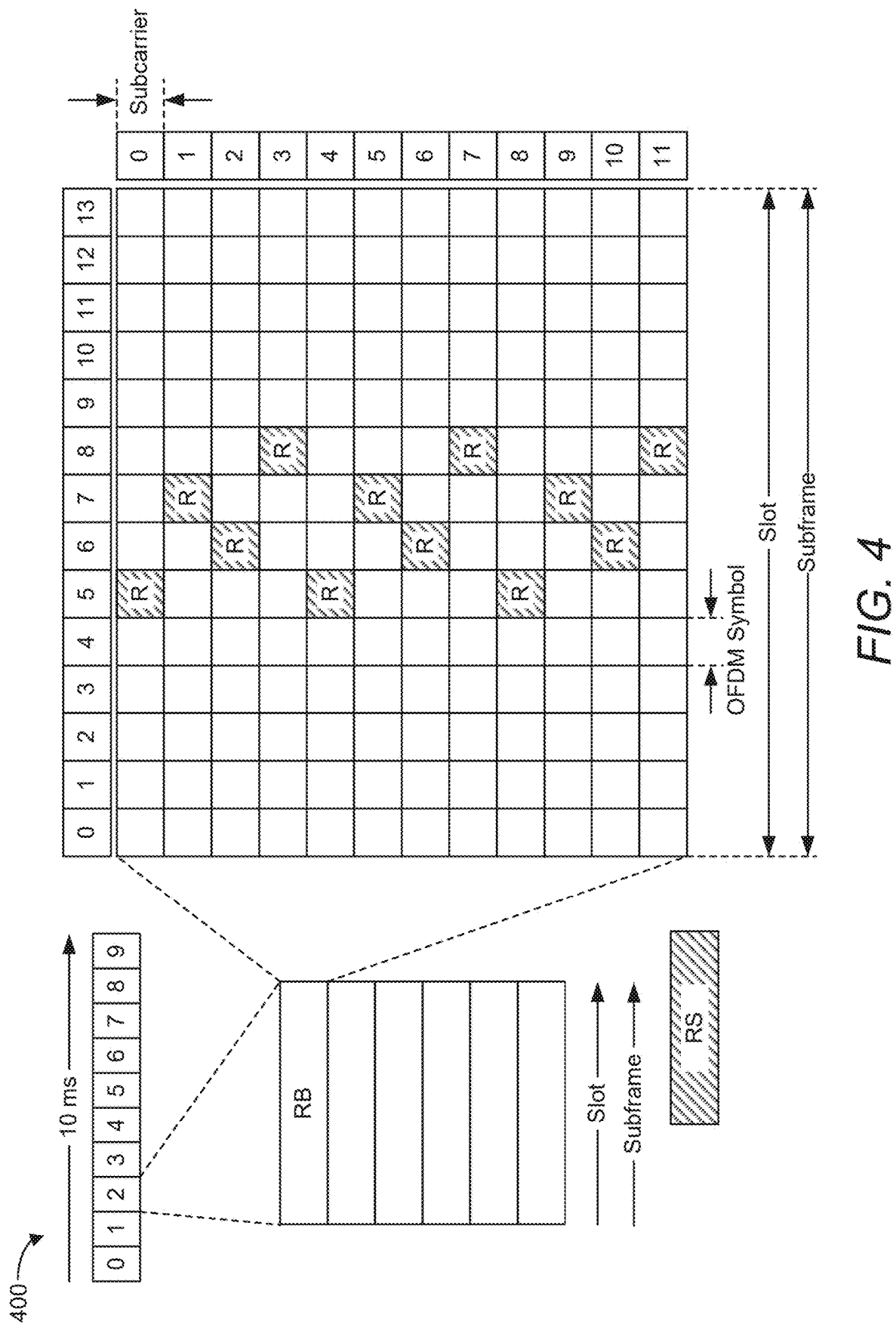


FIG. 3B

*FIG. 3C*



500

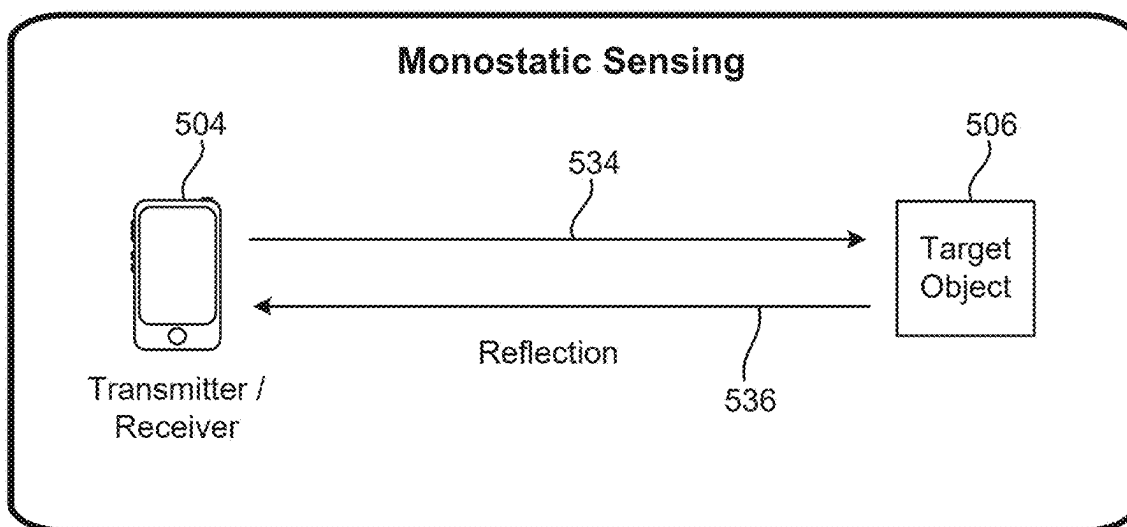


FIG. 5A

530

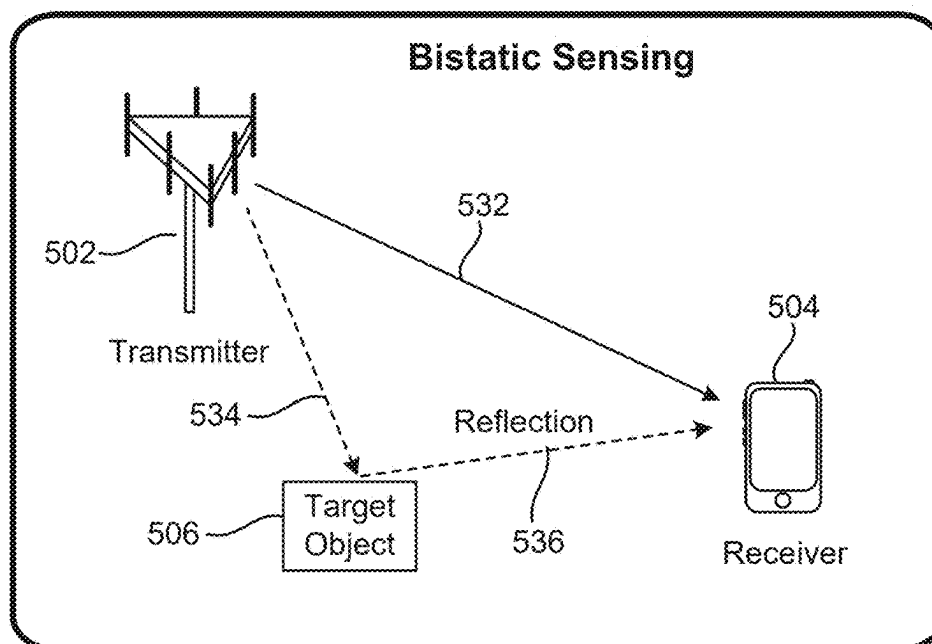
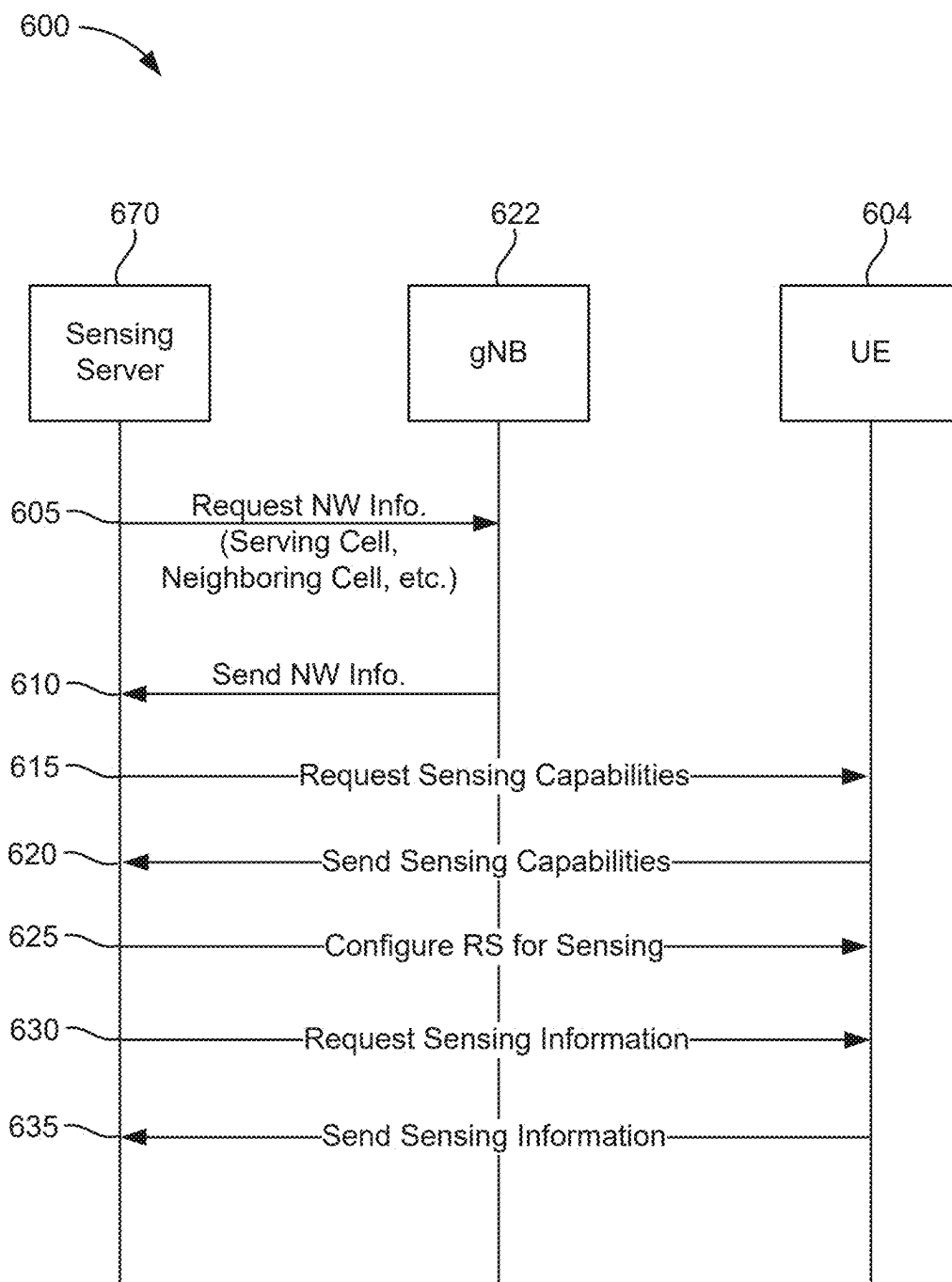


FIG. 5B

**FIG. 6**

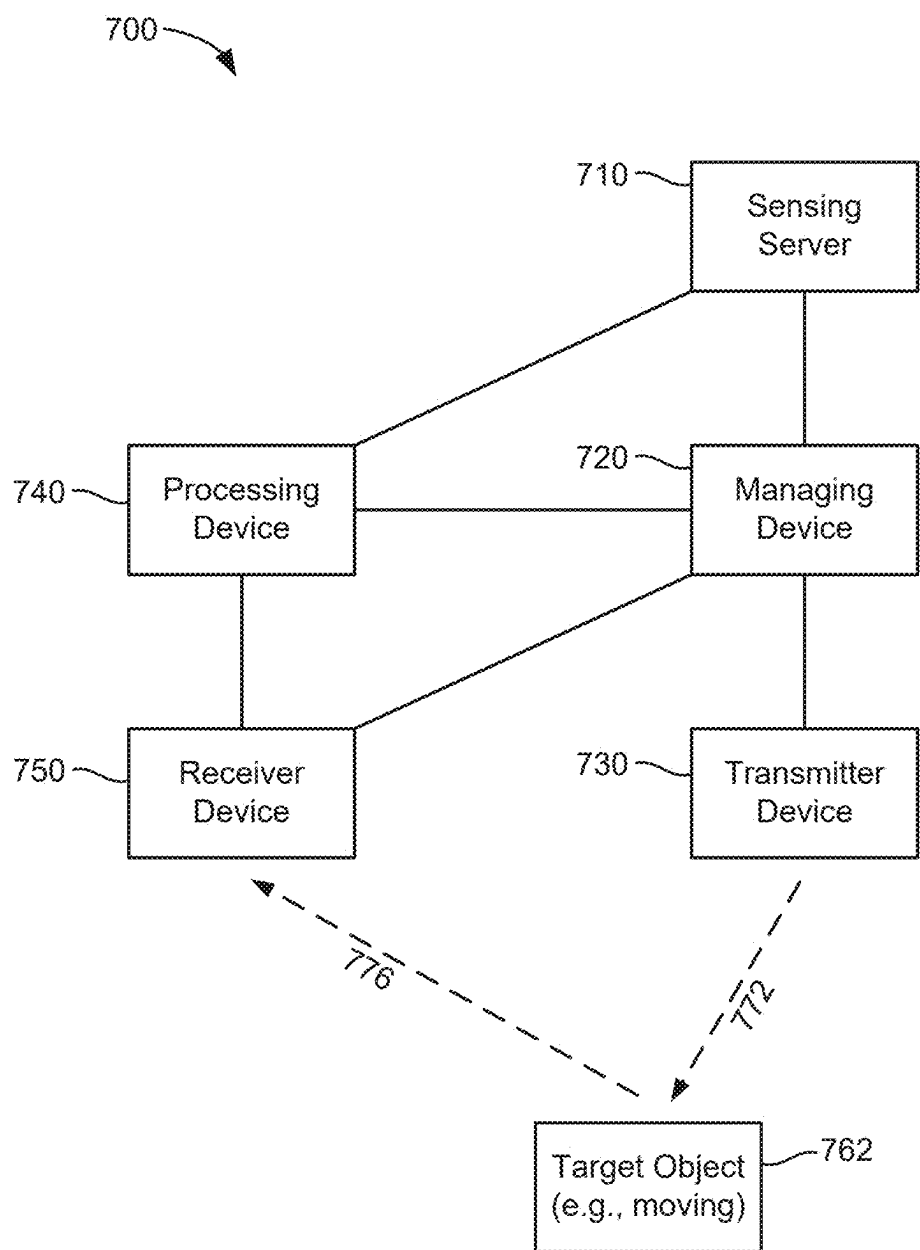


FIG. 7

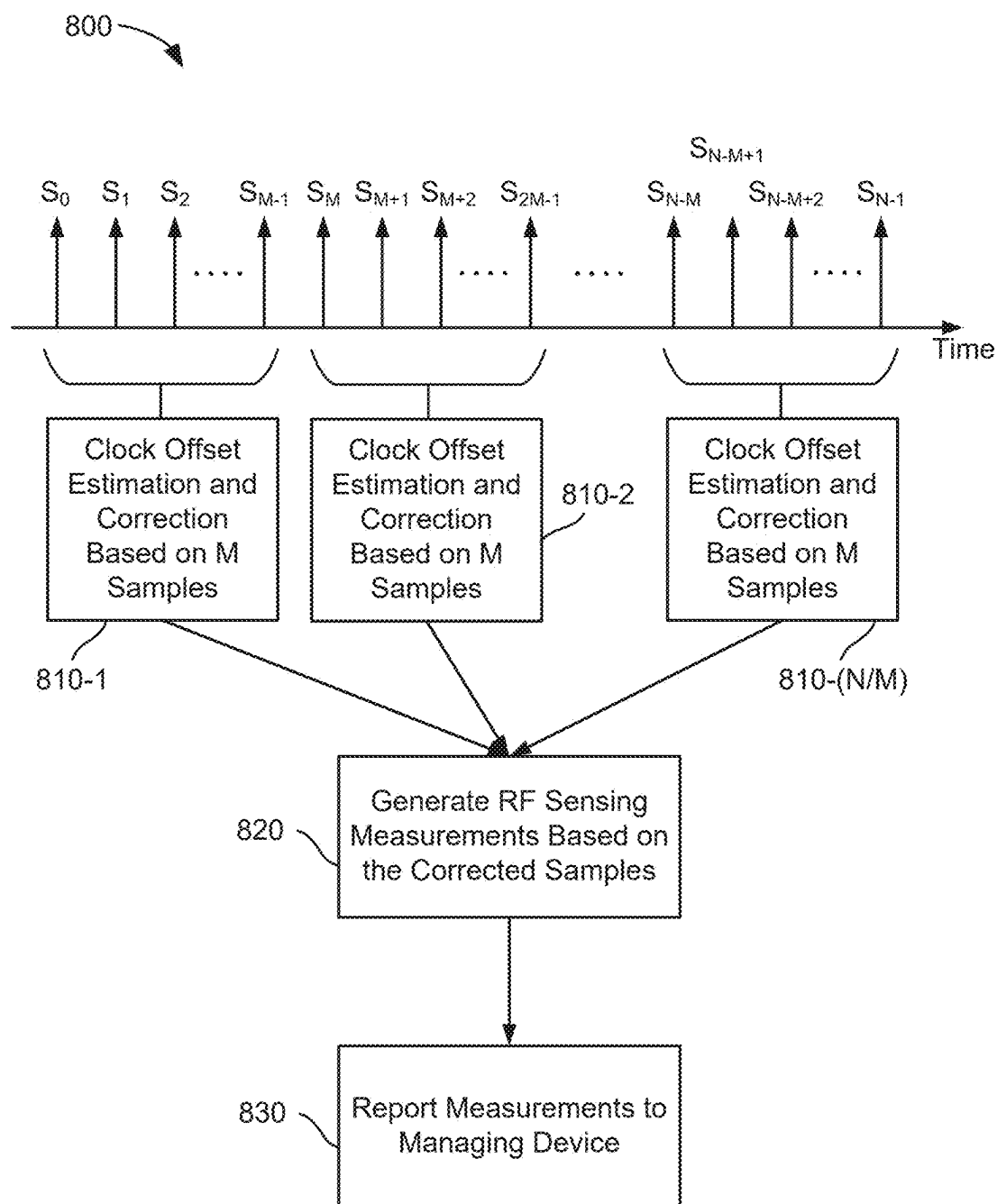


FIG. 8

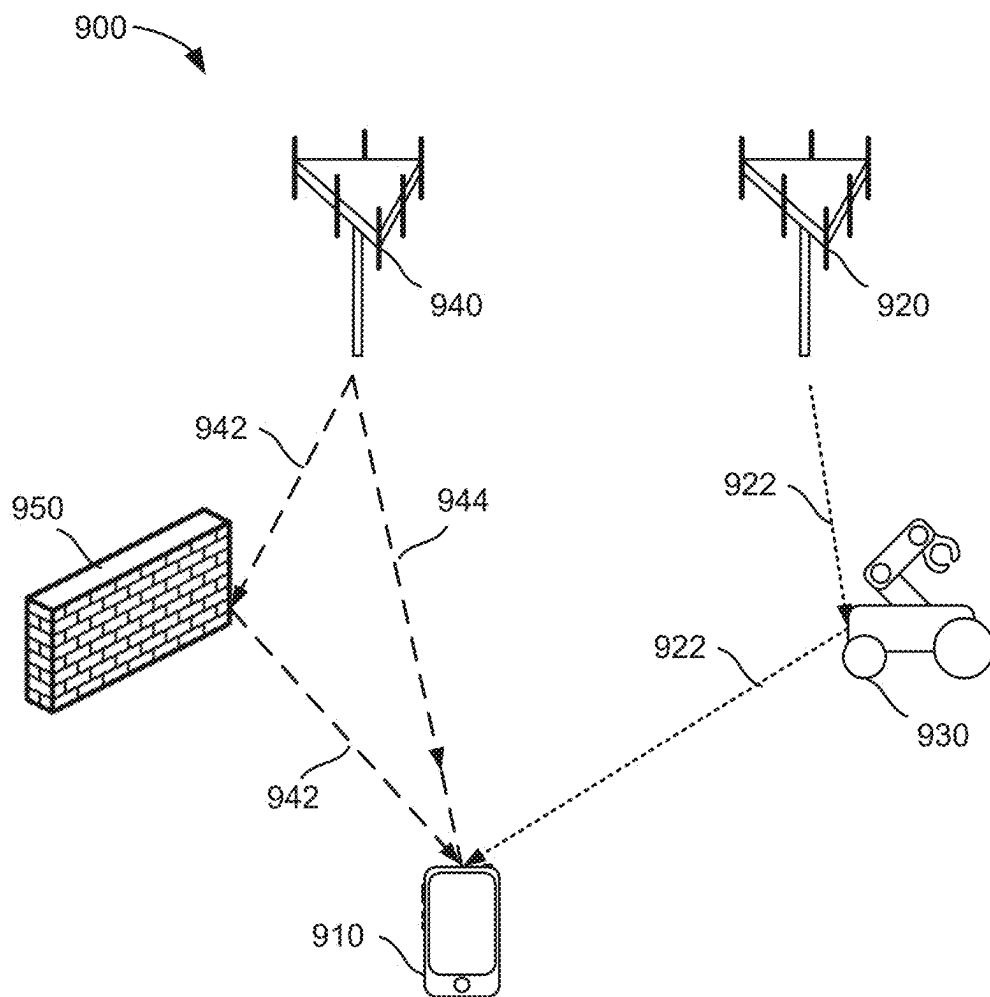


FIG. 9

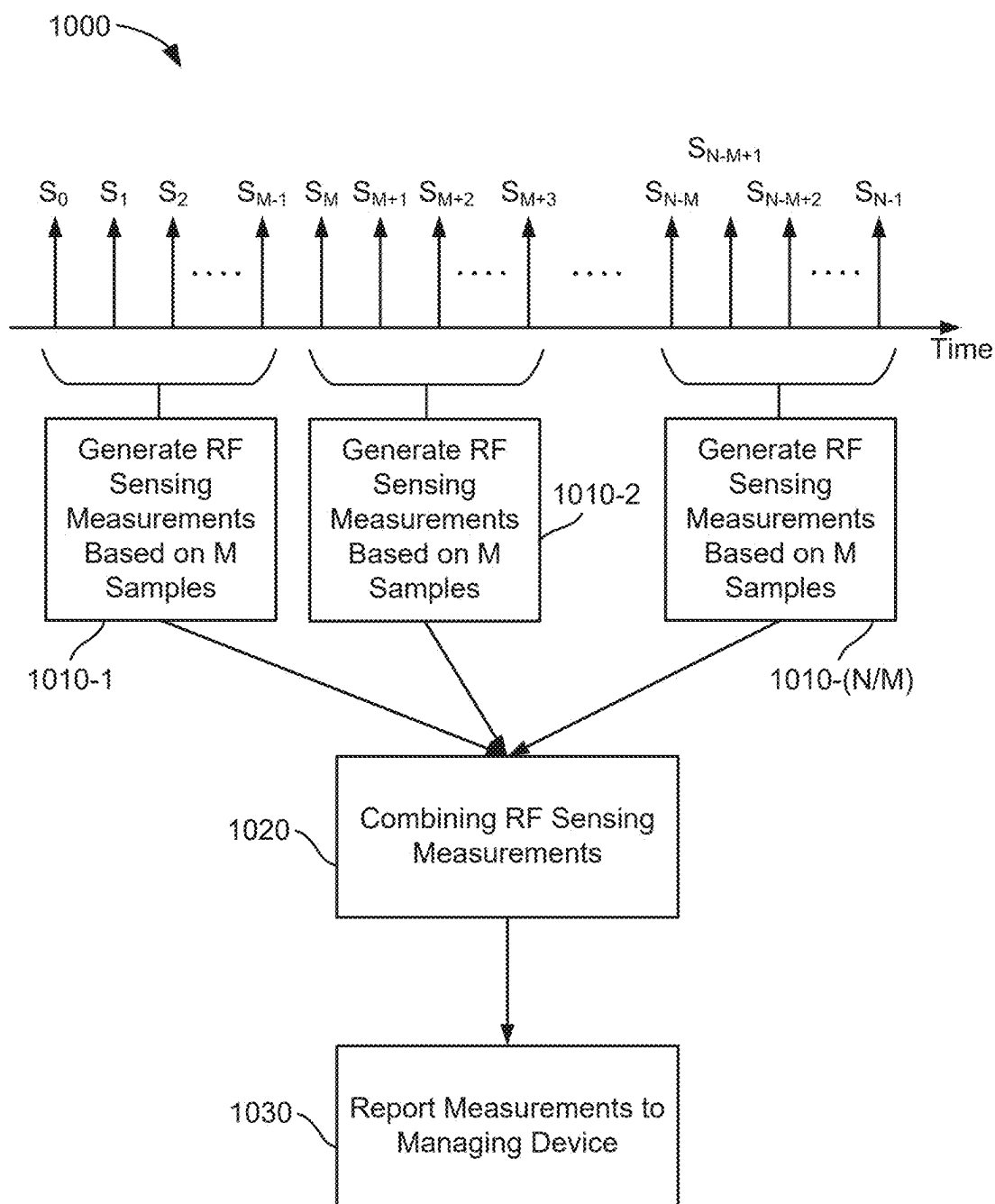


FIG. 10

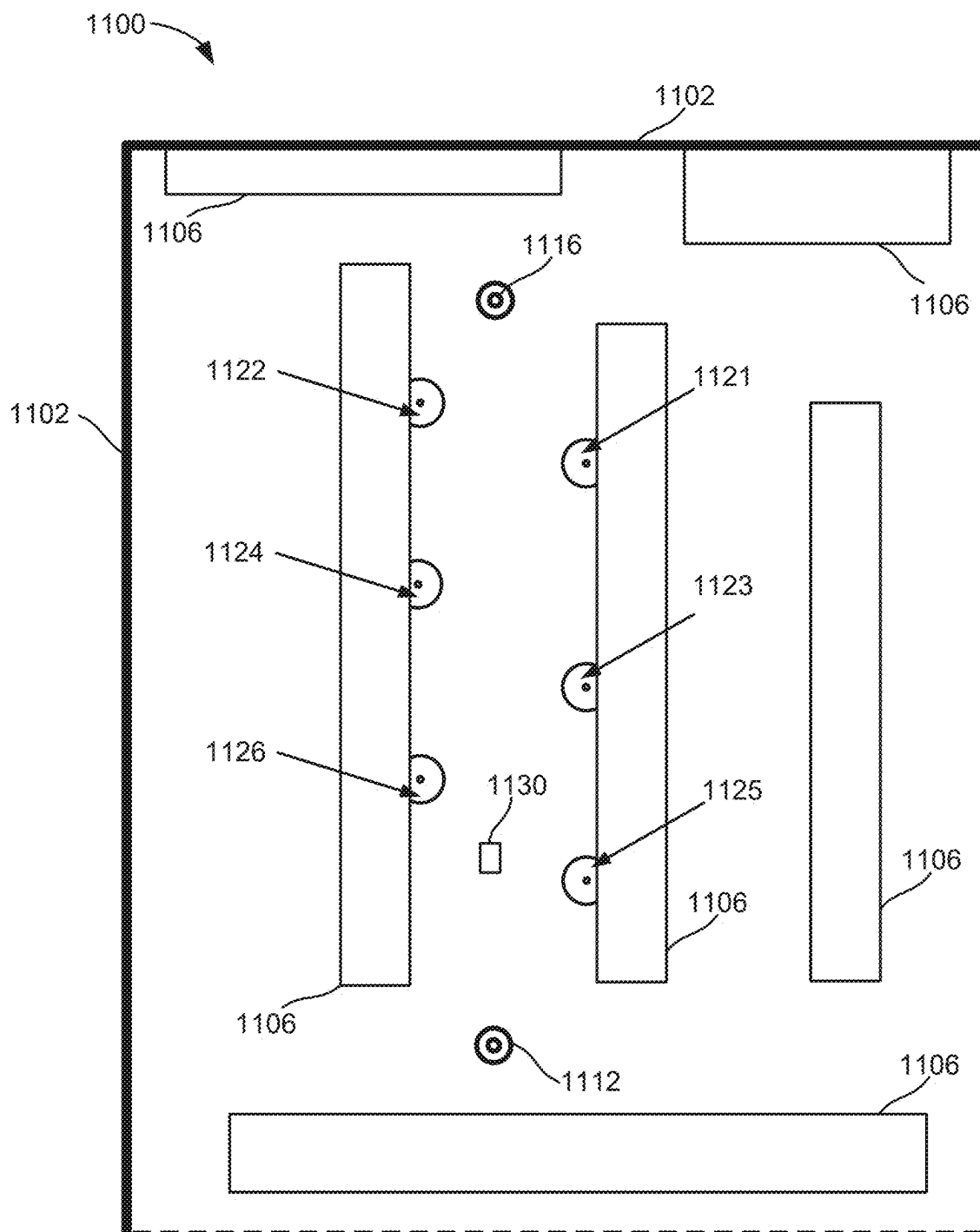


FIG. 11

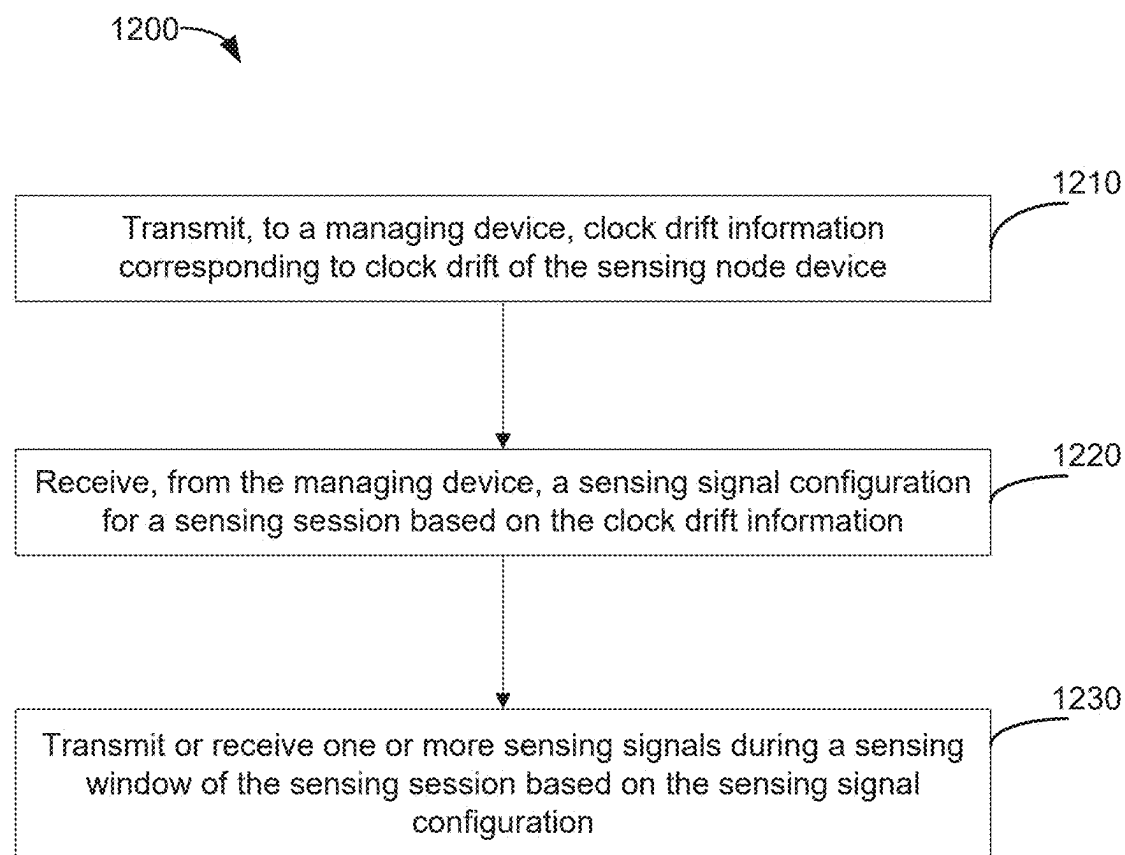
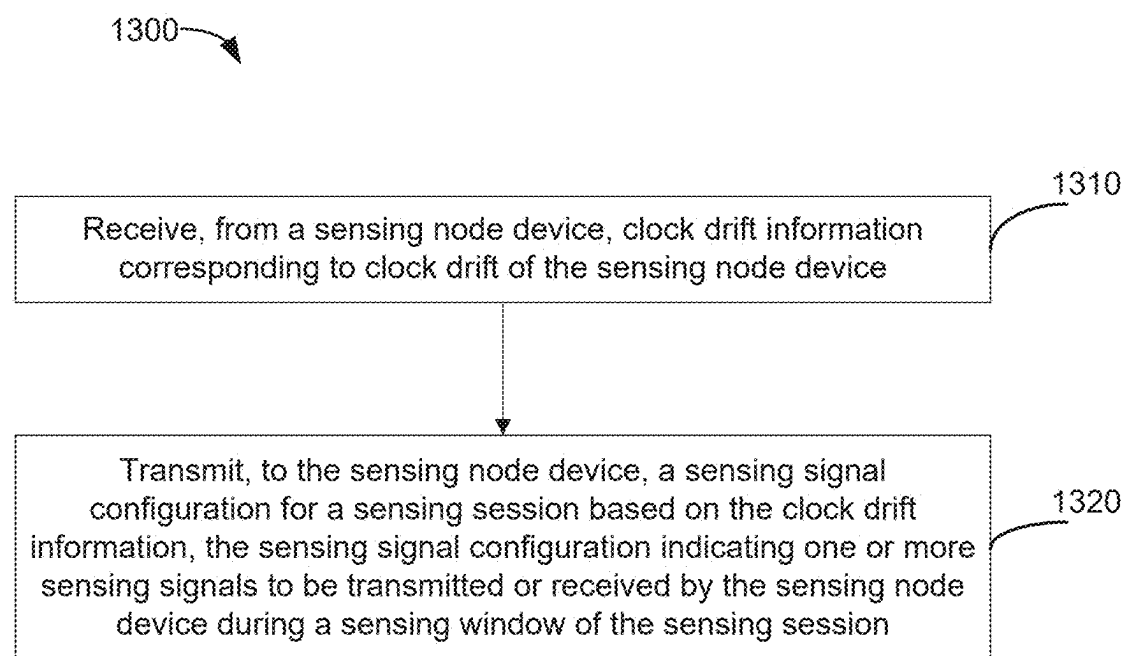


FIG. 12

*FIG. 13*

RADIO FREQUENCY (RF) SENSING SESSION BASED ON CLOCK DRIFT INFORMATION

BACKGROUND OF THE DISCLOSURE

1. Field of the Disclosure

[0001] Aspects of the disclosure relate generally to wireless technologies.

2. Description of the Related Art

[0002] Wireless communication systems have developed through various generations, including a first-generation analog wireless phone service (1G), a second-generation (2G) digital wireless phone service (including interim 2.5G and 2.75G networks), a third-generation (3G) high speed data, Internet-capable wireless service and a fourth-generation (4G) service (e.g., Long Term Evolution (LTE) or WiMax). There are presently many different types of wireless communication systems in use, including cellular and personal communications service (PCS) systems. Examples of known cellular systems include the cellular analog advanced mobile phone system (AMPS), and digital cellular systems based on code division multiple access (CDMA), frequency division multiple access (FDMA), time division multiple access (TDMA), the Global System for Mobile communications (GSM), etc.

[0003] A fifth generation (5G) wireless standard, referred to as New Radio (NR), enables higher data transfer speeds, greater numbers of connections, and better coverage, among other improvements. The 5G standard, according to the Next Generation Mobile Networks Alliance, is designed to provide higher data rates as compared to previous standards, more accurate positioning (e.g., based on reference signals for positioning (RS-P), such as downlink, uplink, or sidelink positioning reference signals (PRS)), and other technical enhancements. These enhancements, as well as the use of higher frequency bands, advances in PRS processes and technology, and high-density deployments for 5G, enable highly accurate 5G-based positioning.

SUMMARY

[0004] The following presents a simplified summary relating to one or more aspects disclosed herein. Thus, the following summary should not be considered an extensive overview relating to all contemplated aspects, nor should the following summary be considered to identify key or critical elements relating to all contemplated aspects or to delineate the scope associated with any particular aspect. Accordingly, the following summary has the sole purpose to present certain concepts relating to one or more aspects relating to the mechanisms disclosed herein in a simplified form to precede the detailed description presented below.

[0005] In an aspect, a method of operating a sensing node device includes transmitting, to a managing device, clock drift information corresponding to clock drift of the sensing node device; receiving, from the managing device, a sensing signal configuration for a sensing session based on the clock drift information; and transmitting or receiving one or more sensing signals during a sensing window of the sensing session based on the sensing signal configuration.

[0006] In an aspect, a method of operating a managing device includes receiving, from a sensing node device, clock

drift information corresponding to clock drift of the sensing node device; and transmitting, to the sensing node device, a sensing signal configuration for a sensing session based on the clock drift information, the sensing signal configuration indicating one or more sensing signals to be transmitted or received by the sensing node device during a sensing window of the sensing session.

[0007] In an aspect, a sensing node device includes one or more memories; one or more transceivers; and one or more processors communicatively coupled to the one or more memories and the one or more transceivers, the one or more processors, either alone or in combination, configured to: transmit, to a managing device via the one or more transceivers, clock drift information corresponding to clock drift of the sensing node device; receive, from the managing device via the one or more transceivers, a sensing signal configuration for a sensing session based on the clock drift information; and transmit or receive, via the one or more transceivers, one or more sensing signals during a sensing window of the sensing session based on the sensing signal configuration.

[0008] In an aspect, a managing device includes one or more memories; one or more transceivers; and one or more processors communicatively coupled to the one or more memories and the one or more transceivers, the one or more processors, either alone or in combination, configured to: receive, from a sensing node device via the one or more transceivers, clock drift information corresponding to clock drift of the sensing node device; and transmit, to the sensing node device via the one or more transceivers, a sensing signal configuration for a sensing session based on the clock drift information, the sensing signal configuration indicating one or more sensing signals to be transmitted or received by the sensing node device during a sensing window of the sensing session.

[0009] In an aspect, a sensing node device includes means for transmitting, to a managing device, clock drift information corresponding to clock drift of the sensing node device; means for receiving, from the managing device, a sensing signal configuration for a sensing session based on the clock drift information; and means for transmitting or receiving one or more sensing signals during a sensing window of the sensing session based on the sensing signal configuration.

[0010] In an aspect, a managing device includes means for receiving, from a sensing node device, clock drift information corresponding to clock drift of the sensing node device; and means for transmitting, to the sensing node device, a sensing signal configuration for a sensing session based on the clock drift information, the sensing signal configuration indicating one or more sensing signals to be transmitted or received by the sensing node device during a sensing window of the sensing session.

[0011] In an aspect, a non-transitory computer-readable medium stores computer-executable instructions that, when executed by a sensing node device, cause the sensing node device to: transmit, to a managing device, clock drift information corresponding to clock drift of the sensing node device; receive, from the managing device, a sensing signal configuration for a sensing session based on the clock drift information; and transmit or receive one or more sensing signals during a sensing window of the sensing session based on the sensing signal configuration.

[0012] In an aspect, a non-transitory computer-readable medium stores computer-executable instructions that, when

executed by a managing device, cause the managing device to: receive, from a sensing node device, clock drift information corresponding to clock drift of the sensing node device; and transmit, to the sensing node device, a sensing signal configuration for a sensing session based on the clock drift information, the sensing signal configuration indicating one or more sensing signals to be transmitted or received by the sensing node device during a sensing window of the sensing session.

[0013] Other objects and advantages associated with the aspects disclosed herein will be apparent to those skilled in the art based on the accompanying drawings and detailed description.

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] The accompanying drawings are presented to aid in the description of various aspects of the disclosure and are provided solely for illustration of the aspects and not limitation thereof.

[0015] FIG. 1 illustrates an example wireless communications system, according to aspects of the disclosure.

[0016] FIGS. 2A, 2B, and 2C illustrate example wireless network structures, according to aspects of the disclosure.

[0017] FIGS. 3A, 3B, and 3C are simplified block diagrams of several sample aspects of components that may be employed in a user equipment (UE), a base station, and a network entity, respectively, and configured to support communications as taught herein.

[0018] FIG. 4 is a diagram illustrating an example frame structure, according to aspects of the disclosure.

[0019] FIGS. 5A and 5B illustrate different types of wireless sensing, according to aspects of the disclosure.

[0020] FIG. 6 illustrates an example call flow for a New Radio (NR)-based sensing procedure in which the network configures the sensing parameters, according to aspects of the disclosure.

[0021] FIG. 7 is a block diagram illustrating a system for performing a sensing session, according to aspects of the disclosure.

[0022] FIG. 8 illustrates a processing flow for processing signal samples by a sensing node device based on a second example approach, according to aspects of the disclosure.

[0023] FIG. 9 illustrates an example scenario for determining a clock drift correction estimation, according to aspects of the disclosure.

[0024] FIG. 10 illustrates a processing flow for processing signal samples by a sensing node device based on a third example approach, according to aspects of the disclosure.

[0025] FIG. 11 illustrates an example application of performing a sensing session in a factory-like environment, according to aspects of the disclosure.

[0026] FIG. 12 is a flowchart illustrating a method of operating a sensing node device, according to aspects of the disclosure.

[0027] FIG. 13 is a flowchart illustrating a method of operating a managing device, according to aspects of the disclosure.

DETAILED DESCRIPTION

[0028] Aspects of the disclosure are provided in the following description and related drawings directed to various examples provided for illustration purposes. Alternate aspects may be devised without departing from the scope of

the disclosure. Additionally, well-known elements of the disclosure will not be described in detail or will be omitted so as not to obscure the relevant details of the disclosure.

[0029] Various aspects relate generally to configuring a sensing session based on clock drift information of a sensing node device. Some aspects more specifically relate to a sensing node device reporting its clock drift information to a managing device, and the managing device configuring a sensing session based on the clock drift information.

[0030] Particular aspects of the subject matter described in this disclosure can be implemented to realize one or more of the following potential advantages. In some examples, by a sensing node device reporting its clock drift information to a managing device, the described techniques can be used to configure a sensing signal configuration and/or a measurement configuration for a sensing session such that the range resolution or the velocity resolution of the sensing session may be kept within a desirable level by taking the clock drift of the sensing node device into consideration.

[0031] The words “exemplary” and/or “example” are used herein to mean “serving as an example, instance, or illustration.” Any aspect described herein as “exemplary” and/or “example” is not necessarily to be construed as preferred or advantageous over other aspects. Likewise, the term “aspects of the disclosure” does not require that all aspects of the disclosure include the discussed feature, advantage or mode of operation.

[0032] Those of skill in the art will appreciate that the information and signals described below may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the description below may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof, depending in part on the particular application, in part on the desired design, in part on the corresponding technology, etc.

[0033] Further, many aspects are described in terms of sequences of actions to be performed by, for example, elements of a computing device. It will be recognized that various actions described herein can be performed by specific circuits (e.g., application specific integrated circuits (ASICs)), by program instructions being executed by one or more processors, or by a combination of both. Additionally, the sequence(s) of actions described herein can be considered to be embodied entirely within any form of non-transitory computer-readable storage medium having stored therein a corresponding set of computer instructions that, upon execution, would cause or instruct an associated processor of a device to perform the functionality described herein. Thus, the various aspects of the disclosure may be embodied in a number of different forms, all of which have been contemplated to be within the scope of the claimed subject matter. In addition, for each of the aspects described herein, the corresponding form of any such aspects may be described herein as, for example, “logic configured to” perform the described action.

[0034] As used herein, the terms “user equipment” (UE) and “base station” are not intended to be specific or otherwise limited to any particular radio access technology (RAT), unless otherwise noted. In general, a UE may be any wireless communication device (e.g., a mobile phone, router, tablet computer, laptop computer, consumer asset

locating device, wearable (e.g., smartwatch, glasses, augmented reality (AR)/virtual reality (VR) headset, etc.), vehicle (e.g., automobile, motorcycle, bicycle, etc.), Internet of Things (IoT) device, etc.) used by a user to communicate over a wireless communications network. A UE may be mobile or may (e.g., at certain times) be stationary, and may communicate with a radio access network (RAN). As used herein, the term “UE” may be referred to interchangeably as an “access terminal” or “AT,” a “client device,” a “wireless device,” a “subscriber device,” a “subscriber terminal,” a “subscriber station,” a “user terminal” or “UT,” a “mobile device,” a “mobile terminal,” a “mobile station,” or variations thereof. Generally, UEs can communicate with a core network via a RAN, and through the core network the UEs can be connected with external networks such as the Internet and with other UEs. Of course, other mechanisms of connecting to the core network and/or the Internet are also possible for the UEs, such as over wired access networks, wireless local area network (WLAN) networks (e.g., based on the Institute of Electrical and Electronics Engineers (IEEE) 802.11 specification, etc.) and so on.

[0035] A base station may operate according to one of several RATs in communication with UEs depending on the network in which it is deployed, and may be alternatively referred to as an access point (AP), a network node, a NodeB, an evolved NodeB (eNB), a next generation eNB (ng-eNB), a New Radio (NR) Node B (also referred to as a gNB or gNodeB), etc. A base station may be used primarily to support wireless access by UEs, including supporting data, voice, and/or signaling connections for the supported UEs. In some systems a base station may provide purely edge node signaling functions while in other systems it may provide additional control and/or network management functions. A communication link through which UEs can send signals to a base station is called an uplink (UL) channel (e.g., a reverse traffic channel, a reverse control channel, an access channel, etc.). A communication link through which the base station can send signals to UEs is called a downlink (DL) or forward link channel (e.g., a paging channel, a control channel, a broadcast channel, a forward traffic channel, etc.). As used herein the term traffic channel (TCH) can refer to either an uplink/reverse or downlink/forward traffic channel.

[0036] The term “base station” may refer to a single physical transmission-reception point (TRP) or to multiple physical TRPs that may or may not be co-located. For example, where the term “base station” refers to a single physical TRP, the physical TRP may be an antenna of the base station corresponding to a cell (or several cell sectors) of the base station. Where the term “base station” refers to multiple co-located physical TRPs, the physical TRPs may be an array of antennas (e.g., as in a multiple-input multiple-output (MIMO) system or where the base station employs beamforming) of the base station. Where the term “base station” refers to multiple non-co-located physical TRPs, the physical TRPs may be a distributed antenna system (DAS) (a network of spatially separated antennas connected to a common source via a transport medium) or a remote radio head (RRH) (a remote base station connected to a serving base station). Alternatively, the non-co-located physical TRPs may be the serving base station receiving the measurement report from the UE and a neighbor base station whose reference radio frequency (RF) signals the UE is measuring. Because a TRP is the point from which a base

station transmits and receives wireless signals, as used herein, references to transmission from or reception at a base station are to be understood as referring to a particular TRP of the base station.

[0037] In some implementations that support positioning of UEs, a base station may not support wireless access by UEs (e.g., may not support data, voice, and/or signaling connections for UEs), but may instead transmit reference signals to UEs to be measured by the UEs, and/or may receive and measure signals transmitted by the UEs. Such a base station may be referred to as a positioning beacon (e.g., when transmitting signals to UEs) and/or as a location measurement unit (e.g., when receiving and measuring signals from UEs).

[0038] An “RF signal” comprises an electromagnetic wave of a given frequency that transports information through the space between a transmitter and a receiver. As used herein, a transmitter may transmit a single “RF signal” or multiple “RF signals” to a receiver. However, the receiver may receive multiple “RF signals” corresponding to each transmitted RF signal due to the propagation characteristics of RF signals through multipath channels. The same transmitted RF signal on different paths between the transmitter and receiver may be referred to as a “multipath” RF signal. As used herein, an RF signal may also be referred to as a “wireless signal” or simply a “signal” where it is clear from the context that the term “signal” refers to a wireless signal or an RF signal.

[0039] FIG. 1 illustrates an example wireless communications system 100, according to aspects of the disclosure. The wireless communications system 100 (which may also be referred to as a wireless wide area network (WWAN)) may include various base stations 102 (labeled “BS”) and various UEs 104. The base stations 102 may include macro cell base stations (high power cellular base stations) and/or small cell base stations (low power cellular base stations). In an aspect, the macro cell base stations may include eNBs and/or ng-eNBs where the wireless communications system 100 corresponds to an LTE network, or gNBs where the wireless communications system 100 corresponds to a NR network, or a combination of both, and the small cell base stations may include femtocells, picocells, microcells, etc.

[0040] The base stations 102 may collectively form a RAN and interface with a core network 170 (e.g., an evolved packet core (EPC) or a 5G core (5GC)) through backhaul links 122, and through the core network 170 to one or more location servers 172 (e.g., a location management function (LMF) or a secure user plane location (SUPL) location platform (SLP)). The location server(s) 172 may be part of core network 170 or may be external to core network 170. A location server 172 may be integrated with a base station 102. A UE 104 may communicate with a location server 172 directly or indirectly. For example, a UE 104 may communicate with a location server 172 via the base station 102 that is currently serving that UE 104. A UE 104 may also communicate with a location server 172 through another path, such as via an application server (not shown), via another network, such as via a wireless local area network (WLAN) access point (AP) (e.g., AP 150 described below), and so on. For signaling purposes, communication between a UE 104 and a location server 172 may be represented as an indirect connection (e.g., through the core network 170, etc.) or a direct connection (e.g., as shown via direct

connection **128**), with the intervening nodes (if any) omitted from a signaling diagram for clarity.

[0041] In addition to other functions, the base stations **102** may perform functions that relate to one or more of transferring user data, radio channel ciphering and deciphering, integrity protection, header compression, mobility control functions (e.g., handover, dual connectivity), inter-cell interference coordination, connection setup and release, load balancing, distribution for non-access stratum (NAS) messages, NAS node selection, synchronization, RAN sharing, multimedia broadcast multicast service (MBMS), subscriber and equipment trace, RAN information management (RIM), paging, positioning, and delivery of warning messages. The base stations **102** may communicate with each other directly or indirectly (e.g., through the EPC/5GC) over backhaul links **134**, which may be wired or wireless.

[0042] The base stations **102** may wirelessly communicate with the UEs **104**. Each of the base stations **102** may provide communication coverage for a respective geographic coverage area **110**. In an aspect, one or more cells may be supported by a base station **102** in each geographic coverage area **110**. A “cell” is a logical communication entity used for communication with a base station (e.g., over some frequency resource, referred to as a carrier frequency, component carrier, carrier, band, or the like), and may be associated with an identifier (e.g., a physical cell identifier (PCI), an enhanced cell identifier (ECI), a virtual cell identifier (VCI), a cell global identifier (CGI), etc.) for distinguishing cells operating via the same or a different carrier frequency. In some cases, different cells may be configured according to different protocol types (e.g., machine-type communication (MTC), narrowband IoT (NB-IoT), enhanced mobile broadband (eMBB), or others) that may provide access for different types of UEs. Because a cell is supported by a specific base station, the term “cell” may refer to either or both of the logical communication entity and the base station that supports it, depending on the context. In addition, because a TRP is typically the physical transmission point of a cell, the terms “cell” and “TRP” may be used interchangeably. In some cases, the term “cell” may also refer to a geographic coverage area of a base station (e.g., a sector), insofar as a carrier frequency can be detected and used for communication within some portion of geographic coverage areas **110**.

[0043] While neighboring macro cell base station **102** geographic coverage areas **110** may partially overlap (e.g., in a handover region), some of the geographic coverage areas **110** may be substantially overlapped by a larger geographic coverage area **110**. For example, a small cell base station **102'** (labeled “SC” for “small cell”) may have a geographic coverage area **110'** that substantially overlaps with the geographic coverage area **110** of one or more macro cell base stations **102**. A network that includes both small cell and macro cell base stations may be known as a heterogeneous network. A heterogeneous network may also include home eNBs (HeNBs), which may provide service to a restricted group known as a closed subscriber group (CSG).

[0044] The communication links **120** between the base stations **102** and the UEs **104** may include uplink (also referred to as reverse link) transmissions from a UE **104** to a base station **102** and/or downlink (DL) (also referred to as forward link) transmissions from a base station **102** to a UE **104**. The communication links **120** may use MIMO antenna technology, including spatial multiplexing, beamforming, and/or transmit diversity. The communication links **120** may

be through one or more carrier frequencies. Allocation of carriers may be asymmetric with respect to downlink and uplink (e.g., more or less carriers may be allocated for downlink than for uplink).

[0045] The wireless communications system **100** may further include a wireless local area network (WLAN) access point (AP) **150** in communication with WLAN stations (STAs) **152** via communication links **154** in an unlicensed frequency spectrum (e.g., 5 GHz). When communicating in an unlicensed frequency spectrum, the WLAN STAs **152** and/or the WLAN AP **150** may perform a clear channel assessment (CCA) or listen before talk (LBT) procedure prior to communicating in order to determine whether the channel is available.

[0046] The small cell base station **102'** may operate in a licensed and/or an unlicensed frequency spectrum. When operating in an unlicensed frequency spectrum, the small cell base station **102'** may employ LTE or NR technology and use the same 5 GHz unlicensed frequency spectrum as used by the WLAN AP **150**. The small cell base station **102'**, employing LTE/5G in an unlicensed frequency spectrum, may boost coverage to and/or increase capacity of the access network. NR in unlicensed spectrum may be referred to as NR-U. LTE in an unlicensed spectrum may be referred to as LTE-U, licensed assisted access (LAA), or MULTIFIRE®.

[0047] The wireless communications system **100** may further include a millimeter wave (mmW) base station **180** that may operate in mmW frequencies and/or near mmW frequencies in communication with a UE **182**. Extremely high frequency (EHF) is part of the RF in the electromagnetic spectrum. EHF has a range of 30 GHz to 300 GHz and a wavelength between 1 millimeter and 10 millimeters. Radio waves in this band may be referred to as a millimeter wave. Near mmW may extend down to a frequency of 3 GHz with a wavelength of 100 millimeters. The super high frequency (SHF) band extends between 3 GHz and 30 GHz, also referred to as centimeter wave. Communications using the mmW/near mmW radio frequency band have high path loss and a relatively short range. The mmW base station **180** and the UE **182** may utilize beamforming (transmit and/or receive) over a mmW communication link **184** to compensate for the extremely high path loss and short range. Further, it will be appreciated that in alternative configurations, one or more base stations **102** may also transmit using mmW or near mmW and beamforming. Accordingly, it will be appreciated that the foregoing illustrations are merely examples and should not be construed to limit the various aspects disclosed herein.

[0048] Transmit beamforming is a technique for focusing an RF signal in a specific direction. Traditionally, when a network node (e.g., a base station) broadcasts an RF signal, it broadcasts the signal in all directions (omni-directionally). With transmit beamforming, the network node determines where a given target device (e.g., a UE) is located (relative to the transmitting network node) and projects a stronger downlink RF signal in that specific direction, thereby providing a faster (in terms of data rate) and stronger RF signal for the receiving device(s). To change the directionality of the RF signal when transmitting, a network node can control the phase and relative amplitude of the RF signal at each of the one or more transmitters that are broadcasting the RF signal. For example, a network node may use an array of antennas (referred to as a “phased array” or an “antenna array”) that creates a beam of RF waves that can be

“steered” to point in different directions, without actually moving the antennas. Specifically, the RF current from the transmitter is fed to the individual antennas with the correct phase relationship so that the radio waves from the separate antennas add together to increase the radiation in a desired direction, while cancelling to suppress radiation in undesired directions.

[0049] Transmit beams may be quasi-co-located, meaning that they appear to the receiver (e.g., a UE) as having the same parameters, regardless of whether or not the transmitting antennas of the network node themselves are physically co-located. In NR, there are four types of quasi-co-location (QCL) relations. Specifically, a QCL relation of a given type means that certain parameters about a second reference RF signal on a second beam can be derived from information about a source reference RF signal on a source beam. Thus, if the source reference RF signal is QCL Type A, the receiver can use the source reference RF signal to estimate the Doppler shift, Doppler spread, average delay, and delay spread of a second reference RF signal transmitted on the same channel. If the source reference RF signal is QCL Type B, the receiver can use the source reference RF signal to estimate the Doppler shift and Doppler spread of a second reference RF signal transmitted on the same channel. If the source reference RF signal is QCL Type C, the receiver can use the source reference RF signal to estimate the Doppler shift and average delay of a second reference RF signal transmitted on the same channel. If the source reference RF signal is QCL Type D, the receiver can use the source reference RF signal to estimate the spatial receive parameter of a second reference RF signal transmitted on the same channel.

[0050] In receive beamforming, the receiver uses a receive beam to amplify RF signals detected on a given channel. For example, the receiver can increase the gain setting and/or adjust the phase setting of an array of antennas in a particular direction to amplify (e.g., to increase the gain level of) the RF signals received from that direction. Thus, when a receiver is said to beamform in a certain direction, it means the beam gain in that direction is high relative to the beam gain along other directions, or the beam gain in that direction is the highest compared to the beam gain in that direction of all other receive beams available to the receiver. This results in a stronger received signal strength (e.g., reference signal received power (RSRP), reference signal received quality (RSRQ), signal-to-interference-plus-noise ratio (SINR), etc.) of the RF signals received from that direction.

[0051] Transmit and receive beams may be spatially related. A spatial relation means that parameters for a second beam (e.g., a transmit or receive beam) for a second reference signal can be derived from information about a first beam (e.g., a receive beam or a transmit beam) for a first reference signal. For example, a UE may use a particular receive beam to receive a reference downlink reference signal (e.g., synchronization signal block (SSB)) from a base station. The UE can then form a transmit beam for sending an uplink reference signal (e.g., sounding reference signal (SRS)) to that base station based on the parameters of the receive beam.

[0052] Note that a “downlink” beam may be either a transmit beam or a receive beam, depending on the entity forming it. For example, if a base station is forming the downlink beam to transmit a reference signal to a UE, the downlink beam is a transmit beam. If the UE is forming the

downlink beam, however, it is a receive beam to receive the downlink reference signal. Similarly, an “uplink” beam may be either a transmit beam or a receive beam, depending on the entity forming it. For example, if a base station is forming the uplink beam, it is an uplink receive beam, and if a UE is forming the uplink beam, it is an uplink transmit beam.

[0053] The electromagnetic spectrum is often subdivided, based on frequency/wavelength, into various classes, bands, channels, etc. In 5G NR two initial operating bands have been identified as frequency range designations FR1 (410 MHz-7.125 GHz) and FR2 (24.25 GHz-52.6 GHz). It should be understood that although a portion of FR1 is greater than 6 GHz, FR1 is often referred to (interchangeably) as a “Sub-6 GHz” band in various documents and articles. A similar nomenclature issue sometimes occurs with regard to FR2, which is often referred to (interchangeably) as a “millimeter wave” band in documents and articles, despite being different from the extremely high frequency (EHF) band (30 GHz-300 GHz) which is identified by the INTERNATIONAL TELECOMMUNICATION UNION® as a “millimeter wave” band.

[0054] The frequencies between FR1 and FR2 are often referred to as mid-band frequencies. Recent 5G NR studies have identified an operating band for these mid-band frequencies as frequency range designation FR3 (7.125 GHz-24.25 GHz). Frequency bands falling within FR3 may inherit FR1 characteristics and/or FR2 characteristics, and thus may effectively extend features of FR1 and/or FR2 into mid-band frequencies. In addition, higher frequency bands are currently being explored to extend 5G NR operation beyond 52.6 GHz. For example, three higher operating bands have been identified as frequency range designations FR4a or FR4-1 (52.6 GHz-71 GHz), FR4 (52.6 GHz-114.25 GHz), and FR5 (114.25 GHz-300 GHz). Each of these higher frequency bands falls within the EHF band.

[0055] With the above aspects in mind, unless specifically stated otherwise, it should be understood that the term “sub-6 GHz” or the like if used herein may broadly represent frequencies that may be less than 6 GHz, may be within FR1, or may include mid-band frequencies. Further, unless specifically stated otherwise, it should be understood that the term “millimeter wave” or the like if used herein may broadly represent frequencies that may include mid-band frequencies, may be within FR2, FR4, FR4-a or FR4-1, and/or FR5, or may be within the EHF band.

[0056] In a multi-carrier system, such as 5G, one of the carrier frequencies is referred to as the “primary carrier” or “anchor carrier” or “primary serving cell” or “PCell,” and the remaining carrier frequencies are referred to as “secondary carriers” or “secondary serving cells” or “SCells.” In carrier aggregation, the anchor carrier is the carrier operating on the primary frequency (e.g., FR1) utilized by a UE **104/182** and the cell in which the UE **104/182** either performs the initial radio resource control (RRC) connection establishment procedure or initiates the RRC connection re-establishment procedure. The primary carrier carries all common and UE-specific control channels, and may be a carrier in a licensed frequency (however, this is not always the case). A secondary carrier is a carrier operating on a second frequency (e.g., FR2) that may be configured once the RRC connection is established between the UE **104** and the anchor carrier and that may be used to provide additional radio resources. In some cases, the secondary carrier may be

a carrier in an unlicensed frequency. The secondary carrier may contain only necessary signaling information and signals, for example, those that are UE-specific may not be present in the secondary carrier, since both primary uplink and downlink carriers are typically UE-specific. This means that different UEs **104/182** in a cell may have different downlink primary carriers. The same is true for the uplink primary carriers. The network is able to change the primary carrier of any UE **104/182** at any time. This is done, for example, to balance the load on different carriers. Because a “serving cell” (whether a PCell or an SCell) corresponds to a carrier frequency/component carrier over which some base station is communicating, the term “cell,” “serving cell,” “component carrier,” “carrier frequency,” and the like can be used interchangeably.

[0057] For example, still referring to FIG. 1, one of the frequencies utilized by the macro cell base stations **102** may be an anchor carrier (or “PCell”) and other frequencies utilized by the macro cell base stations **102** and/or the mmW base station **180** may be secondary carriers (“SCells”). The simultaneous transmission and/or reception of multiple carriers enables the UE **104/182** to significantly increase its data transmission and/or reception rates. For example, two 20 MHz aggregated carriers in a multi-carrier system would theoretically lead to a two-fold increase in data rate (i.e., 40 MHz), compared to that attained by a single 20 MHz carrier.

[0058] The wireless communications system **100** may further include a UE **164** that may communicate with a macro cell base station **102** over a communication link **120** and/or the mmW base station **180** over a mmW communication link **184**. For example, the macro cell base station **102** may support a PCell and one or more SCells for the UE **164** and the mmW base station **180** may support one or more SCells for the UE **164**.

[0059] In some cases, the UE **164** and the UE **182** may be capable of sidelink communication. Sidelink-capable UEs (SL-UEs) may communicate with base stations **102** over communication links **120** using the Uu interface (i.e., the air interface between a UE and a base station). SL-UEs (e.g., UE **164**, UE **182**) may also communicate directly with each other over a wireless sidelink **160** using the PC5 interface (i.e., the air interface between sidelink-capable UEs). A wireless sidelink (or just “sidelink”) is an adaptation of the core cellular (e.g., LTE, NR) standard that allows direct communication between two or more UEs without the communication needing to go through a base station. Sidelink communication may be unicast or multicast, and may be used for device-to-device (D2D) media-sharing, vehicle-to-vehicle (V2V) communication, vehicle-to-everything (V2X) communication (e.g., cellular V2X (cV2X) communication, enhanced V2X (eV2X) communication, etc.), emergency rescue applications, etc. One or more of a group of SL-UEs utilizing sidelink communications may be within the geographic coverage area **110** of a base station **102**. Other SL-UEs in such a group may be outside the geographic coverage area **110** of a base station **102** or be otherwise unable to receive transmissions from a base station **102**. In some cases, groups of SL-UEs communicating via sidelink communications may utilize a one-to-many (1:M) system in which each SL-UE transmits to every other SL-UE in the group. In some cases, a base station **102** facilitates the scheduling of resources for sidelink communications. In

other cases, sidelink communications are carried out between SL-UEs without the involvement of a base station **102**.

[0060] In an aspect, the sidelink **160** may operate over a wireless communication medium of interest, which may be shared with other wireless communications between other vehicles and/or infrastructure access points, as well as other RATs. A “medium” may be composed of one or more time, frequency, and/or space communication resources (e.g., encompassing one or more channels across one or more carriers) associated with wireless communication between one or more transmitter/receiver pairs. In an aspect, the medium of interest may correspond to at least a portion of an unlicensed frequency band shared among various RATs. Although different licensed frequency bands have been reserved for certain communication systems (e.g., by a government entity such as the Federal Communications Commission (FCC) in the United States), these systems, in particular those employing small cell access points, have recently extended operation into unlicensed frequency bands such as the Unlicensed National Information Infrastructure (U-NII) band used by wireless local area network (WLAN) technologies, most notably IEEE 802.11x WLAN technologies generally referred to as “Wi-Fi.” Example systems of this type include different variants of CDMA systems, TDMA systems, FDMA systems, orthogonal FDMA (OFDMA) systems, single-carrier FDMA (SC-FDMA) systems, and so on.

[0061] Note that although FIG. 1 only illustrates two of the UEs as SL-UEs (i.e., UEs **164** and **182**), any of the illustrated UEs may be SL-UEs. Further, although only UE **182** was described as being capable of beamforming, any of the illustrated UEs, including UE **164**, may be capable of beamforming. Where SL-UEs are capable of beamforming, they may beamform towards each other (i.e., towards other SL-UEs), towards other UEs (e.g., UEs **104**), towards base stations (e.g., base stations **102**, **180**, small cell **102'**, access point **150**), etc. Thus, in some cases, UEs **164** and **182** may utilize beamforming over sidelink **160**.

[0062] In the example of FIG. 1, any of the illustrated UEs (shown in FIG. 1 as a single UE **104** for simplicity) may receive signals **124** from one or more Earth orbiting space vehicles (SVs) **112** (e.g., satellites). In an aspect, the SVs **112** may be part of a satellite positioning system that a UE **104** can use as an independent source of location information. A satellite positioning system typically includes a system of transmitters (e.g., SVs **112**) positioned to enable receivers (e.g., UEs **104**) to determine their location on or above the Earth based, at least in part, on positioning signals (e.g., signals **124**) received from the transmitters. Such a transmitter typically transmits a signal marked with a repeating pseudo-random noise (PN) code of a set number of chips. While typically located in SVs **112**, transmitters may sometimes be located on ground-based control stations, base stations **102**, and/or other UEs **104**. A UE **104** may include one or more dedicated receivers specifically designed to receive signals **124** for deriving geo location information from the SVs **112**.

[0063] In a satellite positioning system, the use of signals **124** can be augmented by various satellite-based augmentation systems (SBAS) that may be associated with or otherwise enabled for use with one or more global and/or regional navigation satellite systems. For example an SBAS may include an augmentation system(s) that provides integ-

ity information, differential corrections, etc., such as the Wide Area Augmentation System (WAAS), the European Geostationary Navigation Overlay Service (EGNOS), the Multi-functional Satellite Augmentation System (MSAS), the Global Positioning System (GPS) Aided Geo Augmented Navigation or GPS and Geo Augmented Navigation system (GAGAN), and/or the like. Thus, as used herein, a satellite positioning system may include any combination of one or more global and/or regional navigation satellites associated with such one or more satellite positioning systems.

[0064] In an aspect, SVs 112 may additionally or alternatively be part of one or more non-terrestrial networks (NTNs). In an NTN, an SV 112 is connected to an earth station (also referred to as a ground station, NTN gateway, or gateway), which in turn is connected to an element in a 5G network, such as a modified base station 102 (without a terrestrial antenna) or a network node in a 5GC. This element would in turn provide access to other elements in the 5G network and ultimately to entities external to the 5G network, such as Internet web servers and other user devices. In that way, a UE 104 may receive communication signals (e.g., signals 124) from an SV 112 instead of, or in addition to, communication signals from a terrestrial base station 102.

[0065] The wireless communications system 100 may further include one or more UEs, such as UE 190, that connects indirectly to one or more communication networks via one or more device-to-device (D2D) peer-to-peer (P2P) links (referred to as “sidelinks”). In the example of FIG. 1, UE 190 has a D2D P2P link 192 with one of the UEs 104 connected to one of the base stations 102 (e.g., through which UE 190 may indirectly obtain cellular connectivity) and a D2D P2P link 194 with WLAN STA 152 connected to the WLAN AP 150 (through which UE 190 may indirectly obtain WLAN-based Internet connectivity). In an example, the D2D P2P links 192 and 194 may be supported with any well-known D2D RAT, such as LTE Direct (LTE-D), WI-FI DIRECT®, BLUETOOTH®, and so on.

[0066] FIG. 2A illustrates an example wireless network structure 200. For example, a 5GC 210 (also referred to as a Next Generation Core (NGC)) can be viewed functionally as control plane (C-plane) functions 214 (e.g., UE registration, authentication, network access, gateway selection, etc.) and user plane (U-plane) functions 212, (e.g., UE gateway function, access to data networks, IP routing, etc.) which operate cooperatively to form the core network. User plane interface (NG-U) 213 and control plane interface (NG-C) 215 connect the gNB 222 to the 5GC 210 and specifically to the user plane functions 212 and control plane functions 214, respectively. In an additional configuration, an ng-eNB 224 may also be connected to the 5GC 210 via NG-C 215 to the control plane functions 214 and NG-U 213 to user plane functions 212. Further, ng-eNB 224 may directly communicate with gNB 222 via a backhaul connection 223. In some configurations, a Next Generation RAN (NG-RAN) 220 may have one or more gNBs 222, while other configurations include one or more of both ng-eNBs 224 and gNBs 222. Either (or both) gNB 222 or ng-eNB 224 may communicate with one or more UEs 204 (e.g., any of the UEs described herein).

[0067] Another optional aspect may include a location server 230, which may be in communication with the 5GC 210 to provide location assistance for UE(s) 204. The

location server 230 can be implemented as a plurality of separate servers (e.g., physically separate servers, different software modules on a single server, different software modules spread across multiple physical servers, etc.), or alternately may each correspond to a single server. The location server 230 can be configured to support one or more location services for UEs 204 that can connect to the location server 230 via the core network, 5GC 210, and/or via the Internet (not illustrated). Further, the location server 230 may be integrated into a component of the core network, or alternatively may be external to the core network (e.g., a third party server, such as an original equipment manufacturer (OEM) server or service server).

[0068] FIG. 2B illustrates another example wireless network structure 240. A 5GC 260 (which may correspond to 5GC 210 in FIG. 2A) can be viewed functionally as control plane functions, provided by an access and mobility management function (AMF) 264, and user plane functions, provided by a user plane function (UPF) 262, which operate cooperatively to form the core network (i.e., 5GC 260). The functions of the AMF 264 include registration management, connection management, reachability management, mobility management, lawful interception, transport for session management (SM) messages between one or more UEs 204 (e.g., any of the UEs described herein) and a session management function (SMF) 266, transparent proxy services for routing SM messages, access authentication and access authorization, transport for short message service (SMS) messages between the UE 204 and the short message service function (SMSF) (not shown), and security anchor functionality (SEAF). The AMF 264 also interacts with an authentication server function (AUSF) (not shown) and the UE 204, and receives the intermediate key that was established as a result of the UE 204 authentication process. In the case of authentication based on a UMTS (universal mobile telecommunications system) subscriber identity module (USIM), the AMF 264 retrieves the security material from the AUSF. The functions of the AMF 264 also include security context management (SCM). The SCM receives a key from the SEAF that it uses to derive access-network specific keys. The functionality of the AMF 264 also includes location services management for regulatory services, transport for location services messages between the UE 204 and a location management function (LMF) 270 (which acts as a location server 230), transport for location services messages between the NG-RAN 220 and the LMF 270, evolved packet system (EPS) bearer identifier allocation for interworking with the EPS, and UE 204 mobility event notification. In addition, the AMF 264 also supports functionalities for non-3GPP® (Third Generation Partnership Project) access networks.

[0069] Functions of the UPF 262 include acting as an anchor point for intra/inter-RAT mobility (when applicable), acting as an external protocol data unit (PDU) session point of interconnect to a data network (not shown), providing packet routing and forwarding, packet inspection, user plane policy rule enforcement (e.g., gating, redirection, traffic steering), lawful interception (user plane collection), traffic usage reporting, quality of service (QoS) handling for the user plane (e.g., uplink/downlink rate enforcement, reflective QoS marking in the downlink), uplink traffic verification (service data flow (SDF) to QoS flow mapping), transport level packet marking in the uplink and downlink, downlink packet buffering and downlink data notification triggering,

and sending and forwarding of one or more “end markers” to the source RAN node. The UPF 262 may also support transfer of location services messages over a user plane between the UE 204 and a location server, such as an SLP 272.

[0070] The functions of the SMF 266 include session management, UE Internet protocol (IP) address allocation and management, selection and control of user plane functions, configuration of traffic steering at the UPF 262 to route traffic to the proper destination, control of part of policy enforcement and QoS, and downlink data notification. The interface over which the SMF 266 communicates with the AMF 264 is referred to as the N11 interface.

[0071] Another optional aspect may include an LMF 270, which may be in communication with the 5GC 260 to provide location assistance for UEs 204. The LMF 270 can be implemented as a plurality of separate servers (e.g., physically separate servers, different software modules on a single server, different software modules spread across multiple physical servers, etc.), or alternately may each correspond to a single server. The LMF 270 can be configured to support one or more location services for UEs 204 that can connect to the LMF 270 via the core network, 5GC 260, and/or via the Internet (not illustrated). The SLP 272 may support similar functions to the LMF 270, but whereas the LMF 270 may communicate with the AMF 264, NG-RAN 220, and UEs 204 over a control plane (e.g., using interfaces and protocols intended to convey signaling messages and not voice or data), the SLP 272 may communicate with UEs 204 and external clients (e.g., third-party server 274) over a user plane (e.g., using protocols intended to carry voice and/or data like the transmission control protocol (TCP) and/or IP).

[0072] Yet another optional aspect may include a third-party server 274, which may be in communication with the LMF 270, the SLP 272, the 5GC 260 (e.g., via the AMF 264 and/or the UPF 262), the NG-RAN 220, and/or the UE 204 to obtain location information (e.g., a location estimate) for the UE 204. As such, in some cases, the third-party server 274 may be referred to as a location services (LCS) client or an external client. The third-party server 274 can be implemented as a plurality of separate servers (e.g., physically separate servers, different software modules on a single server, different software modules spread across multiple physical servers, etc.), or alternately may each correspond to a single server.

[0073] User plane interface 263 and control plane interface 265 connect the 5GC 260, and specifically the UPF 262 and AMF 264, respectively, to one or more gNBs 222 and/or ng-eNBs 224 in the NG-RAN 220. The interface between gNB(s) 222 and/or ng-eNB(s) 224 and the AMF 264 is referred to as the “N2” interface, and the interface between gNB(s) 222 and/or ng-eNB(s) 224 and the UPF 262 is referred to as the “N3” interface. The gNB(s) 222 and/or ng-eNB(s) 224 of the NG-RAN 220 may communicate directly with each other via backhaul connections 223, referred to as the “Xn-C” interface. One or more of gNBs 222 and/or ng-eNBs 224 may communicate with one or more UEs 204 over a wireless interface, referred to as the “Uu” interface.

[0074] The functionality of a gNB 222 may be divided between a gNB central unit (gNB-CU) 226, one or more gNB distributed units (gNB-DUs) 228, and one or more gNB radio units (gNB-RUs) 229. A gNB-CU 226 is a logical

node that includes the base station functions of transferring user data, mobility control, radio access network sharing, positioning, session management, and the like, except for those functions allocated exclusively to the gNB-DU(s) 228. More specifically, the gNB-CU 226 generally host the radio resource control (RRC), service data adaptation protocol (SDAP), and packet data convergence protocol (PDCP) protocols of the gNB 222. A gNB-DU 228 is a logical node that generally hosts the radio link control (RLC) and medium access control (MAC) layer of the gNB 222. Its operation is controlled by the gNB-CU 226. One gNB-DU 228 can support one or more cells, and one cell is supported by only one gNB-DU 228. The interface 232 between the gNB-CU 226 and the one or more gNB-DUs 228 is referred to as the “F1” interface. The physical (PHY) layer functionality of a gNB 222 is generally hosted by one or more standalone gNB-RUs 229 that perform functions such as power amplification and signal transmission/reception. The interface between a gNB-DU 228 and a gNB-RU 229 is referred to as the “Fx” interface. Thus, a UE 204 communicates with the gNB-CU 226 via the RRC, SDAP, and PDCP layers, with a gNB-DU 228 via the RLC and MAC layers, and with a gNB-RU 229 via the PHY layer.

[0075] Deployment of communication systems, such as 5G NR systems, may be arranged in multiple manners with various components or constituent parts. In a 5G NR system, or network, a network node, a network entity, a mobility element of a network, a RAN node, a core network node, a network element, or a network equipment, such as a base station, or one or more units (or one or more components) performing base station functionality, may be implemented in an aggregated or disaggregated architecture. For example, a base station (such as a Node B (NB), evolved NB (eNB), NR base station, 5G NB, AP, TRP, cell, etc.) may be implemented as an aggregated base station (also known as a standalone base station or a monolithic base station) or a disaggregated base station.

[0076] An aggregated base station may be configured to utilize a radio protocol stack that is physically or logically integrated within a single RAN node. A disaggregated base station may be configured to utilize a protocol stack that is physically or logically distributed among two or more units (such as one or more central or centralized units (CUs), one or more distributed units (DUs), or one or more radio units (RUs)). In some aspects, a CU may be implemented within a RAN node, and one or more DUs may be co-located with the CU, or alternatively, may be geographically or virtually distributed throughout one or multiple other RAN nodes. The DUs may be implemented to communicate with one or more RUs. Each of the CU, DU and RU also can be implemented as virtual units, i.e., a virtual central unit (VCU), a virtual distributed unit (VDU), or a virtual radio unit (VRU).

[0077] Base station-type operation or network design may consider aggregation characteristics of base station functionality. For example, disaggregated base stations may be utilized in an integrated access backhaul (IAB) network, an open radio access network (O-RAN (such as the network configuration sponsored by the O-RAN ALLIANCE®)), or a virtualized radio access network (vRAN, also known as a cloud radio access network (C-RAN)). Disaggregation may include distributing functionality across two or more units at various physical locations, as well as distributing functionality for at least one unit virtually, which can enable flex-

ibility in network design. The various units of the disaggregated base station, or disaggregated RAN architecture, can be configured for wired or wireless communication with at least one other unit.

[0078] FIG. 2C illustrates an example disaggregated base station architecture 250, according to aspects of the disclosure. The disaggregated base station architecture 250 may include one or more central units (CUs) 280 (e.g., gNB-CU 226) that can communicate directly with a core network 267 (e.g., 5GC 210, 5GC 260) via a backhaul link, or indirectly with the core network 267 through one or more disaggregated base station units (such as a Near-Real Time (Near-RT) RAN Intelligent Controller (RIC) 259 via an E2 link, or a Non-Real Time (Non-RT) RIC 257 associated with a Service Management and Orchestration (SMO) Framework 255, or both). A CU 280 may communicate with one or more DUs 285 (e.g., gNB-DUs 228) via respective midhaul links, such as an F1 interface. The DUs 285 may communicate with one or more radio units (RUs) 287 (e.g., gNB-RUs 229) via respective fronthaul links. The RUs 287 may communicate with respective UEs 204 via one or more radio frequency (RF) access links. In some implementations, the UE 204 may be simultaneously served by multiple RUs 287.

[0079] Each of the units, i.e., the CUs 280, the DUs 285, the RUs 287, as well as the Near-RT RICs 259, the Non-RT RICs 257 and the SMO Framework 255, may include one or more interfaces or be coupled to one or more interfaces configured to receive or transmit signals, data, or information (collectively, signals) via a wired or wireless transmission medium. Each of the units, or an associated processor or controller providing instructions to the communication interfaces of the units, can be configured to communicate with one or more of the other units via the transmission medium. For example, the units can include a wired interface configured to receive or transmit signals over a wired transmission medium to one or more of the other units. Additionally, the units can include a wireless interface, which may include a receiver, a transmitter or transceiver (such as a RF transceiver), configured to receive or transmit signals, or both, over a wireless transmission medium to one or more of the other units.

[0080] In some aspects, the CU 280 may host one or more higher layer control functions. Such control functions can include RRC, PDCP, service data adaptation protocol (SDAP), or the like. Each control function can be implemented with an interface configured to communicate signals with other control functions hosted by the CU 280. The CU 280 may be configured to handle user plane functionality (i.e., Central Unit-User Plane (CU-UP)), control plane functionality (i.e., Central Unit-Control Plane (CU-CP)), or a combination thereof. In some implementations, the CU 280 can be logically split into one or more CU-UP units and one or more CU-CP units. The CU-UP unit can communicate bidirectionally with the CU-CP unit via an interface, such as the E1 interface when implemented in an O-RAN configuration. The CU 280 can be implemented to communicate with the DU 285, as necessary, for network control and signaling.

[0081] The DU 285 may correspond to a logical unit that includes one or more base station functions to control the operation of one or more RUs 287. In some aspects, the DU 285 may host one or more of a RLC layer, a MAC layer, and one or more high PHY layers (such as modules for forward error correction (FEC) encoding and decoding, scrambling,

modulation and demodulation, or the like) depending, at least in part, on a functional split, such as those defined by the 3rd Generation Partnership Project (3GPP®). In some aspects, the DU 285 may further host one or more low PHY layers. Each layer (or module) can be implemented with an interface configured to communicate signals with other layers (and modules) hosted by the DU 285, or with the control functions hosted by the CU 280.

[0082] Lower-layer functionality can be implemented by one or more RUs 287. In some deployments, an RU 287, controlled by a DU 285, may correspond to a logical node that hosts RF processing functions, or low-PHY layer functions (such as performing fast Fourier transform (FFT), inverse FFT (iFFT), digital beamforming, physical random access channel (PRACH) extraction and filtering, or the like), or both, based at least in part on the functional split, such as a lower layer functional split. In such an architecture, the RU(s) 287 can be implemented to handle over the air (OTA) communication with one or more UEs 204. In some implementations, real-time and non-real-time aspects of control and user plane communication with the RU(s) 287 can be controlled by the corresponding DU 285. In some scenarios, this configuration can enable the DU(s) 285 and the CU 280 to be implemented in a cloud-based RAN architecture, such as a vRAN architecture.

[0083] The SMO Framework 255 may be configured to support RAN deployment and provisioning of non-virtualized and virtualized network elements. For non-virtualized network elements, the SMO Framework 255 may be configured to support the deployment of dedicated physical resources for RAN coverage requirements which may be managed via an operations and maintenance interface (such as an O1 interface). For virtualized network elements, the SMO Framework 255 may be configured to interact with a cloud computing platform (such as an open cloud (O-Cloud) 269) to perform network element life cycle management (such as to instantiate virtualized network elements) via a cloud computing platform interface (such as an O2 interface). Such virtualized network elements can include, but are not limited to, CUs 280, DUs 285, RUs 287 and Near-RT RICs 259. In some implementations, the SMO Framework 255 can communicate with a hardware aspect of a 4G RAN, such as an open eNB (O-eNB) 261, via an O1 interface. Additionally, in some implementations, the SMO Framework 255 can communicate directly with one or more RUs 287 via an O1 interface. The SMO Framework 255 also may include a Non-RT RIC 257 configured to support functionality of the SMO Framework 255.

[0084] The Non-RT RIC 257 may be configured to include a logical function that enables non-real-time control and optimization of RAN elements and resources, artificial intelligence/machine learning (AI/ML) workflows including model training and updates, or policy-based guidance of applications/features in the Near-RT RIC 259. The Non-RT RIC 257 may be coupled to or communicate with (such as via an A1 interface) the Near-RT RIC 259. The Near-RT RIC 259 may be configured to include a logical function that enables near-real-time control and optimization of RAN elements and resources via data collection and actions over an interface (such as via an E2 interface) connecting one or more CUs 280, one or more DUs 285, or both, as well as an O-eNB, with the Near-RT RIC 259.

[0085] In some implementations, to generate AI/ML models to be deployed in the Near-RT RIC 259, the Non-RT RIC

257 may receive parameters or external enrichment information from external servers. Such information may be utilized by the Near-RT RIC **259** and may be received at the SMO Framework **255** or the Non-RT RIC **257** from non-network data sources or from network functions. In some examples, the Non-RT RIC **257** or the Near-RT RIC **259** may be configured to tune RAN behavior or performance. For example, the Non-RT RIC **257** may monitor long-term trends and patterns for performance and employ AI/ML models to perform corrective actions through the SMO Framework **255** (such as reconfiguration via O1) or via creation of RAN management policies (such as A1 policies).

[0086] FIGS. 3A, 3B, and 3C illustrate several example components (represented by corresponding blocks) that may be incorporated into a UE **302** (which may correspond to any of the UEs described herein), a base station **304** (which may correspond to any of the base stations described herein), and a network entity **306** (which may correspond to or embody any of the network functions described herein, including the location server **230** and the LMF **270**, or alternatively may be independent from the NG-RAN **220** and/or 5GC **210/260** infrastructure depicted in FIGS. 2A and 2B, such as a private network) to support the operations described herein. It will be appreciated that these components may be implemented in different types of apparatuses in different implementations (e.g., in an ASIC, in a system-on-chip (SoC), etc.). The illustrated components may also be incorporated into other apparatuses in a communication system. For example, other apparatuses in a system may include components similar to those described to provide similar functionality. Also, a given apparatus may contain one or more of the components. For example, an apparatus may include multiple transceiver components that enable the apparatus to operate on multiple carriers and/or communicate via different technologies.

[0087] The UE **302** and the base station **304** each include one or more wireless wide area network (WWAN) transceivers **310** and **350**, respectively, providing means for communicating (e.g., means for transmitting, means for receiving, means for measuring, means for tuning, means for refraining from transmitting, etc.) via one or more wireless communication networks (not shown), such as an NR network, an LTE network, a GSM network, and/or the like. The WWAN transceivers **310** and **350** may each be connected to one or more antennas **316** and **356**, respectively, for communicating with other network nodes, such as other UEs, access points, base stations (e.g., eNBs, gNBs), etc., via at least one designated RAT (e.g., NR, LTE, GSM, etc.) over a wireless communication medium of interest (e.g., some set of time/frequency resources in a particular frequency spectrum). The WWAN transceivers **310** and **350** may be variously configured for transmitting and encoding signals **318** and **358** (e.g., messages, indications, information, and so on), respectively, and, conversely, for receiving and decoding signals **318** and **358** (e.g., messages, indications, information, pilots, and so on), respectively, in accordance with the designated RAT. Specifically, the WWAN transceivers **310** and **350** include one or more transmitters **314** and **354**, respectively, for transmitting and encoding signals **318** and **358**, respectively, and one or more receivers **312** and **352**, respectively, for receiving and decoding signals **318** and **358**, respectively.

[0088] The UE **302** and the base station **304** each also include, at least in some cases, one or more short-range wireless transceivers **320** and **360**, respectively. The short-

range wireless transceivers **320** and **360** may be connected to one or more antennas **326** and **366**, respectively, and provide means for communicating (e.g., means for transmitting, means for receiving, means for measuring, means for tuning, means for refraining from transmitting, etc.) with other network nodes, such as other UEs, access points, base stations, etc., via at least one designated RAT (e.g., Wi-Fi, LTE Direct, BLUETOOTH®, ZIGBEE®, Z-WAVE®, PC5, dedicated short-range communications (DSRC), wireless access for vehicular environments (WAVE), near-field communication (NFC), ultra-wideband (UWB), etc.) over a wireless communication medium of interest. The short-range wireless transceivers **320** and **360** may be variously configured for transmitting and encoding signals **328** and **368** (e.g., messages, indications, information, and so on), respectively, and, conversely, for receiving and decoding signals **328** and **368** (e.g., messages, indications, information, pilots, and so on), respectively, in accordance with the designated RAT. Specifically, the short-range wireless transceivers **320** and **360** include one or more transmitters **324** and **364**, respectively, for transmitting and encoding signals **328** and **368**, respectively, and one or more receivers **322** and **362**, respectively, for receiving and decoding signals **328** and **368**, respectively. As specific examples, the short-range wireless transceivers **320** and **360** may be Wi-Fi transceivers, BLUETOOTH® transceivers, ZIGBEE® and/or Z-WAVE® transceivers, NFC transceivers, UWB transceivers, or vehicle-to-vehicle (V2V) and/or vehicle-to-everything (V2X) transceivers.

[0089] The UE **302** and the base station **304** also include, at least in some cases, satellite signal interfaces **330** and **370**, which each include one or more satellite signal receivers **332** and **372**, respectively, and may optionally include one or more satellite signal transmitters **334** and **374**, respectively. In some cases, the base station **304** may be a terrestrial base station that may communicate with space vehicles (e.g., space vehicles **112**) via the satellite signal interface **370**. In other cases, the base station **304** may be a space vehicle (or other non-terrestrial entity) that uses the satellite signal interface **370** to communicate with terrestrial networks and/or other space vehicles.

[0090] The satellite signal receivers **332** and **372** may be connected to one or more antennas **336** and **376**, respectively, and may provide means for receiving and/or measuring satellite positioning/communication signals **338** and **378**, respectively. Where the satellite signal receiver(s) **332** and **372** are satellite positioning system receivers, the satellite positioning/communication signals **338** and **378** may be global positioning system (GPS) signals, global navigation satellite system (GLONASS) signals, Galileo signals, Beidou signals, Indian Regional Navigation Satellite System (NAVIC), Quasi-Zenith Satellite System (QZSS) signals, etc. Where the satellite signal receiver(s) **332** and **372** are non-terrestrial network (NTN) receivers, the satellite positioning/communication signals **338** and **378** may be communication signals (e.g., carrying control and/or user data) originating from a 5G network. The satellite signal receiver(s) **332** and **372** may comprise any suitable hardware and/or software for receiving and processing satellite positioning/communication signals **338** and **378**, respectively. The satellite signal receiver(s) **332** and **372** may request information and operations as appropriate from the other systems, and, at least in some cases, perform calculations to determine locations of the UE **302** and the base station **304**,

respectively, using measurements obtained by any suitable satellite positioning system algorithm.

[0091] The optional satellite signal transmitter(s) 334 and 374, when present, may be connected to the one or more antennas 336 and 376, respectively, and may provide means for transmitting satellite positioning/communication signals 338 and 378, respectively. Where the satellite signal transmitter(s) 374 are satellite positioning system transmitters, the satellite positioning/communication signals 378 may be GPS signals, GLONASS® signals, Galileo signals, Beidou signals, NAVIC, QZSS signals, etc. Where the satellite signal transmitter(s) 334 and 374 are NTN transmitters, the satellite positioning/communication signals 338 and 378 may be communication signals (e.g., carrying control and/or user data) originating from a 5G network. The satellite signal transmitter(s) 334 and 374 may comprise any suitable hardware and/or software for transmitting satellite positioning/communication signals 338 and 378, respectively. The satellite signal transmitter(s) 334 and 374 may request information and operations as appropriate from the other systems.

[0092] The base station 304 and the network entity 306 each include one or more network transceivers 380 and 390, respectively, providing means for communicating (e.g., means for transmitting, means for receiving, etc.) with other network entities (e.g., other base stations 304, other network entities 306). For example, the base station 304 may employ the one or more network transceivers 380 to communicate with other base stations 304 or network entities 306 over one or more wired or wireless backhaul links. As another example, the network entity 306 may employ the one or more network transceivers 390 to communicate with one or more base station 304 over one or more wired or wireless backhaul links, or with other network entities 306 over one or more wired or wireless core network interfaces.

[0093] A transceiver may be configured to communicate over a wired or wireless link. A transceiver (whether a wired transceiver or a wireless transceiver) includes transmitter circuitry (e.g., transmitters 314, 324, 354, 364) and receiver circuitry (e.g., receivers 312, 322, 352, 362). A transceiver may be an integrated device (e.g., embodying transmitter circuitry and receiver circuitry in a single device) in some implementations, may comprise separate transmitter circuitry and separate receiver circuitry in some implementations, or may be embodied in other ways in other implementations. The transmitter circuitry and receiver circuitry of a wired transceiver (e.g., network transceivers 380 and 390 in some implementations) may be coupled to one or more wired network interface ports. Wireless transmitter circuitry (e.g., transmitters 314, 324, 354, 364) may include or be coupled to a plurality of antennas (e.g., antennas 316, 326, 356, 366), such as an antenna array, that permits the respective apparatus (e.g., UE 302, base station 304) to perform transmit “beamforming,” as described herein. Similarly, wireless receiver circuitry (e.g., receivers 312, 322, 352, 362) may include or be coupled to a plurality of antennas (e.g., antennas 316, 326, 356, 366), such as an antenna array, that permits the respective apparatus (e.g., UE 302, base station 304) to perform receive beamforming, as described herein. In an aspect, the transmitter circuitry and receiver circuitry may share the same plurality of antennas (e.g., antennas 316, 326, 356, 366), such that the respective apparatus can only receive or transmit at a given time, not both at the same time. A wireless transceiver (e.g., WWAN

transceivers 310 and 350, short-range wireless transceivers 320 and 360) may also include a network listen module (NLM) or the like for performing various measurements.

[0094] As used herein, the various wireless transceivers (e.g., transceivers 310, 320, 350, and 360, and network transceivers 380 and 390 in some implementations) and wired transceivers (e.g., network transceivers 380 and 390 in some implementations) may generally be characterized as “a transceiver,” “at least one transceiver,” or “one or more transceivers.” As such, whether a particular transceiver is a wired or wireless transceiver may be inferred from the type of communication performed. For example, backhaul communication between network devices or servers will generally relate to signaling via a wired transceiver, whereas wireless communication between a UE (e.g., UE 302) and a base station (e.g., base station 304) will generally relate to signaling via a wireless transceiver.

[0095] The UE 302, the base station 304, and the network entity 306 also include other components that may be used in conjunction with the operations as disclosed herein. The UE 302, the base station 304, and the network entity 306 include one or more processors 342, 384, and 394, respectively, for providing functionality relating to, for example, wireless communication, and for providing other processing functionality. The processors 342, 384, and 394 may therefore provide means for processing, such as means for determining, means for calculating, means for receiving, means for transmitting, means for indicating, etc. In an aspect, the processors 342, 384, and 394 may include, for example, one or more general purpose processors, multi-core processors, central processing units (CPUs), ASICs, digital signal processors (DSPs), field programmable gate arrays (FPGAs), other programmable logic devices or processing circuitry, or various combinations thereof.

[0096] The UE 302, the base station 304, and the network entity 306 include memory circuitry implementing memories 340, 386, and 396 (e.g., each including a memory device), respectively, for maintaining information (e.g., information indicative of reserved resources, thresholds, parameters, and so on). The memories 340, 386, and 396 may therefore provide means for storing, means for retrieving, means for maintaining, etc. In some cases, the UE 302, the base station 304, and the network entity 306 may include sensing component 348, 388, and 398, respectively. The sensing component 348, 388, and 398 may be hardware circuits that are part of or coupled to the processors 342, 384, and 394, respectively, that, when executed, cause the UE 302, the base station 304, and the network entity 306 to perform the functionality described herein. In other aspects, the sensing component 348, 388, and 398 may be external to the processors 342, 384, and 394 (e.g., part of a modem processing system, integrated with another processing system, etc.). Alternatively, the sensing component 348, 388, and 398 may be memory modules stored in the memories 340, 386, and 396, respectively, that, when executed by the processors 342, 384, and 394 (or a modem processing system, another processing system, etc.), cause the UE 302, the base station 304, and the network entity 306 to perform the functionality described herein. FIG. 3A illustrates possible locations of the sensing component 348, which may be, for example, part of the one or more WWAN transceivers 310, the memory 340, the one or more processors 342, or any combination thereof, or may be a standalone component. FIG. 3B illustrates possible locations of the sensing

component **388**, which may be, for example, part of the one or more WWAN transceivers **350**, the memory **386**, the one or more processors **384**, or any combination thereof, or may be a standalone component. FIG. 3C illustrates possible locations of the sensing component **398**, which may be, for example, part of the one or more network transceivers **390**, the memory **396**, the one or more processors **394**, or any combination thereof, or may be a standalone component.

[0097] The UE **302** may include one or more sensors **344** coupled to the one or more processors **342** to provide means for sensing or detecting movement and/or orientation information that is independent of motion data derived from signals received by the one or more WWAN transceivers **310**, the one or more short-range wireless transceivers **320**, and/or the satellite signal interface **330**. By way of example, the sensor(s) **344** may include an accelerometer (e.g., a micro-electrical mechanical systems (MEMS) device), a gyroscope, a geomagnetic sensor (e.g., a compass), an altimeter (e.g., a barometric pressure altimeter), and/or any other type of movement detection sensor. Moreover, the sensor(s) **344** may include a plurality of different types of devices and combine their outputs in order to provide motion information. For example, the sensor(s) **344** may use a combination of a multi-axis accelerometer and orientation sensors to provide the ability to compute positions in two-dimensional (2D) and/or three-dimensional (3D) coordinate systems.

[0098] In addition, the UE **302** includes a user interface **346** providing means for providing indications (e.g., audible and/or visual indications) to a user and/or for receiving user input (e.g., upon user actuation of an input device such as a keypad, a touch screen, a microphone, and so on). Although not shown, the base station **304** and the network entity **306** may also include user interfaces.

[0099] Referring to the one or more processors **384** in more detail, in the downlink, IP packets from the network entity **306** may be provided to the processor **384**. The one or more processors **384** may implement functionality for an RRC layer, a packet data convergence protocol (PDCP) layer, a radio link control (RLC) layer, and a medium access control (MAC) layer. The one or more processors **384** may provide RRC layer functionality associated with broadcasting of system information (e.g., master information block (MIB), system information blocks (SIBs)), RRC connection control (e.g., RRC connection paging, RRC connection establishment, RRC connection modification, and RRC connection release), inter-RAT mobility, and measurement configuration for UE measurement reporting; PDCP layer functionality associated with header compression/decompression, security (ciphering, deciphering, integrity protection, integrity verification), and handover support functions; RLC layer functionality associated with the transfer of upper layer PDUs, error correction through automatic repeat request (ARQ), concatenation, segmentation, and reassembly of RLC service data units (SDUs), re-segmentation of RLC data PDUs, and reordering of RLC data PDUs; and MAC layer functionality associated with mapping between logical channels and transport channels, scheduling information reporting, error correction, priority handling, and logical channel prioritization.

[0100] The transmitter **354** and the receiver **352** may implement Layer-1 (L1) functionality associated with various signal processing functions. Layer-1, which includes a physical (PHY) layer, may include error detection on the

transport channels, forward error correction (FEC) coding/decoding of the transport channels, interleaving, rate matching, mapping onto physical channels, modulation/demodulation of physical channels, and MIMO antenna processing. The transmitter **354** handles mapping to signal constellations based on various modulation schemes (e.g., binary phase-shift keying (BPSK), quadrature phase-shift keying (QPSK), M-phase-shift keying (M-PSK), M-quadrature amplitude modulation (M-QAM)). The coded and modulated symbols may then be split into parallel streams. Each stream may then be mapped to an orthogonal frequency division multiplexing (OFDM) subcarrier, multiplexed with a reference signal (e.g., pilot) in the time and/or frequency domain, and then combined together using an inverse fast Fourier transform (IFFT) to produce a physical channel carrying a time domain OFDM symbol stream. The OFDM symbol stream is spatially precoded to produce multiple spatial streams. Channel estimates from a channel estimator may be used to determine the coding and modulation scheme, as well as for spatial processing. The channel estimate may be derived from a reference signal and/or channel condition feedback transmitted by the UE **302**. Each spatial stream may then be provided to one or more different antennas **356**. The transmitter **354** may modulate an RF carrier with a respective spatial stream for transmission.

[0101] At the UE **302**, the receiver **312** receives a signal through its respective antenna(s) **316**. The receiver **312** recovers information modulated onto an RF carrier and provides the information to the one or more processors **342**. The transmitter **314** and the receiver **312** implement Layer-1 functionality associated with various signal processing functions. The receiver **312** may perform spatial processing on the information to recover any spatial streams destined for the UE **302**. If multiple spatial streams are destined for the UE **302**, they may be combined by the receiver **312** into a single OFDM symbol stream. The receiver **312** then converts the OFDM symbol stream from the time-domain to the frequency domain using a fast Fourier transform (FFT). The frequency domain signal comprises a separate OFDM symbol stream for each subcarrier of the OFDM signal. The symbols on each subcarrier, and the reference signal, are recovered and demodulated by determining the most likely signal constellation points transmitted by the base station **304**. These soft decisions may be based on channel estimates computed by a channel estimator. The soft decisions are then decoded and de-interleaved to recover the data and control signals that were originally transmitted by the base station **304** on the physical channel. The data and control signals are then provided to the one or more processors **342**, which implements Layer-3 (L3) and Layer-2 (L2) functionality.

[0102] In the downlink, the one or more processors **342** provides demultiplexing between transport and logical channels, packet reassembly, deciphering, header decompression, and control signal processing to recover IP packets from the core network. The one or more processors **342** are also responsible for error detection.

[0103] Similar to the functionality described in connection with the downlink transmission by the base station **304**, the one or more processors **342** provides RRC layer functionality associated with system information (e.g., MIB, SIBs) acquisition, RRC connections, and measurement reporting; PDCP layer functionality associated with header compression/decompression, and security (ciphering, deciphering, integrity protection, integrity verification); RLC layer func-

tionality associated with the transfer of upper layer PDUs, error correction through ARQ, concatenation, segmentation, and reassembly of RLC SDUs, re-segmentation of RLC data PDUs, and reordering of RLC data PDUs; and MAC layer functionality associated with mapping between logical channels and transport channels, multiplexing of MAC SDUs onto transport blocks (TBs), demultiplexing of MAC SDUs from TBs, scheduling information reporting, error correction through hybrid automatic repeat request (HARQ), priority handling, and logical channel prioritization.

[0104] Channel estimates derived by the channel estimator from a reference signal or feedback transmitted by the base station **304** may be used by the transmitter **314** to select the appropriate coding and modulation schemes, and to facilitate spatial processing. The spatial streams generated by the transmitter **314** may be provided to different antenna(s) **316**. The transmitter **314** may modulate an RF carrier with a respective spatial stream for transmission.

[0105] The uplink transmission is processed at the base station **304** in a manner similar to that described in connection with the receiver function at the UE **302**. The receiver **352** receives a signal through its respective antenna(s) **356**. The receiver **352** recovers information modulated onto an RF carrier and provides the information to the one or more processors **384**.

[0106] In the uplink, the one or more processors **384** provides demultiplexing between transport and logical channels, packet reassembly, deciphering, header decompression, control signal processing to recover IP packets from the UE **302**. IP packets from the one or more processors **384** may be provided to the core network. The one or more processors **384** are also responsible for error detection.

[0107] For convenience, the UE **302**, the base station **304**, and/or the network entity **306** are shown in FIGS. **3A**, **3B**, and **3C** as including various components that may be configured according to the various examples described herein. It will be appreciated, however, that the illustrated components may have different functionality in different designs. In particular, various components in FIGS. **3A** to **3C** are optional in alternative configurations and the various aspects include configurations that may vary due to design choice, costs, use of the device, or other considerations. For example, in case of FIG. **3A**, a particular implementation of UE **302** may omit the WWAN transceiver(s) **310** (e.g., a wearable device or tablet computer or personal computer (PC) or laptop may have Wi-Fi and/or BLUETOOTH® capability without cellular capability), or may omit the short-range wireless transceiver(s) **320** (e.g., cellular-only, etc.), or may omit the satellite signal interface **330**, or may omit the sensor(s) **344**, and so on. In another example, in case of FIG. **3B**, a particular implementation of the base station **304** may omit the WWAN transceiver(s) **350** (e.g., a Wi-Fi “hotspot” access point without cellular capability), or may omit the short-range wireless transceiver(s) **360** (e.g., cellular-only, etc.), or may omit the satellite signal interface **370**, and so on. For brevity, illustration of the various alternative configurations is not provided herein, but would be readily understandable to one skilled in the art.

[0108] The various components of the UE **302**, the base station **304**, and the network entity **306** may be communicatively coupled to each other over data buses **308**, **382**, and **392**, respectively. In an aspect, the data buses **308**, **382**, and **392** may form, or be part of, a communication interface of the UE **302**, the base station **304**, and the network entity **306**,

respectively. For example, where different logical entities are embodied in the same device (e.g., gNB and location server functionality incorporated into the same base station **304**), the data buses **308**, **382**, and **392** may provide communication between them.

[0109] The components of FIGS. **3A**, **3B**, and **3C** may be implemented in various ways. In some implementations, the components of FIGS. **3A**, **3B**, and **3C** may be implemented in one or more circuits such as, for example, one or more processors and/or one or more ASICs (which may include one or more processors). Here, each circuit may use and/or incorporate at least one memory component for storing information or executable code used by the circuit to provide this functionality. For example, some or all of the functionality represented by blocks **310** to **346** may be implemented by processor and memory component(s) of the UE **302** (e.g., by execution of appropriate code and/or by appropriate configuration of processor components). Similarly, some or all of the functionality represented by blocks **350** to **388** may be implemented by processor and memory component(s) of the base station **304** (e.g., by execution of appropriate code and/or by appropriate configuration of processor components). Also, some or all of the functionality represented by blocks **390** to **398** may be implemented by processor and memory component(s) of the network entity **306** (e.g., by execution of appropriate code and/or by appropriate configuration of processor components). For simplicity, various operations, acts, and/or functions are described herein as being performed “by a UE,” “by a base station,” “by a network entity,” etc. However, as will be appreciated, such operations, acts, and/or functions may actually be performed by specific components or combinations of components of the UE **302**, base station **304**, network entity **306**, etc., such as the processors **342**, **384**, **394**, the transceivers **310**, **320**, **350**, and **360**, the memories **340**, **386**, and **396**, the sensing component **348**, **388**, and **398**, etc.

[0110] In some designs, the network entity **306** may be implemented as a core network component. In other designs, the network entity **306** may be distinct from a network operator or operation of the cellular network infrastructure (e.g., NG RAN **220** and/or 5GC **210/260**). For example, the network entity **306** may be a component of a private network that may be configured to communicate with the UE **302** via the base station **304** or independently from the base station **304** (e.g., over a non-cellular communication link, such as Wi-Fi).

[0111] Various frame structures may be used to support downlink and uplink transmissions between network nodes (e.g., base stations and UEs). FIG. **4** is a diagram **400** illustrating an example frame structure, according to aspects of the disclosure. The frame structure may be a downlink or uplink frame structure. Other wireless communications technologies may have different frame structures and/or different channels.

[0112] LTE, and in some cases NR, utilizes orthogonal frequency-division multiplexing (OFDM) on the downlink and single-carrier frequency division multiplexing (SC-FDM) on the uplink. Unlike LTE, however, NR has an option to use OFDM on the uplink as well. OFDM and SC-FDM partition the system bandwidth into multiple (K) orthogonal subcarriers, which are also commonly referred to as tones, bins, etc. Each subcarrier may be modulated with data. In general, modulation symbols are sent in the frequency domain with OFDM and in the time domain with

SC-FDM. The spacing between adjacent subcarriers may be fixed, and the total number of subcarriers (K) may be dependent on the system bandwidth. For example, the spacing of the subcarriers may be 15 kilohertz (kHz) and the minimum resource allocation (resource block) may be 12 subcarriers (or 180 kHz). Consequently, the nominal fast Fourier transform (FFT) size may be equal to 128, 256, 512, 1024, or 2048 for system bandwidth of 1.25, 2.5, 5, 10, or 20 megahertz (MHz), respectively. The system bandwidth may also be partitioned into subbands. For example, a subband may cover 1.08 MHz (i.e., 6 resource blocks), and there may be 1, 2, 4, 8, or 16 subbands for system bandwidth of 1.25, 2.5, 5, 10, or 20 MHz, respectively.

[0113] LTE supports a single numerology (subcarrier spacing (SCS), symbol length, etc.). In contrast, NR may support multiple numerologies (μ), for example, subcarrier spacings of 15 kHz ($\mu=0$), 30 kHz ($\mu=1$), 60 kHz ($\mu=2$), 120 kHz ($\mu=3$), and 240 kHz ($\mu=4$) or greater may be available. In each subcarrier spacing, there are 14 symbols per slot. For 15 kHz SCS ($\mu=0$), there is one slot per subframe, 10 slots per frame, the slot duration is 1 millisecond (ms), the symbol duration is 66.7 microseconds (μ s), and the maximum nominal system bandwidth (in MHz) with a 4K FFT size is 50. For 30 kHz SCS ($\mu=1$), there are two slots per subframe, 20 slots per frame, the slot duration is 0.5 ms, the symbol duration is 33.3 μ s, and the maximum nominal system bandwidth (in MHz) with a 4K FFT size is 100. For 60 kHz SCS ($\mu=2$), there are four slots per subframe, 40 slots per frame, the slot duration is 0.25 ms, the symbol duration is 16.7 μ s, and the maximum nominal system bandwidth (in MHz) with a 4K FFT size is 200. For 120 kHz SCS ($\mu=3$), there are eight slots per subframe, 80 slots per frame, the slot duration is 0.125 ms, the symbol duration is 8.33 μ s, and the maximum nominal system bandwidth (in MHz) with a 4K FFT size is 400. For 240 kHz SCS ($\mu=4$), there are 16 slots per subframe, 160 slots per frame, the slot duration is 0.0625 ms, the symbol duration is 4.17 μ s, and the maximum nominal system bandwidth (in MHz) with a 4K FFT size is 800.

[0114] In the example of FIG. 4, a numerology of 15 kHz is used. Thus, in the time domain, a 10 ms frame is divided into 10 equally sized subframes of 1 ms each, and each subframe includes one time slot. In FIG. 4, time is represented horizontally (on the X axis) with time increasing from left to right, while frequency is represented vertically (on the Y axis) with frequency increasing (or decreasing) from bottom to top.

[0115] A resource grid may be used to represent time slots, each time slot including one or more time-concurrent resource blocks (RBs) (also referred to as physical RBs (PRBs)) in the frequency domain. The resource grid is further divided into multiple resource elements (REs). An RE may correspond to one symbol length in the time domain and one subcarrier in the frequency domain. In the numerology of FIG. 4, for a normal cyclic prefix, an RB may contain 12 consecutive subcarriers in the frequency domain and seven consecutive symbols in the time domain, for a total of 84 REs. For an extended cyclic prefix, an RB may contain 12 consecutive subcarriers in the frequency domain and six consecutive symbols in the time domain, for a total of 72 REs. The number of bits carried by each RE depends on the modulation scheme.

[0116] Some of the REs may carry reference (pilot) signals (RS). The reference signals may include positioning refer-

ence signals (PRS), tracking reference signals (TRS), phase tracking reference signals (PTRS), cell-specific reference signals (CRS), channel state information reference signals (CSI-RS), demodulation reference signals (DMRS), primary synchronization signals (PSS), secondary synchronization signals (SSS), synchronization signal blocks (SSBs), sounding reference signals (SRS), etc., depending on whether the illustrated frame structure is used for uplink or downlink communication. FIG. 4 illustrates example locations of REs carrying a reference signal (labeled "R").

[0117] A collection of resource elements (REs) that are used for transmission of PRS is referred to as a "PRS resource." The collection of resource elements can span multiple PRBs in the frequency domain and 'N' (such as 1 or more) consecutive symbol(s) within a slot in the time domain. In a given OFDM symbol in the time domain, a PRS resource occupies consecutive PRBs in the frequency domain.

[0118] The transmission of a PRS resource within a given PRB has a particular comb size (also referred to as the "comb density"). A comb size 'N' represents the subcarrier spacing (or frequency/tone spacing) within each symbol of a PRS resource configuration. Specifically, for a comb size 'N,' PRS are transmitted in every Nth subcarrier of a symbol of a PRB. For example, for comb-4, for each symbol of the PRS resource configuration, REs corresponding to every fourth subcarrier (such as subcarriers 0, 4, 8) are used to transmit PRS of the PRS resource. Currently, comb sizes of comb-2, comb-4, comb-6, and comb-12 are supported for DL-PRS. FIG. 4 illustrates an example PRS resource configuration for comb-4 (which spans four symbols). That is, the locations of the shaded REs (labeled "R") indicate a comb-4 PRS resource configuration.

[0119] Currently, a DL-PRS resource may span 2, 4, 6, or 12 consecutive symbols within a slot with a fully frequency-domain staggered pattern. A DL-PRS resource can be configured in any higher layer configured downlink or flexible (FL) symbol of a slot. There may be a constant energy per resource element (EPRE) for all REs of a given DL-PRS resource. The following are the frequency offsets from symbol to symbol for comb sizes 2, 4, 6, and 12 over 2, 4, 6, and 12 symbols. 2-symbol comb-2: {0, 1}; 4-symbol comb-2: {0, 1, 0, 1}; 6-symbol comb-2: {0, 1, 0, 1, 0, 1}; 12-symbol comb-2: {0, 1, 0, 1, 0, 1, 0, 1, 0, 1, 0, 1}; 4-symbol comb-4: {0, 2, 1, 3} (as in the example of FIG. 4); 12-symbol comb-4: {0, 2, 1, 3, 0, 2, 1, 3, 0, 2, 1, 3}; 6-symbol comb-6: {0, 3, 1, 4, 2, 5}; 12-symbol comb-6: {0, 3, 1, 4, 2, 5, 0, 3, 1, 4, 2, 5}; and 12-symbol comb-12: {0, 6, 3, 9, 1, 7, 4, 10, 2, 8, 5, 11}.

[0120] A "PRS resource set" is a set of PRS resources used for the transmission of PRS signals, where each PRS resource has a PRS resource ID. In addition, the PRS resources in a PRS resource set are associated with the same TRP. A PRS resource set is identified by a PRS resource set ID and is associated with a particular TRP (identified by a TRP ID). In addition, the PRS resources in a PRS resource set have the same periodicity, a common muting pattern configuration, and the same repetition factor (such as "PRS-ResourceRepetitionFactor") across slots. The periodicity is the time from the first repetition of the first PRS resource of a first PRS instance to the same first repetition of the same first PRS resource of the next PRS instance. The periodicity may have a length selected from $2^{\mu} \cdot \{4, 5, 8, 10, 16, 20, 32, 40, 64, 80, 160, 320, 640, 1280, 2560, 5120, 10240\}$ slots,

with $\mu=0, 1, 2, 3$. The repetition factor may have a length selected from $\{1, 2, 4, 6, 8, 16, 32\}$ slots.

[0121] A PRS resource ID in a PRS resource set is associated with a single beam (or beam ID) transmitted from a single TRP (where a TRP may transmit one or more beams). That is, each PRS resource of a PRS resource set may be transmitted on a different beam, and as such, a “PRS resource,” or simply “resource,” also can be referred to as a “beam.” Note that this does not have any implications on whether the TRPs and the beams on which PRS are transmitted are known to the UE.

[0122] A “PRS instance” or “PRS occasion” is one instance of a periodically repeated time window (such as a group of one or more consecutive slots) where PRS are expected to be transmitted. A PRS occasion also may be referred to as a “PRS positioning occasion,” a “PRS positioning instance,” a “positioning occasion,” a “positioning instance,” a “positioning repetition,” or simply an “occasion,” an “instance,” or a “repetition.”

[0123] A “positioning frequency layer” (also referred to simply as a “frequency layer”) is a collection of one or more PRS resource sets across one or more TRPs that have the same values for certain parameters. Specifically, the collection of PRS resource sets has the same subcarrier spacing and cyclic prefix (CP) type (meaning all numerologies supported for the physical downlink shared channel (PDSCH) are also supported for PRS), the same Point A, the same value of the downlink PRS bandwidth, the same start PRB (and center frequency), and the same comb-size. The Point A parameter takes the value of the parameter “ARFCN-ValueNR” (where “ARFCN” stands for “absolute radio-frequency channel number”) and is an identifier/code that specifies a pair of physical radio channel used for transmission and reception. The downlink PRS bandwidth may have a granularity of four PRBs, with a minimum of 24 PRBs and a maximum of 272 PRBs. Currently, up to four frequency layers have been defined, and up to two PRS resource sets may be configured per TRP per frequency layer.

[0124] The concept of a frequency layer is somewhat like the concept of component carriers and bandwidth parts (BWPs), but different in that component carriers and BWPs are used by one base station (or a macro cell base station and a small cell base station) to transmit data channels, while frequency layers are used by several (usually three or more) base stations to transmit PRS. A UE may indicate the number of frequency layers it can support when it sends the network its positioning capabilities, such as during an LTE positioning protocol (LPP) session. For example, a UE may indicate whether it can support one or four positioning frequency layers.

[0125] Note that the terms “positioning reference signal” and “PRS” generally refer to specific reference signals that are used for positioning in NR and LTE systems. However, as used herein, the terms “positioning reference signal” and “PRS” may also refer to any type of reference signal that can be used for positioning, such as but not limited to, PRS as defined in LTE and NR, TRS, PTRS, CRS, CSI-RS, DMRS, PSS, SSS, SSB, SRS, UL-PRS, etc. In addition, the terms “positioning reference signal” and “PRS” may refer to downlink, uplink, or sidelink positioning reference signals, unless otherwise indicated by the context. If needed to further distinguish the type of PRS, a downlink positioning reference signal may be referred to as a “DL-PRS,” an

uplink positioning reference signal (e.g., an SRS-for-positioning, PTRS) may be referred to as an “UL-PRS,” and a sidelink positioning reference signal may be referred to as an “SL-PRS.” In addition, for signals that may be transmitted in the downlink, uplink, and/or sidelink (e.g., DMRS), the signals may be prepended with “DL,” “UL,” or “SL” to distinguish the direction. For example, “UL-DMRS” is different from “DL-DMRS.”

[0126] Wireless communication signals (e.g., radio frequency (RF) signals configured to carry orthogonal frequency division multiplexing (OFDM) symbols in accordance with a wireless communications standard, such as LTE, NR, etc.) transmitted between a UE and a base station can be used for environment sensing (also referred to as “RF sensing” or “radar”). Using wireless communication signals for environment sensing can be regarded as consumer-level radar with advanced detection capabilities that enable, among other things, touchless/device-free interaction with a device/system. The wireless communication signals may be cellular communication signals, such as LTE or NR signals, WLAN signals, such as Wi-Fi signals, etc. As a particular example, the wireless communication signals may be an OFDM waveform as utilized in LTE and NR. High-frequency communication signals, such as millimeter wave (mmW) RF signals, are especially beneficial to use as sensing signals because the higher frequency provides, at least, more accurate range (distance) detection.

[0127] Possible use cases of RF sensing include health monitoring use cases, such as heartbeat detection, respiration rate monitoring, and the like, gesture recognition use cases, such as human activity recognition, keystroke detection, sign language recognition, and the like, contextual information acquisition use cases, such as location detection/tracking, direction finding, range estimation, and the like, and automotive sensing use cases, such as smart cruise control, collision avoidance, and the like.

[0128] There are different types of sensing, including monostatic sensing (also referred to as “active sensing”) and bistatic sensing (also referred to as “passive sensing”). FIGS. 5A and 5B illustrate these different types of sensing. Specifically, FIG. 5A is a diagram 500 illustrating a monostatic sensing scenario and FIG. 5B is a diagram 530 illustrating a bistatic sensing scenario. In FIG. 5A, the transmitter (Tx) and receiver (Rx) are co-located in the same sensing device 504 (e.g., a UE). The sensing device 504 transmits one or more RF sensing signals 534 (e.g., uplink or sidelink positioning reference signals (PRS) where the sensing device 504 is a UE), and some of the RF sensing signals 534 reflect off a target object 506. The sensing device 504 can measure various properties (e.g., times of arrival (ToAs), angles of arrival (AoAs), phase shift, etc.) of the reflections 536 of the RF sensing signals 534 to determine characteristics of the target object 506 (e.g., size, shape, speed, motion state, etc.).

[0129] In FIG. 5B, the transmitter (Tx) and receiver (Rx) are not co-located, that is, they are separate devices (e.g., a UE and a base station). Note that while FIG. 5B illustrates using a downlink RF signal as the RF sensing signal 532, uplink RF signals or sidelink RF signals can also be used as RF sensing signals 532. In a downlink scenario, as shown, the transmitter is a base station and the receiver is a UE, whereas in an uplink scenario, the transmitter is a UE and the receiver is a base station.

[0130] Referring to FIG. 5B in greater detail, the transmitter device 502 transmits RF sensing signals 532 and 534 (e.g., positioning reference signals (PRS)) to the sensing device 504, but some of the RF sensing signals 534 reflect off a target object 506. The sensing device 504 (also referred to as the “sensing device”) can measure the times of arrival (ToAs) of the RF sensing signals 532 received directly from the transmitter device and the ToAs of the reflections 536 of the RF sensing signals 534 reflected from the target object 506.

[0131] More specifically, as described above, a transmitter device (e.g., a base station) may transmit a single RF signal or multiple RF signals to a sensing device (e.g., a UE). However, the receiver may receive multiple RF signals corresponding to each transmitted RF signal due to the propagation characteristics of RF signals through multipath channels. Each path may be associated with a cluster of one or more channel taps. Generally, the time at which the receiver detects the first cluster of channel taps is considered the ToA of the RF signal on the line-of-sight (LOS) path (i.e., the shortest path between the transmitter and the receiver). Later clusters of channel taps are considered to have reflected off objects between the transmitter and the receiver and therefore to have followed non-LOS (NLOS) paths between the transmitter and the receiver.

[0132] Thus, referring back to FIG. 5B, the RF sensing signals 532 followed the LOS path between the transmitter device 502 and the sensing device 504, and the RF sensing signals 534 followed an NLOS path between the transmitter device 502 and the sensing device 504 due to reflecting off the target object 506. The transmitter device 502 may have transmitted multiple RF sensing signals 532, 534, some of which followed the LOS path and others of which followed the NLOS path. Alternatively, the transmitter device 502 may have transmitted a single RF sensing signal in a broad enough beam that a portion of the RF sensing signal followed the LOS path (RF sensing signal 532) and a portion of the RF sensing signal followed the NLOS path (RF sensing signal 534).

[0133] Based on the ToA of the LOS path, the ToA of the NLOS path, and the speed of light, the sensing device 504 can determine the distance to the target object(s). For example, the sensing device 504 can calculate the distance to the target object as the difference between the ToA of the LOS path and the ToA of the NLOS path multiplied by the speed of light. In addition, if the sensing device 504 is capable of receive beamforming, the sensing device 504 may be able to determine the general direction to a target object as the direction (angle) of the receive beam on which the RF sensing signal following the NLOS path was received. That is, the sensing device 504 may determine the direction to the target object as the angle of arrival (AoA) of the RF sensing signal, which is the angle of the receive beam used to receive the RF sensing signal. The sensing device 504 may then optionally report this information to the transmitter device 502, its serving base station, an application server associated with the core network, an external client, a third-party application, or some other sensing entity. Alternatively, the sensing device 504 may report the ToA measurements to the transmitter device 502, or other sensing entity (e.g., if the sensing device 504 does not have the processing capability to perform the calculations itself), and the transmitter device 502 may determine the distance and, optionally, the direction to the target object 506.

[0134] Note that if the RF sensing signals are uplink RF signals transmitted by a UE to a base station, the base station would perform object detection based on the uplink RF signals just like the UE does based on the downlink RF signals.

[0135] Like conventional radar, wireless communication-based sensing signals can be used to estimate the range (distance), velocity (Doppler), and angle (AoA) of a target object. However, the performance (e.g., resolution and maximum values of range, velocity, and angle) may depend on the design of the reference signal.

[0136] FIG. 6 illustrates an example call flow 600 for an NR-based sensing procedure (e.g., a bistatic sensing procedure) in which the network configures the sensing parameters, according to aspects of the disclosure. Although FIG. 6 illustrates a network-coordinated sensing procedure, the sensing procedure could be coordinated over sidelink channels.

[0137] At stage 605, a sensing server 670 (e.g., inside or outside the core network) sends a request for network (NW) information to a gNB 622 (e.g., the serving gNB of a UE 604). The request may be for a list of the UE's 604 serving cell and any neighboring cells. At stage 610, the gNB 622 sends the requested information to the sensing server 670. At stage 615, the sensing server 670 sends a request for sensing capabilities to the UE 604. At stage 620, the UE 604 provides its sensing capabilities to the sensing server 670.

[0138] At stage 625, the sensing server 670 sends a configuration to the UE 604 indicating one or more reference signal (RS) resources that will be transmitted for sensing. The reference signal resources may be transmitted by the serving and/or neighboring cells identified at stage 610. In some cases, the NR-based sensing procedure illustrated in FIG. 6 may be a sensing-only procedure or a joint communication and sensing (JCS) procedure. In the case of a sensing-only procedure, the reference signal resources may be reference signal resources specifically configured for sensing purposes. In the case of a JCS procedure, the reference signal resources may be reference signal resources for communication that can also be used for sensing purposes. Alternatively, the reference signal resources for sensing may be multiplexed (e.g., time-division multiplexed) with reference signal resources for communication. For example, the reference signal resources for communication may be an orthogonal frequency division multiplexing (OFDM) waveform, while the reference signal resources for sensing may be a frequency modulation continuous wave (FMCW) waveform.

[0139] At stage 630, the sensing server 670 sends a request for sensing information to the UE 604. The UE 604 then measures the transmitted reference signals and, at stage 635, sends the measurements, or any sensing results determined from the measurements, to the sensing server 670.

[0140] In an aspect, the communication between the UE 604 and the sensing server 670 may be via the LTE positioning protocol (LPP). The communication between the sensing server 670 and the gNB may be via NR positioning protocol type A (NRPPa).

[0141] FIG. 7 is a block diagram illustrating a system 700 for performing a sensing session (e.g., an RF sensing session), according to aspects of the disclosure. In some aspects, a sensing session may correspond to a use case for detecting a moving target object. In some aspects, some use case examples may include use cases on intruder detection

in the surroundings of a smart home, use cases on automated guided vehicle (AGV) detection and tracking in a factory or warehouse, use cases of integrated sensing and positioning in a factory hall, or the like.

[0142] In this non-limiting example, the system 700 may include a sensing server 710, a managing device 720, a transmitter device 730, a processing device 740, and a receiver device 750. In some aspects, the sensing server 710 may send a request to the managing device 720 requesting for performing a sensing session. In some aspects, the managing device 720 may be configured to manage and coordinate the sensing session. In some aspects, the managing device may also be referred to as a sensing entity of the sensing session.

[0143] In some aspects, the managing device 720 may instruct the transmitter device 730 to transmit one or more sensing signals for the sensing session. For example, in an environment that has a target object 762 (which may be a moving object), the transmitter device 730 may transmit an incident portion 772 of a sensing signal. In some aspects, the incident portion 772 of the sensing signal may interact with the target object 762 to cause a reflection portion 776 of the sensing signal. In some aspects, the receiver device 750 may receive the reflection portion 776 of a sensing signal and convert the received reflection portion of the sensing signals into a set of measurement samples associated with the object sensing session. In some aspects, one or more wireless devices that incorporate the transmitter device 730, the receiver device 750, or both, may also be referred to as one or more sensing nodes (or sensing node devices) of the sensing session.

[0144] In some aspects, the managing device 720 may inform the receiver device 750 and/or the processing device 740 how the one or more sensing signals for the sensing session are configured. The receiver device 750 may provide the set of measurement samples to the processing device 740. In some aspects, the processing device 740 may receive the set of measurement samples and provide a measurement report to the managing device 720 and/or the sensing server 710.

[0145] In some aspects, the sensing server 710 may be part of a core network or outside the core network. In some aspects, the sensing server 710 may be a location server described herein or a proprietary server. In some aspects, the managing device 720 may be part of the core network, and the transmitter device 730 may be a base station (or TRP) or an anchor UE (e.g., a roadside unit, RSU). In some aspects, the managing device 720 (as part of the core network) may be an entity different from the sensing server 710 or may be a part of (or merged into) the sensing server 710. In some aspects, the managing device 720 may be a base station or an anchor UE; and the transmitter device 730 may be part of the managing device 720. In some aspects, the sensing signal may be a reference signal that may include PRS, SSB, DMRS, CSI-RS, or PTRS. In some aspects, the sensing signal may be a dedicated signal different from any of the above-mentioned reference signal examples.

[0146] In some aspects, the processing device 740 may be a standalone entity, and the receiver device 750 may be a UE or a base station (or TRP or AP). In some aspects, the processing device 740 may be a UE or a base station (or TRP or AP); and the receiver device 750 may be part of the processing device 740. In some aspects, the processing device 740 may be a UE or a base station (or TRP or AP);

and the receiver device 750 as well as the transmitter device 730 may be part of the processing device 740 for monostatic sensing.

[0147] In some aspects, the sensing time window of a sensing session (or also referred to as a sensing integration time window or a sensing window) may be set to be sufficiently large in order to meet a desirable accuracy of the velocity estimation from the sensing session. In some aspects, while base stations can have stable clocks during the sensing window, UEs typically suffer from clock drift which may impact the sensing performance. Clock drift may impact the sensing measurements (e.g., a range-velocity map based on a Doppler analysis) as additional undesired phase terms. In some aspects, while the clock drift of a UE based on a practical implementation may be sufficient for communication purposes (e.g., no greater than 0.1 parts per million (ppm)), such clock drift may impact the resolution of the velocity estimation or range estimation when the UE is configured as a sensing node device (either to function as a transmitter device, a receiver device, or both) with a large sensing window.

[0148] For example, for an OFDM-based RF sensing, the velocity resolution may be determined based on an equation of $\Delta v = \lambda / 2T_{frm}$, where T_{frm} represents the duration of the sensing window, and λ represents the wavelength of the operating frequency. In an example, to achieve a velocity resolution of 0.5 m/s with an operating frequency 6 GHz, the duration of the sensing window T_{frm} according to the equation may be at least 50 ms. In this example, assuming that a UE configured as a sensing node device has clock drift of 0.1 ppm, the total clock drift during the duration of the sensing window T_{frm} (e.g., 50 ms) may be about 5 ns, which may correspond to a range error of 1.5 m. In some aspects, such range error may be too large for the use cases of RF sensing as illustrated above.

[0149] In some aspects, a sensing node device (e.g., a UE) may experience both clock drift and a clock synchronization offset with the network. In some aspects, a UE may synchronize its clock to the network using reference signals (e.g., SSB). However, there will still be synchronization offsets which may impact the performance of a sensing session. In this disclosure, the proposed embodiments focus on primarily reduce the impact of the clock drift, with the clock synchronization offset issue being addressed together or separately by one or more other approaches.

[0150] In some aspects, in order to address the issue of undesirable performance degradation in an RF sensing session caused by the clock drift as illustrated above, the sensing entity (e.g., the managing device 720) may configure the sensing session by taking the clock drift of the sensing node device (e.g., a UE incorporating the transmitter device 730, the receiver device 750, or a combination thereof) into consideration. In some aspects, the sensing node device may transmit, to a managing device, clock drift information corresponding to clock drift of the sensing node device. In some aspects, the managing device may adopt a sensing signal configuration for a sensing session based on the clock drift information. In some aspects, the managing device may transmit, to the sensing node device, a measurement configuration for the sensing session based on the clock drift information.

[0151] In some aspects, the sensing node device (e.g., a UE incorporating the transmitter device 730, the receiver device 750, or a combination thereof) may share its clock

drift information with the managing device **720**. In some aspects, the managing device **720** may be a sensing entity of a sensing session and may be a dedicated core network entity, or a part of an existing core network entity (gNB, LMF), or a UE (e.g., a roadside UE). In some aspects, the clock drift information may include a maximum value of the clock drift of the sensing node device, a median value of the clock drift of the sensing node device, a statistical description of the clock drift of the sensing node device (e.g., a distribution and/or range of possible values), variability of the clock drift as a function of one or more environmental parameters (e.g., the temperature and/or pressure), a maximum duration of the sensing window supported by the sensing node device, or a combination thereof.

[0152] In some aspects, the sensing node device may be a UE. In some aspects, the clock drift information may be transmitted to the managing device based on LPP signaling in a case that the managing device is a location server. In some aspects, the clock drift information may be transmitted to the managing device based on MAC control element (MAC CE) signaling or downlink control information (DCI) signaling in a case that the managing device is a base station or a TRP. In some aspects, the clock drift information may be transmitted to the managing device based on sidelink LPP signaling, sidelink RRC signaling, sidelink MAC CE signaling, or sidelink control information (SCI) signaling (e.g., SCI stage 1, SCI stage 2, or both) in a case that the managing device is another UE (e.g., a roadside UE).

[0153] In some aspects, the clock drift information may be transmitted to the managing device as a solicited transmission in response to a request from the managing device. In some aspects, the clock drift information may be transmitted to the managing device as an unsolicited transmission initiated by the sensing node device.

[0154] In some aspects, based on the clock drift information from the sensing node device, the managing device may configure the sensing node device with adapted sensing signal transmissions for a sensing session. In some aspects, the adapted sensing signal transmissions may correspond to one or more sensing signals to be transmitted or received by the sensing node device during a sensing window of the sensing session based on a sensing signal configuration for the sensing session. In some aspects, the one or more sensing signals may be transmitted by the sensing node device, such as SRS or SL-PRS. In some aspects, the one or more sensing signals may be received by the sensing node device, such as DL-PRS or SL-PRS.

[0155] In some aspects, the sensing signal configuration may be configured such that the clock drift during the sensing window of the sensing session is expected to be negligible (with respect to the required resolution of the velocity estimation or range estimation), and thus may not impact the sensing measurements and results. In some aspects, the sensing signal configuration may indicate one or more parameters based at least in part on the clock drift information, and the one or more parameters may include a numerology of the one or more sensing signals (e.g., sub-carrier spacing), an operating frequency of the one or more sensing signals, an inter-symbol duration of the one or more sensing signals (e.g., periodicity or a time interval between consecutive sensing signal transmissions within the sensing window), a duration of the sensing window, or a combination thereof. In some aspects, the configuration may reduce an impact of the clock drift of the sensing node device.

[0156] In some aspects, the managing device may select the sensing node devices for a sensing session from a set of candidate sensing node devices based on the corresponding clock drift information reported from the candidate sensing node devices. For example, the managing device may select a subset of available sensing node devices that the configured sensing signal transmissions within the sensing window may not be impacted by the clock drift happened at the selected subset of the available sensing node device. In some aspects, the managing device may transmit, to a selected sensing node device, a configuration to participate in the sensing session based on the clock drift information of the selected sensing node device.

[0157] In some aspects, as an alternative of or in addition to the adapted sensing signal transmissions, the managing device may configure a sensing node device (e.g., a UE incorporating the processing device **740** and the receiver device **750**) with adapted measurement processing. In some aspects, the adapted measurement processing may be indicated by a measurement configuration for the sensing session based on the clock drift information.

[0158] Assuming that the sensing node device receives the one or more sensing signals during the sensing window of the sensing session and obtains N signal samples based on the one or more sensing signals, N being a positive integer, the managing device may configure the sensing node device to process the N signal samples based on different approaches.

[0159] In some aspects, based on a first example approach, if the clock drift of the sensing node device within the sensing window is negligible, the measurement configuration may indicate that the sensing measurements (e.g., a range map or a range-velocity map based on a Doppler analysis) may be generated based on processing the N signal samples as a single batch to obtain the sensing measurements.

[0160] FIG. **8** illustrates a processing flow **800** for processing signal samples by a sensing node device based on a second example approach, according to aspects of the disclosure. In some aspects, based on the second example approach, the measurement configuration may indicate that the sensing measurements may be generated based on dividing the N signal samples into a plurality of batches, adjusting the N signal samples based on a clock drift correction estimation on a per-batch basis to obtain N adjusted signal samples, and processing the N adjusted signal samples as a single batch to obtain the sensing measurements.

[0161] For example, the processing flow **800** shows N samples $S_0 \sim S_{N-1}$ received by a sensing node device within a sensing window. The N samples $S_0 \sim S_{N-1}$ may be divided into a plurality of batches, where each batch of the plurality of batches may have M signal samples and M is a positive integer less than N . In some aspects, the monitoring device may include the value M and/or the value N in the measurement configuration to the sensing node device. In the example shown in FIG. **8**, the first batch may include samples $S_0, S_1, \dots, S_{M-1}$, the second batch may include samples $S_M, S_{M+1}, \dots, S_{2M-1}$, and the (N/M) -th batch may include samples $S_{N-M}, S_{N-M+1}, \dots, S_{N-1}$ (assuming N/M is an integer in this example).

[0162] In some aspects, for each of the batch, a clock offset estimation and correction process (e.g., blocks **810-1**, **810-2**, and **810-(N/M)**) may be performed based on the M samples of the corresponding batch to obtain M respective

corrected samples (and N corrected samples in total form all the batches). In some aspects, the clock offset estimation and correction process may be based on a clock drift correction estimation that may address the clock drift of the sensing node device, or both the clock drift and clock synchronization offset of the sensing node device. In some aspects, the sensing node device may estimate the clock drift correction estimation τ_{offset} for each batch (i.e., on a per-batch basis). In some aspects, the sensing node device may apply the clock drift correction estimation τ_{offset} to the M samples of a batch in the frequency domain by multiplying the observed channel frequency response with a term of $\exp(-j2\pi\Delta f\tau_{offset}k)$, $k=0, \dots, K-1$, where Δf represents the subcarrier spacing, and K represents the number of subcarriers.

[0163] In some aspects, after obtaining the N corrected samples based on the respective clock offset estimation and correction processes (e.g., blocks 810-1, 810-2, and 810-(N/M)), the sensing node device may generate RF sensing measurements (e.g., a range map or a range-velocity map) based on performing a Doppler analysis on the N corrected samples (e.g., block 820). Afterwards, the sensing node device may report the sensing measurements to the managing device (e.g., block 830).

[0164] FIG. 9 illustrates an example scenario 900 for determining a clock drift correction estimation (e.g., the clock drift correction estimation τ_{offset} for each batch in FIG. 8), according to aspects of the disclosure. In this example scenario, a UE 910 is configured as a sensing node device for a sensing session that is based on a sensing signal from a TRP 920. In this sensing session, the sensing signal may be along a sensing signal path 922 from the TRP 920, reflected by a target object 930, and then reaching the UE 910. Also, to clock drift correction estimation, the UE 910 may receive a reference signal from a reference TRP 940 along an LOS reference signal path 944 or an NLOS reference signal path 942 reflected by an obstacle 950.

[0165] In some aspects, the reference signal path 942 and/or the reference signal path 944 may be configured by the managing device. In some aspects, the reference signal path 942 and/or the reference signal path 944 may be specifically for the UE 910, and other UEs may determine the corresponding clock drift correction estimations based on other reference signal paths and/or other reference TRPs. In some aspects, the TRP 920 may also be used as the reference TRP 940. In some aspects, the reference TRP 940 may be different from the TRP 920. In some aspects, the identity of the reference TRP 940 and/or the location of the reference TRP 940 may be known to the managing device and may be signaled to the UE 910 by the managing device (e.g., included in the measurement configuration). In some aspects, the location of the UE 910 may be known to the managing device and may be signaled to the UE 910 by the managing device (e.g., included in the measurement configuration). In some aspects, the UE 910 may already know its own location.

[0166] In some aspects, in the case that the reference signal path configured for the UE 910 corresponds to the NLOS reference signal path 942 that involves the obstacle 950, the location of the obstacle 950 may be known to the managing device and may be signaled to the UE 910 (e.g., included in the measurement configuration). In some aspects, the obstacle 950 may be a building in the environment or a reflecting intelligent surface (RIS). In some aspects, the UE 910 may determine the clock drift correction

estimation to be used in the process of FIG. 8 based on the received reference signal, the location of the UE 910, the location of the reference TRP 940, and the location of the obstacle 950 (if the reference signal is along the NLOS reference signal path 942). In some aspects, the clock drift correction estimation may take the synchronization clock offset of the UE 910 and the clock drift of the UE 910 into consideration.

[0167] FIG. 10 illustrates a processing flow 1000 for processing signal samples by a sensing node device based on a third example approach, according to aspects of the disclosure. In some aspects, based on the third example approach, the measurement configuration may indicate that the sensing measurements may be generated based on dividing the N signal samples into the plurality of batches, processing the N signal samples on a per-batch basis to obtain a plurality of per-batch measurements (e.g., per-batch range maps or per-batch range-velocity maps), and combining the plurality of per-batch measurements to obtain the sensing measurements (e.g., a collective range map or a collective range-velocity map).

[0168] In this example, the processing flow 1000 shows N samples $S_0 \sim S_{N-1}$ received by a sensing node device within a sensing window, where the N samples may be divided into (N/M) batches with each batch having M samples, as illustrated in FIG. 8 (assuming N/M is an integer in this example). In some aspects, the monitoring device may signal the value M and/or the value N to the sensing node device (e.g., included in a measurement configuration).

[0169] In some aspects, for each of the batch, a RF sensing measurements generation process may be performed (e.g., blocks 1010-1, 1010-2, and 1010-(N/M)) based on the M signal samples of the corresponding batch to obtain respective sets of sensing measurements (e.g., per-batch respective range maps or per-batch range-velocity maps for each of the batches based on respective Doppler analyses on a per-batch basis). In some aspects, the monitoring device may determine the value M in a manner that the clock drift of the sensing node device within each batch may be negligible.

[0170] In some aspects, after obtaining the respective sets of sensing measurements for individual batches, the sensing node device may combine the sensing measurements from blocks 1010-1, 1010-2, and 1010-(N/M) (e.g., block 1020). Afterwards, the sensing node device may report the sensing measurements to the managing device (e.g., block 1030).

[0171] In some aspects, the first example approach may be deemed as a special case of the third example approach with the M is set to N (i.e., all the N samples belong to a single batch).

[0172] In some aspects, any of the first, second, and third example approaches and the corresponding parameters may be configured by the managing device for each sensing node device (e.g., the corresponding transmitter devices and/or the corresponding receiver devices) in a sensing session individually. For example, multiple sensing node devices may be measuring the same DL-PRS resources for a sensing session, but may process the samples differently based on respective clock drift information.

[0173] In some aspects, for a sensing session where a UE is configured as a sensing node device, the UE may transmit an on-demand sensing request to a base station or a TRP requesting resource allocation for the one or more sensing signals (either as uplink signals or downlink signals). In some aspects, the on-demand sensing request may include

the clock drift information of the sensing node device. In some aspects, the on-demand sensing request may include other parameters including a desired bandwidth, a target velocity resolution, a target maximum velocity, a target maximum range, or any combination thereof.

[0174] FIG. 11 illustrates an example application of performing a sensing session in a factory-like environment 1100, according to aspects of the disclosure. In this example, the factory-like environment 1100 may include a portion of an indoor space defined by the boundaries 1102 (e.g., walls). The factory-like environment 1100 may include obstacles 1106, which may be shelves, machines, and/or storages placed in the indoor space of the factory-like environment 1100. In this example, the factory-like environment 1100 may include base station devices 1112 and 1116 (e.g., APs or TRPs) and sensing reference unit (SRU) devices 1121, 1122, 1123, 1124, 1125, and 1126 (e.g., UEs or RSUs).

[0175] In some aspects, a sensing session for detecting a target object 1130 (e.g., an AGV) may be performed based on configuring base station device 1116 as a transmitter device to transmit one or more sensing signals, and configuring the SRU devices 1121, 1122, 1123, 1124, 1125, and 1126 as receiver devices to receive various reflection signals and convert the received signals into a set of signal samples. In some aspects, the SRU devices 1121, 1122, 1123, 1124, 1125, and 1126 may also be configured processing devices to process the signal samples in order to obtain the corresponding sensing measurements (e.g., range maps or range-velocity maps). In some aspects, the base station device 1116 may correspond to a sensing node device that is configured to transmit the one or more sensing signals. In some aspects, the SRU devices 1121, 1122, 1123, 1124, 1125, and 1126 may correspond to sensing node devices that are configured to receive the one or more sensing signals.

[0176] In the non-limiting example illustrated in FIG. 11, the operating frequency of the one or more sensing signals may be set to 6 GHz with a channel of a bandwidth of 100 MHz. In some aspects, the SRU devices 1121, 1122, 1123, 1124, 1125, and 1126 may be configured to sample the channel every 10 ms, and the sensing window may be set to 60 ms.

[0177] In some aspects, experiments are conducted using the factory-like environment 1100 considering the following scenarios where the clock drift of the SRU devices 1121, 1122, 1123, 1124, 1125, and 1126 may be 0 ppm, 0.01 ppm, 0.1 ppm, or 1 ppm. The experiments are based on configuring the processing of the signal samples based on the first example approach and based on the second example approach as illustrated above, with the base station device 1112 being configured as a reference TRP (with LOS signal paths) for the second example approach. Based on the experiments, the first example approach may have better performance when the clock drift is about 0 ppm or 0.01 ppm; and the second example approach may have better performance when the clock drift is about 0.1 ppm or 1 ppm. As discussed above, the managing device may configure the sensing session based on suitable approaches for individual sensing node devices in order to reduce the positioning errors.

[0178] FIG. 12 is a flowchart illustrating a method 1200 of operating a sensing node device, according to aspects of the disclosure. In some aspects, the sensing node device in the method 1200 may correspond to a device that incorporates the receiver device 750 or the transmitter device 730 in FIG.

7. In some aspects, the sensing node device may be a base station, a TRP, or an AP (such as any of the base station, TRP, or AP described herein); and the method 1200 may be performed by the one or more WWAN transceivers 350, the one or more network transceivers 380, the one or more processors 384, the memory 386, and/or the sensing component 388, any or all of which may be considered means for performing one or more of the following operations of method 1200. In some aspects, the sensing node device may correspond to a UE (such as any of the UE described herein); and the method 1200 may be performed by the one or more WWAN transceivers 310, the one or more processors 342, the memory 340, and/or the sensing component 348, any or all of which may be considered means for performing one or more of the following operations of method 1200.

[0179] At operation 1210, the sensing node device may transmit, to a managing device, clock drift information corresponding to clock drift of the sensing node device. In some aspects, the clock drift information may indicate a maximum value of the clock drift of the sensing node device, a median value of the clock drift of the sensing node device, a statistical description of the clock drift of the sensing node device, variability of the clock drift as a function of one or more environmental parameters, a maximum duration of the sensing window supported by the sensing node device, or a combination thereof. In some aspects, the clock drift information may be transmitted as a solicited transmission in response to a request from the managing device. In some aspects, the clock drift information may be transmitted as an unsolicited transmission initiated by the sensing node device.

[0180] In some aspects, operation 1210 may be performed by the one or more WWAN transceivers 350, the one or more processors 384, the memory 386, and/or the sensing component 388, any or all of which may be considered means for performing operation 1210. In some aspects, operation 1210 may be performed by the one or more WWAN transceivers 310, the one or more processors 342, the memory 340, and/or the sensing component 348, any or all of which may be considered means for performing operation 1210.

[0181] In some aspects, the sensing node device may be a UE, and the clock drift information may be transmitted based on: (i) LPP signaling in a case that the managing device is a location server; (ii) RRC signaling, MAC CE signaling, or DCI signaling in a case that the managing device is a base station or a TRP; or (iii) sidelink LPP signaling, sidelink RRC signaling, sidelink MAC CE signaling, or SCI signaling in a case that the managing device is another UE.

[0182] At operation 1220, the sensing node device may receive, from the managing device, a sensing signal configuration for a sensing session based on the clock drift information. In some aspects, the sensing signal configuration may indicate one or more parameters based at least in part on the clock drift information, the one or more parameters may include a numerology of the one or more sensing signals, an operating frequency of the one or more sensing signals, an inter-symbol duration of the one or more sensing signals, a duration of the sensing window, or a combination thereof. In some aspects, the sensing signal configuration may reduce an impact of the clock drift of the sensing node device.

[0183] In some aspects, operation 1220 may be performed by the one or more WWAN transceivers 350, the one or more processors 384, the memory 386, and/or the sensing component 388, any or all of which may be considered means for performing operation 1220. In some aspects, operation 1220 may be performed by the one or more WWAN transceivers 310, the one or more processors 342, the memory 340, and/or the sensing component 348, any or all of which may be considered means for performing operation 1220.

[0184] At operation 1230, the sensing node device may transmit or receive one or more sensing signals during a sensing window of the sensing session based on the sensing signal configuration. In some aspects, operation 1230 may be performed by the one or more WWAN transceivers 350, the one or more processors 384, the memory 386, and/or the sensing component 388, any or all of which may be considered means for performing operation 1230. In some aspects, operation 1230 may be performed by the one or more WWAN transceivers 310, the one or more processors 342, the memory 340, and/or the sensing component 348, any or all of which may be considered means for performing operation 1230.

[0185] In some aspects, the sensing node device may transmit a request to a base station or a TRP requesting resource allocation for the one or more sensing signals, where the request may include the clock drift information. In some aspects, the sensing node device may receive, from the managing device, a configuration to participate in the sensing session based on the clock drift information.

[0186] In some aspects, the sensing node device may be configured to incorporate the processing device 740 and thus may receive, from the managing device, a measurement configuration for the sensing session based on the clock drift information. In some aspects, the sensing node device may transmit, to the managing device, sensing measurements (e.g., a range map or a range-velocity map) based on the one or more sensing signals received by the sensing node device and the measurement configuration (e.g., based on performing a Doppler analysis on signal samples of the one or more sensing signals).

[0187] In some aspects, the sensing node device may obtain N signal samples based on the one or more sensing signals, where N is a positive integer. In some aspects, the measurement configuration may indicate that the sensing measurements may be generated based on processing the N signal samples as a single batch to obtain the sensing measurements (e.g., the first example approach illustrated above). In some aspects, the measurement configuration may indicate that the sensing measurements may be generated based on dividing the N signal samples into a plurality of batches, adjusting the N signal samples based on a clock drift correction estimation on a per-batch basis to obtain N adjusted signal samples, and processing the N adjusted signal samples as a single batch to obtain the sensing measurements (e.g., the second example approach illustrated above). In some aspects, the measurement configuration may indicate that the sensing measurements may be generated based on dividing the N signal samples into the plurality of batches, processing the N signal samples on a per-batch basis to obtain a plurality of per-batch measurements, and combining the plurality of per-batch measurements to obtain the sensing measurements (e.g., the third example approach

illustrated above). In some aspects, each batch of the plurality of batches may have M signal samples, where M is a positive integer less than N.

[0188] In some aspects, the sensing node device may determine a clock drift correction estimation based on measuring a reference signal along a reference signal path indicated in the measurement configuration. In some aspects, the sensing node device may generate the sensing measurements further based on the one or more sensing signals received by the sensing node device, the clock drift correction estimation, and the measurement configuration.

[0189] As will be appreciated, a technical advantage of the method 1200 is allowing a sensing node device to report its clock drift information to a managing device such that a sensing session may be configured based on the clock drift information. In some aspects, the one or more sensing signals for the sensing session may be based on a sensing signal configuration from the managing device that reduce an impact of the clock drift of the sensing node device. In some aspects, the measurements of the sensing session may be generated as indicated by a measurement configuration from the managing device based on the clock drift information. Accordingly, the range resolution or the velocity resolution of the sensing session may be kept within a desirable level by taking the clock drift of the sensing node device into consideration.

[0190] FIG. 13 is a flowchart illustrating a method 1300 of operating a managing device, according to aspects of the disclosure. In some aspects, the managing device in the method 1300 may correspond to the managing device 720 in FIG. 7. In some aspects, the managing device 720 may be a base station, a TRP, or an AP (such as any of the base station, TRP, or AP described herein); and the method 1300 may be performed by the one or more WWAN transceivers 350, the one or more network transceivers 380, the one or more processors 384, the memory 386, and/or the sensing component 388, any or all of which may be considered means for performing one or more of the following operations of method 1300. In some aspects, the managing device 720 may correspond to a UE (such as any of the UE or RSU described herein); and the method 1300 may be performed by the one or more WWAN transceivers 310, the one or more processors 342, the memory 340, and/or the sensing component 348, any or all of which may be considered means for performing one or more of the following operations of method 1300. In some aspects, the managing device 720 may correspond to a network entity (such as any of the network entity, server, or LMF described herein); and the method 1300 may be performed by the one or more network transceivers 390, the one or more processors 394, the memory 396, and/or the sensing component 398, any or all of which may be considered means for performing one or more of the following operations of method 1300.

[0191] At operation 1310, the managing device may receive, from a sensing node device, clock drift information corresponding to clock drift of the sensing node device. In some aspects, the clock drift information may indicate a maximum value of the clock drift of the sensing node device, a median value of the clock drift of the sensing node device, a statistical description of the clock drift of the sensing node device, variability of the clock drift as a function of one or more environmental parameters, a maximum duration of the sensing window supported by the sensing node device, or a combination thereof. In some

aspects, the managing device may transmit a request to the sensing node device requesting the clock drift information, such that the clock drift information may be transmitted as a solicited transmission in response to the request from the managing device. In some aspects, the clock drift information may be transmitted to the managing device as an unsolicited transmission initiated by the sensing node device.

[0192] In some aspects, operation 1310 may be performed by the one or more WWAN transceivers 350, the one or more processors 384, the memory 386, and/or the sensing component 388, any or all of which may be considered means for performing operation 1310. In some aspects, operation 1310 may be performed by the one or more WWAN transceivers 310, the one or more processors 342, the memory 340, and/or the sensing component 348, any or all of which may be considered means for performing operation 1310. In some aspects, operation 1310 may be performed by one or more network transceivers 390, the one or more processors 394, the memory 396, and/or the sensing component 398, any or all of which may be considered means for performing operation 1310.

[0193] In some aspects, the sensing node device may be a UE, and the clock drift information may be transmitted based on: (i) LPP signaling in a case that the managing device is a location server; (ii) RRC signaling, MAC CE signaling, or DCI signaling in a case that the managing device is a base station or a TRP; or (iii) sidelink LPP signaling, sidelink RRC signaling, sidelink MAC CE signaling, or SCI signaling in a case that the managing device is another UE.

[0194] At operation 1320, the managing device may transmit, to the sensing node device, a sensing signal configuration for a sensing session based on the clock drift information, where the sensing signal configuration may indicate one or more sensing signals to be transmitted or received by the sensing node device during a sensing window of the sensing session. In some aspects, the managing device may determine the sensing signal configuration indicative of one or more parameters based at least in part on the clock drift information. In some aspects, the one or more parameters may include a numerology of the one or more sensing signals, an operating frequency of the one or more sensing signals, an inter-symbol duration of the one or more sensing signals, a duration of the sensing window, or a combination thereof. In some aspects, the sensing signal configuration may reduce an impact of the clock drift of the sensing node device.

[0195] In some aspects, operation 1320 may be performed by the one or more WWAN transceivers 350, the one or more processors 384, the memory 386, and/or the sensing component 388, any or all of which may be considered means for performing operation 1320. In some aspects, operation 1320 may be performed by the one or more WWAN transceivers 310, the one or more processors 342, the memory 340, and/or the sensing component 348, any or all of which may be considered means for performing operation 1320. In some aspects, operation 1320 may be performed by one or more network transceivers 390, the one or more processors 394, the memory 396, and/or the sensing component 398, any or all of which may be considered means for performing operation 1320.

[0196] In some aspects, the managing device may select the sensing node devices for a sensing session from a set of

candidate sensing node devices based on the corresponding clock drift information reported from the candidate sensing node devices. In some aspects, the managing device may transmit, to the sensing node device that is selected for the sensing session, a configuration to participate in the sensing session based on the clock drift information.

[0197] In some aspects, the managing device may transmit, to the sensing node device, a measurement configuration for the sensing session based on the clock drift information. In some aspects, the measurement configuration may indicate that the sensing measurements may be generated based on processing N signal samples obtained based on the one or more sensing signals as a single batch to obtain the sensing measurements (e.g., the first example approach illustrated above). In some aspects, the measurement configuration may indicate that the sensing measurements may be generated based on dividing the N signal samples into a plurality of batches, adjusting the N signal samples based on a clock drift correction estimation on a per-batch basis to obtain N adjusted signal samples, and processing the N adjusted signal samples as a single batch to obtain the sensing measurements (e.g., the second example approach illustrated above). In some aspects, the measurement configuration may indicate that the sensing measurements may be generated based on dividing the N signal samples into the plurality of batches, processing the N signal samples on a per-batch basis to obtain a plurality of per-batch measurements, and combining the plurality of per-batch measurements to obtain the sensing measurements (e.g., the third example approach illustrated above). In some aspects, each batch of the plurality of batches may have M signal samples, where M is a positive integer less than N.

[0198] In some aspects, the measurement configuration may indicate a value of M that corresponds to a size of each batch of signal samples on which sensing measurements (e.g., a range map or a range-velocity map) may be generated (e.g., based on performing a Doppler analysis on signal samples of the one or more sensing signals). In some aspects, the measurement configuration may further indicate a reference signal path, and the clock drift correction estimation may be determinable based on measuring a reference signal along the reference signal path.

[0199] As will be appreciated, a technical advantage of the method 1300 is allowing a managing device to receive the clock drift information of a sensing node device such that the managing device may configure a sensing session based on the clock drift information. In some aspects, the managing device may provide a sensing signal configuration based on the clock drift information such that the one or more sensing signals for the sensing session based on the sensing signal configuration may reduce an impact of the clock drift of the sensing node device. In some aspects, the managing device may provide a measurement configuration based on the clock drift information such that the measurements of the sensing session may be generated as indicated by the clock drift information. Accordingly, the range resolution or the velocity resolution of the sensing session may be kept within a desirable level by taking the clock drift of the sensing node device into consideration.

[0200] In the detailed description above it can be seen that different features are grouped together in examples. This manner of disclosure should not be understood as an intention that the example clauses have more features than are explicitly mentioned in each clause. Rather, the various

aspects of the disclosure may include fewer than all features of an individual example clause disclosed. Therefore, the following clauses should hereby be deemed to be incorporated in the description, wherein each clause by itself can stand as a separate example. Although each dependent clause can refer in the clauses to a specific combination with one of the other clauses, the aspect(s) of that dependent clause are not limited to the specific combination. It will be appreciated that other example clauses can also include a combination of the dependent clause aspect(s) with the subject matter of any other dependent clause or independent clause or a combination of any feature with other dependent and independent clauses. The various aspects disclosed herein expressly include these combinations, unless it is explicitly expressed or can be readily inferred that a specific combination is not intended (e.g., contradictory aspects, such as defining an element as both an electrical insulator and an electrical conductor). Furthermore, it is also intended that aspects of a clause can be included in any other independent clause, even if the clause is not directly dependent on the independent clause.

[0201] Implementation examples are described in the following numbered clauses:

[0202] Clause 1. A method of operating a sensing node device, the method comprising: transmitting, to a managing device, clock drift information corresponding to clock drift of the sensing node device; receiving, from the managing device, a sensing signal configuration for a sensing session based on the clock drift information; and transmitting or receiving one or more sensing signals during a sensing window of the sensing session based on the sensing signal configuration.

[0203] Clause 2. The method of clause 1, wherein the sensing signal configuration indicates one or more parameters based at least in part on the clock drift information, the one or more parameters comprising: a numerology of the one or more sensing signals; an operating frequency of the one or more sensing signals; an inter-symbol duration of the one or more sensing signals; a duration of the sensing window; or a combination thereof.

[0204] Clause 3. The method of any of clauses 1 to 2, further comprising: receiving, from the managing device, a configuration to participate in the sensing session based on the clock drift information.

[0205] Clause 4. The method of any of clauses 1 to 3, further comprising: receiving, from the managing device, a measurement configuration for the sensing session based on the clock drift information; and transmitting, to the managing device, sensing measurements based on the one or more sensing signals received by the sensing node device and the measurement configuration.

[0206] Clause 5. The method of clause 4, wherein: the receiving the one or more sensing signals during the sensing window of the sensing session comprises obtaining N signal samples based on the one or more sensing signals, and the measurement configuration indicates that the sensing measurements are generated based on: processing the N signal samples as a single batch to obtain the sensing measurements; dividing the N signal samples into a plurality of batches, adjusting the N signal samples based on a clock drift correction estimation on a per-batch basis to obtain N adjusted signal samples, and processing the N adjusted signal samples as a single batch to obtain the sensing measurements; or dividing the N signal samples into the

plurality of batches, processing the N signal samples on a per-batch basis to obtain a plurality of per-batch measurements, and combining the plurality of per-batch measurements to obtain the sensing measurements, each batch of the plurality of batches has M signal samples, N is a positive integer, and M is a positive integer less than N.

[0207] Clause 6. The method of clause 4, further comprising: determining a clock drift correction estimation based on measuring a reference signal along a reference signal path indicated in the measurement configuration, wherein the sensing measurements are generated based on the one or more sensing signals received by the sensing node device, the clock drift correction estimation, and the measurement configuration.

[0208] Clause 7. The method of any of clauses 1 to 6, wherein the clock drift information indicates: a maximum value of the clock drift of the sensing node device, a median value of the clock drift of the sensing node device, a statistical description of the clock drift of the sensing node device, variability of the clock drift as a function of one or more environmental parameters, a maximum duration of the sensing window supported by the sensing node device, or a combination thereof.

[0209] Clause 8. The method of any of clauses 1 to 7, wherein: the sensing node device is a user equipment (UE), and the clock drift information is transmitted based on: long term evolution (LTE) positioning protocol (LPP) signaling in a case that the managing device is a location server, radio resource control (RRC) signaling, medium access control (MAC) control element (MAC CE) signaling, or downlink control information (DCI) signaling in a case that the managing device is a base station or a transmission/reception point (TRP), or sidelink LPP signaling, sidelink RRC signaling, sidelink MAC CE signaling, or sidelink control information (SCI) signaling in a case that the managing device is another UE.

[0210] Clause 9. The method of any of clauses 1 to 8, wherein the clock drift information is transmitted as: a solicited transmission in response to a request from the managing device; or an unsolicited transmission initiated by the sensing node device.

[0211] Clause 10. The method of any of clauses 1 to 9, further comprising: transmitting a request to a base station or a transmission/reception point (TRP) requesting resource allocation for the one or more sensing signals, wherein the request includes the clock drift information.

[0212] Clause 11. A method of operating a managing device, the method comprising: receiving, from a sensing node device, clock drift information corresponding to clock drift of the sensing node device; and transmitting, to the sensing node device, a sensing signal configuration for a sensing session based on the clock drift information, the sensing signal configuration indicating one or more sensing signals to be transmitted or received by the sensing node device during a sensing window of the sensing session.

[0213] Clause 12. The method of clause 11, further comprising determining the sensing signal configuration indicative of one or more parameters based at least in part on the clock drift information, the one or more parameters comprising: a numerology of the one or more sensing signals; an operating frequency of the one or more sensing signals; an inter-symbol duration of the one or more sensing signals; a duration of the sensing window; or a combination thereof.

[0214] Clause 13. The method of any of clauses 11 to 12, further comprising: transmitting, to the sensing node device, a configuration to participate in the sensing session based on the clock drift information.

[0215] Clause 14. The method of any of clauses 11 to 13, further comprising: transmitting, to the sensing node device, a measurement configuration for the sensing session based on the clock drift information, wherein: the measurement configuration indicates that sensing measurements are generated based on: processing N signal samples obtained based on the one or more sensing signals as a single batch to obtain the sensing measurements; dividing the N signal samples into a plurality of batches, adjusting the N signal samples based on a clock drift correction estimation on a per-batch basis to obtain N adjusted signal samples, and processing the N adjusted signal samples as a single batch to obtain the sensing measurements; or dividing the N signal samples into the plurality of batches, processing the N signal samples on a per-batch basis to obtain a plurality of per-batch measurements, and combining the plurality of per-batch measurements to obtain the sensing measurements, each batch of the plurality of batches has M signal samples, N is a positive integer, and M is a positive integer less than N.

[0216] Clause 15. The method of clause 14, wherein: the measurement configuration indicates a value of M that corresponds to a size of each batch of signal samples on which sensing measurements are generated.

[0217] Clause 16. The method of any of clauses 14 to 15, wherein: the measurement configuration further indicates a reference signal path, and the clock drift correction estimation is determinable based on measuring a reference signal along the reference signal path.

[0218] Clause 17. The method of any of clauses 11 to 16, wherein the clock drift information indicates: a maximum value of the clock drift of the sensing node device, a median value of the clock drift of the sensing node device, a statistical description of the clock drift of the sensing node device, variability of the clock drift as a function of one or more environmental parameters, a maximum duration of the sensing window supported by the sensing node device, or a combination thereof.

[0219] Clause 18. The method of any of clauses 11 to 17, wherein: the sensing node device is a user equipment (UE), and the clock drift information is received based on: long term evolution (LTE) positioning protocol (LPP) signaling in a case that the managing device is a location server, radio resource control (RRC) signaling, medium access control (MAC) control element (MAC CE) signaling, or downlink control information (DCI) signaling in a case that the managing device is a base station or a transmission/reception point (TRP), or sidelink LPP signaling, sidelink RRC signaling, sidelink MAC CE signaling, or sidelink control information (SCI) signaling in a case that the managing device is another UE.

[0220] Clause 19. The method of any of clauses 11 to 18, further comprising: transmitting a request to the sensing node device requesting the clock drift information such that the clock drift information is transmitted as a solicited transmission in response to the request from the managing device.

[0221] Clause 20. A sensing node device, comprising: one or more memories; one or more transceivers; and one or more processors communicatively coupled to the one or more memories and the one or more transceivers, the one or

more processors, either alone or in combination, configured to: transmit, to a managing device via the one or more transceivers, clock drift information corresponding to clock drift of the sensing node device; receive, from the managing device via the one or more transceivers, a sensing signal configuration for a sensing session based on the clock drift information; and transmit or receive, via the one or more transceivers, one or more sensing signals during a sensing window of the sensing session based on the sensing signal configuration.

[0222] Clause 21. The sensing node device of clause 20, wherein the sensing signal configuration indicates one or more parameters based at least in part on the clock drift information, the one or more parameters comprising: a numerology of the one or more sensing signals; an operating frequency of the one or more sensing signals; an inter-symbol duration of the one or more sensing signals; a duration of the sensing window; or a combination thereof.

[0223] Clause 22. The sensing node device of any of clauses 20 to 21, wherein the one or more processors, either alone or in combination, are further configured to: receive, from the managing device via the one or more transceivers, a configuration to participate in the sensing session based on the clock drift information.

[0224] Clause 23. The sensing node device of any of clauses 20 to 22, wherein the one or more processors, either alone or in combination, are further configured to: receive, from the managing device via the one or more transceivers, a measurement configuration for the sensing session based on the clock drift information; and transmit, to the managing device via the one or more transceivers, sensing measurements based on the one or more sensing signals received by the sensing node device and the measurement configuration.

[0225] Clause 24. The sensing node device of clause 23, wherein: the one or more processors, either alone or in combination, are further configured to obtain N signal samples based on the one or more sensing signals, and the measurement configuration indicates that the sensing measurements are generated based on: processing the N signal samples as a single batch to obtain the sensing measurements; dividing the N signal samples into a plurality of batches, adjusting the N signal samples based on a clock drift correction estimation on a per-batch basis to obtain N adjusted signal samples, and processing the N adjusted signal samples as a single batch to obtain the sensing measurements; or dividing the N signal samples into the plurality of batches, processing the N signal samples on a per-batch basis to obtain a plurality of per-batch measurements, and combining the plurality of per-batch measurements to obtain the sensing measurements, each batch of the plurality of batches has M signal samples, N is a positive integer, and M is a positive integer less than N.

[0226] Clause 25. The sensing node device of clause 23, wherein the one or more processors, either alone or in combination, are further configured to: determine a clock drift correction estimation based on measuring a reference signal along a reference signal path indicated in the measurement configuration, wherein the sensing measurements are generated based on the one or more sensing signals received by the sensing node device, the clock drift correction estimation, and the measurement configuration.

[0227] Clause 26. The sensing node device of any of clauses 20 to 25, wherein the clock drift information indicates: a maximum value of the clock drift of the sensing

node device, a median value of the clock drift of the sensing node device, a statistical description of the clock drift of the sensing node device, variability of the clock drift as a function of one or more environmental parameters, a maximum duration of the sensing window supported by the sensing node device, or a combination thereof.

[0228] Clause 27. The sensing node device of any of clauses 20 to 26, wherein: the sensing node device is a user equipment (UE), and the clock drift information is transmitted based on: long term evolution (LTE) positioning protocol (LPP) signaling in a case that the managing device is a location server, radio resource control (RRC) signaling, medium access control (MAC) control element (MAC CE) signaling, or downlink control information (DCI) signaling in a case that the managing device is a base station or a transmission/reception point (TRP), or sidelink LPP signaling, sidelink RRC signaling, sidelink MAC CE signaling, or sidelink control information (SCI) signaling in a case that the managing device is another UE.

[0229] Clause 28. The sensing node device of any of clauses 20 to 27, wherein the clock drift information is transmitted as: a solicited transmission in response to a request from the managing device; or an unsolicited transmission initiated by the sensing node device.

[0230] Clause 29. The sensing node device of any of clauses 20 to 28, wherein the one or more processors, either alone or in combination, are further configured to: transmit, via the one or more transceivers, a request to a base station or a transmission/reception point (TRP) requesting resource allocation for the one or more sensing signals, wherein the request includes the clock drift information.

[0231] Clause 30. A managing device, comprising: one or more memories; one or more transceivers; and one or more processors communicatively coupled to the one or more memories and the one or more transceivers, the one or more processors, either alone or in combination, configured to: receive, from a sensing node device via the one or more transceivers, clock drift information corresponding to clock drift of the sensing node device; and transmit, to the sensing node device via the one or more transceivers, a sensing signal configuration for a sensing session based on the clock drift information, the sensing signal configuration indicating one or more sensing signals to be transmitted or received by the sensing node device during a sensing window of the sensing session.

[0232] Clause 31. The managing device of clause 30, wherein the one or more processors, either alone or in combination, are further configured to determine the sensing signal configuration indicative of one or more parameters based at least in part on the clock drift information, the one or more parameters comprising: a numerology of the one or more sensing signals; an operating frequency of the one or more sensing signals; an inter-symbol duration of the one or more sensing signals; a duration of the sensing window; or a combination thereof.

[0233] Clause 32. The managing device of any of clauses 30 to 31, wherein the one or more processors, either alone or in combination, are further configured to: transmit, to the sensing node device via the one or more transceivers, a configuration to participate in the sensing session based on the clock drift information.

[0234] Clause 33. The managing device of any of clauses 30 to 32, wherein the one or more processors, either alone or in combination, are further configured to: transmit, to the

sensing node device, a measurement configuration for the sensing session based on the clock drift information, wherein: the measurement configuration indicates that sensing measurements are generated based on: processing N signal samples obtained based on the one or more sensing signals as a single batch to obtain the sensing measurements; dividing the N signal samples into a plurality of batches, adjusting the N signal samples based on a clock drift correction estimation on a per-batch basis to obtain N adjusted signal samples, and processing the N adjusted signal samples as a single batch to obtain the sensing measurements; or dividing the N signal samples into the plurality of batches, processing the N signal samples on a per-batch basis to obtain a plurality of per-batch measurements, and combining the plurality of per-batch measurements to obtain the sensing measurements, each batch of the plurality of batches has M signal samples, N is a positive integer, and M is a positive integer less than N.

[0235] Clause 34. The managing device of clause 33, wherein: the measurement configuration indicates a value of M that corresponds to a size of each batch of signal samples on which sensing measurements are generated.

[0236] Clause 35. The managing device of any of clauses 33 to 34, wherein: the measurement configuration further indicates a reference signal path, and the clock drift correction estimation is determinable based on measuring a reference signal along the reference signal path.

[0237] Clause 36. The managing device of any of clauses 30 to 35, wherein the clock drift information indicates: a maximum value of the clock drift of the sensing node device, a median value of the clock drift of the sensing node device, a statistical description of the clock drift of the sensing node device, variability of the clock drift as a function of one or more environmental parameters, a maximum duration of the sensing window supported by the sensing node device, or a combination thereof.

[0238] Clause 37. The managing device of any of clauses 30 to 36, wherein: the sensing node device is a user equipment (UE), and the clock drift information is received based on: long term evolution (LTE) positioning protocol (LPP) signaling in a case that the managing device is a location server, radio resource control (RRC) signaling, medium access control (MAC) control element (MAC CE) signaling, or downlink control information (DCI) signaling in a case that the managing device is a base station or a transmission/reception point (TRP), or sidelink LPP signaling, sidelink RRC signaling, sidelink MAC CE signaling, or sidelink control information (SCI) signaling in a case that the managing device is another UE.

[0239] Clause 38. The managing device of any of clauses 30 to 37, wherein the one or more processors, either alone or in combination, are further configured to: transmit, via the one or more transceivers, a request to the sensing node device requesting the clock drift information such that the clock drift information is transmitted as a solicited transmission in response to the request from the managing device.

[0240] Clause 39. A sensing node device, comprising: means for transmitting, to a managing device, clock drift information corresponding to clock drift of the sensing node device; means for receiving, from the managing device, a sensing signal configuration for a sensing session based on the clock drift information; and means for transmitting or

receiving one or more sensing signals during a sensing window of the sensing session based on the sensing signal configuration.

[0241] Clause 40. The sensing node device of clause 39, wherein the sensing signal configuration indicates one or more parameters based at least in part on the clock drift information, the one or more parameters comprising: a numerology of the one or more sensing signals; an operating frequency of the one or more sensing signals; an inter-symbol duration of the one or more sensing signals; a duration of the sensing window; or a combination thereof.

[0242] Clause 41. The sensing node device of any of clauses 39 to 40, further comprising: means for receiving, from the managing device, a configuration to participate in the sensing session based on the clock drift information.

[0243] Clause 42. The sensing node device of any of clauses 39 to 41, further comprising: means for receiving, from the managing device, a measurement configuration for the sensing session based on the clock drift information; and means for transmitting, to the managing device, sensing measurements based on the one or more sensing signals received by the sensing node device and the measurement configuration.

[0244] Clause 43. The sensing node device of clause 42, wherein: the means for receiving the one or more sensing signals during the sensing window of the sensing session comprises means for obtaining N signal samples based on the one or more sensing signals, and the measurement configuration indicates that the sensing measurements are generated based on: processing the N signal samples as a single batch to obtain the sensing measurements; dividing the N signal samples into a plurality of batches, adjusting the N signal samples based on a clock drift correction estimation on a per-batch basis to obtain N adjusted signal samples, and processing the N adjusted signal samples as a single batch to obtain the sensing measurements; or dividing the N signal samples into the plurality of batches, processing the N signal samples on a per-batch basis to obtain a plurality of per-batch measurements, and combining the plurality of per-batch measurements to obtain the sensing measurements, each batch of the plurality of batches has M signal samples, N is a positive integer, and M is a positive integer less than N.

[0245] Clause 44. The sensing node device clause 42, further comprising: means for determining a clock drift correction estimation based on measuring a reference signal along a reference signal path indicated in the measurement configuration, wherein the sensing measurements are generated based on the one or more sensing signals received by the sensing node device, the clock drift correction estimation, and the measurement configuration.

[0246] Clause 45. The sensing node device of any of clauses 39 to 44, wherein the clock drift information indicates: a maximum value of the clock drift of the sensing node device, a median value of the clock drift of the sensing node device, a statistical description of the clock drift of the sensing node device, variability of the clock drift as a function of one or more environmental parameters, a maximum duration of the sensing window supported by the sensing node device, or a combination thereof.

[0247] Clause 46. The sensing node device of any of clauses 39 to 45, wherein: the sensing node device is a user equipment (UE), and the clock drift information is transmitted based on: long term evolution (LTE) positioning

protocol (LPP) signaling in a case that the managing device is a location server, radio resource control (RRC) signaling, medium access control (MAC) control element (MAC CE) signaling, or downlink control information (DCI) signaling in a case that the managing device is a base station or a transmission/reception point (TRP), or sidelink LPP signaling, sidelink RRC signaling, sidelink MAC CE signaling, or sidelink control information (SCI) signaling in a case that the managing device is another UE.

[0248] Clause 47. The sensing node device of any of clauses 39 to 46, wherein the clock drift information is transmitted as: a solicited transmission in response to a request from the managing device; or an unsolicited transmission initiated by the sensing node device.

[0249] Clause 48. The sensing node device of any of clauses 39 to 47, further comprising: means for transmitting a request to a base station or a transmission/reception point (TRP) requesting resource allocation for the one or more sensing signals, wherein the request includes the clock drift information.

[0250] Clause 49. A managing device, comprising: means for receiving, from a sensing node device, clock drift information corresponding to clock drift of the sensing node device; and means for transmitting, to the sensing node device, a sensing signal configuration for a sensing session based on the clock drift information, the sensing signal configuration indicating one or more sensing signals to be transmitted or received by the sensing node device during a sensing window of the sensing session.

[0251] Clause 50. The managing device of clause 49, further comprising means for determining the sensing signal configuration indicative of one or more parameters based at least in part on the clock drift information, the one or more parameters comprising: a numerology of the one or more sensing signals; an operating frequency of the one or more sensing signals; an inter-symbol duration of the one or more sensing signals; a duration of the sensing window; or a combination thereof.

[0252] Clause 51. The managing device of any of clauses 49 to 50, further comprising: means for transmitting, to the sensing node device, a configuration to participate in the sensing session based on the clock drift information.

[0253] Clause 52. The managing device of any of clauses 49 to 51, further comprising: means for transmitting, to the sensing node device, a measurement configuration for the sensing session based on the clock drift information, wherein: the measurement configuration indicates that sensing measurements are generated based on: processing N signal samples obtained based on the one or more sensing signals as a single batch to obtain the sensing measurements; dividing the N signal samples into a plurality of batches, adjusting the N signal samples based on a clock drift correction estimation on a per-batch basis to obtain N adjusted signal samples, and processing the N adjusted signal samples as a single batch to obtain the sensing measurements; or dividing the N signal samples into the plurality of batches, processing the N signal samples on a per-batch basis to obtain a plurality of per-batch measurements, and combining the plurality of per-batch measurements to obtain the sensing measurements, each batch of the plurality of batches has M signal samples, N is a positive integer, and M is a positive integer less than N.

[0254] Clause 53. The managing device of clause 52, wherein: the measurement configuration indicates a value of

M that corresponds to a size of each batch of signal samples on which sensing measurements are generated.

[0255] Clause 54. The managing device of any of clauses 52 to 53, wherein: the measurement configuration further indicates a reference signal path, and the clock drift correction estimation is determinable based on measuring a reference signal along the reference signal path.

[0256] Clause 55. The managing device of any of clauses 49 to 54, wherein the clock drift information indicates: a maximum value of the clock drift of the sensing node device, a median value of the clock drift of the sensing node device, a statistical description of the clock drift of the sensing node device, variability of the clock drift as a function of one or more environmental parameters, a maximum duration of the sensing window supported by the sensing node device, or a combination thereof.

[0257] Clause 56. The managing device of any of clauses 49 to 55, wherein: the sensing node device is a user equipment (UE), and the clock drift information is received based on: long term evolution (LTE) positioning protocol (LPP) signaling in a case that the managing device is a location server, radio resource control (RRC) signaling, medium access control (MAC) control element (MAC CE) signaling, or downlink control information (DCI) signaling in a case that the managing device is a base station or a transmission/reception point (TRP), or sidelink LPP signaling, sidelink RRC signaling, sidelink MAC CE signaling, or sidelink control information (SCI) signaling in a case that the managing device is another UE.

[0258] Clause 57. The managing device of any of clauses 49 to 56, further comprising: means for transmitting a request to the sensing node device requesting the clock drift information such that the clock drift information is transmitted as a solicited transmission in response to the request from the managing device.

[0259] Clause 58. A non-transitory computer-readable medium storing computer-executable instructions that, when executed by a sensing node device, cause the sensing node device to: transmit, to a managing device, clock drift information corresponding to clock drift of the sensing node device; receive, from the managing device, a sensing signal configuration for a sensing session based on the clock drift information; and transmit or receiving one or more sensing signals during a sensing window of the sensing session based on the sensing signal configuration.

[0260] Clause 59. The non-transitory computer-readable medium of clause 58, wherein the sensing signal configuration indicates one or more parameters based at least in part on the clock drift information, the one or more parameters comprising: a numerology of the one or more sensing signals; an operating frequency of the one or more sensing signals; an inter-symbol duration of the one or more sensing signals; a duration of the sensing window; or a combination thereof.

[0261] Clause 60. The non-transitory computer-readable medium of any of clauses 58 to 59, further comprising computer-executable instructions that, when executed by the sensing node device, cause the sensing node device to: receive, from the managing device, a configuration to participate in the sensing session based on the clock drift information.

[0262] Clause 61. The non-transitory computer-readable medium of any of clauses 58 to 60, further comprising computer-executable instructions that, when executed by the

sensing node device, cause the sensing node device to: receive, from the managing device, a measurement configuration for the sensing session based on the clock drift information; and transmit, to the managing device, sensing measurements based on the one or more sensing signals received by the sensing node device and the measurement configuration.

[0263] Clause 62. The non-transitory computer-readable medium of clause 61, further comprising computer-executable instructions that, when executed by the sensing node device, cause the sensing node device to: obtain N signal samples based on the one or more sensing signals, wherein: the measurement configuration indicates that the sensing measurements are generated based on: processing the N signal samples as a single batch to obtain the sensing measurements; dividing the N signal samples into a plurality of batches, adjusting the N signal samples based on a clock drift correction estimation on a per-batch basis to obtain N adjusted signal samples, and processing the N adjusted signal samples as a single batch to obtain the sensing measurements; or dividing the N signal samples into the plurality of batches, processing the N signal samples on a per-batch basis to obtain a plurality of per-batch measurements, and combining the plurality of per-batch measurements to obtain the sensing measurements, each batch of the plurality of batches has M signal samples, N is a positive integer, and M is a positive integer less than N.

[0264] Clause 63. The non-transitory computer-readable medium of clause 61, further comprising computer-executable instructions that, when executed by the sensing node device, cause the sensing node device to: determine a clock drift correction estimation based on measuring a reference signal along a reference signal path indicated in the measurement configuration, wherein the sensing measurements are generated based on the one or more sensing signals received by the sensing node device, the clock drift correction estimation, and the measurement configuration.

[0265] Clause 64. The non-transitory computer-readable medium of any of clauses 58 to 63, wherein the clock drift information indicates: a maximum value of the clock drift of the sensing node device, a median value of the clock drift of the sensing node device, a statistical description of the clock drift of the sensing node device, variability of the clock drift as a function of one or more environmental parameters, a maximum duration of the sensing window supported by the sensing node device, or a combination thereof.

[0266] Clause 65. The non-transitory computer-readable medium of any of clauses 58 to 64, wherein: the sensing node device is a user equipment (UE), and the clock drift information is transmitted based on: long term evolution (LTE) positioning protocol (LPP) signaling in a case that the managing device is a location server, radio resource control (RRC) signaling, medium access control (MAC) control element (MAC CE) signaling, or downlink control information (DCI) signaling in a case that the managing device is a base station or a transmission/reception point (TRP), or sidelink LPP signaling, sidelink RRC signaling, sidelink MAC CE signaling, or sidelink control information (SCI) signaling in a case that the managing device is another UE.

[0267] Clause 66. The non-transitory computer-readable medium of any of clauses 58 to 65, wherein the clock drift information is transmitted as: a solicited transmission in

response to a request from the managing device; or an unsolicited transmission initiated by the sensing node device.

[0268] Clause 67. The non-transitory computer-readable medium of any of clauses 58 to 66, further comprising computer-executable instructions that, when executed by the sensing node device, cause the sensing node device to: transmit a request to a base station or a transmission/reception point (TRP) requesting resource allocation for the one or more sensing signals, wherein the request includes the clock drift information.

[0269] Clause 68. A non-transitory computer-readable medium storing computer-executable instructions that, when executed by a managing device, cause the managing device to: receive, from a sensing node device, clock drift information corresponding to clock drift of the sensing node device; and transmit, to the sensing node device, a sensing signal configuration for a sensing session based on the clock drift information, the sensing signal configuration indicating one or more sensing signals to be transmitted or received by the sensing node device during a sensing window of the sensing session.

[0270] Clause 69. The non-transitory computer-readable medium of clause 68, further comprising computer-executable instructions that, when executed by the managing device, cause the managing device to determine the sensing signal configuration indicative of one or more parameters based at least in part on the clock drift information, the one or more parameters comprising: a numerology of the one or more sensing signals; an operating frequency of the one or more sensing signals; an inter-symbol duration of the one or more sensing signals; a duration of the sensing window; or a combination thereof.

[0271] Clause 70. The non-transitory computer-readable medium of any of clauses 68 to 69, further comprising computer-executable instructions that, when executed by the managing device, cause the managing device to: transmit, to the sensing node device, a configuration to participate in the sensing session based on the clock drift information.

[0272] Clause 71. The non-transitory computer-readable medium of any of clauses 68 to 70, further comprising computer-executable instructions that, when executed by the managing device, cause the managing device to: transmit, to the sensing node device, a measurement configuration for the sensing session based on the clock drift information, wherein: the measurement configuration indicates that sensing measurements are generated based on: processing N signal samples obtained based on the one or more sensing signals as a single batch to obtain the sensing measurements; dividing the N signal samples into a plurality of batches, adjusting the N signal samples based on a clock drift correction estimation on a per-batch basis to obtain N adjusted signal samples, and processing the N adjusted signal samples as a single batch to obtain the sensing measurements; or dividing the N signal samples into the plurality of batches, processing the N signal samples on a per-batch basis to obtain a plurality of per-batch measurements, and combining the plurality of per-batch measurements to obtain the sensing measurements, each batch of the plurality of batches has M signal samples, N is a positive integer, and M is a positive integer less than N .

[0273] Clause 72. The non-transitory computer-readable medium of clause 71, wherein: the measurement configu-

ration indicates a value of M that corresponds to a size of each batch of signal samples on which sensing measurements are generated.

[0274] Clause 73. The non-transitory computer-readable medium of any of clauses 71 to 72, wherein: the measurement configuration further indicates a reference signal path, and the clock drift correction estimation is determinable based on measuring a reference signal along the reference signal path.

[0275] Clause 74. The non-transitory computer-readable medium of any of clauses 68 to 73, wherein the clock drift information indicates: a maximum value of the clock drift of the sensing node device, a median value of the clock drift of the sensing node device, a statistical description of the clock drift of the sensing node device, variability of the clock drift as a function of one or more environmental parameters, a maximum duration of the sensing window supported by the sensing node device, or a combination thereof.

[0276] Clause 75. The non-transitory computer-readable medium of any of clauses 68 to 74, wherein: the sensing node device is a user equipment (UE), and the clock drift information is received based on: long term evolution (LTE) positioning protocol (LPP) signaling in a case that the managing device is a location server, radio resource control (RRC) signaling, medium access control (MAC) control element (MAC CE) signaling, or downlink control information (DCI) signaling in a case that the managing device is a base station or a transmission/reception point (TRP), or sidelink LPP signaling, sidelink RRC signaling, sidelink MAC CE signaling, or sidelink control information (SCI) signaling in a case that the managing device is another UE.

[0277] Clause 76. The non-transitory computer-readable medium of any of clauses 68 to 75, further comprising computer-executable instructions that, when executed by the managing device, cause the managing device to: transmit a request to the sensing node device requesting the clock drift information such that the clock drift information is transmitted as a solicited transmission in response to the request from the managing device.

[0278] Those of skill in the art will appreciate that information and signals may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the above description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof.

[0279] Further, those of skill in the art will appreciate that the various illustrative logical blocks, modules, circuits, and algorithm steps described in connection with the aspects disclosed herein may be implemented as electronic hardware, computer software, or combinations of both. To clearly illustrate this interchangeability of hardware and software, various illustrative components, blocks, modules, circuits, and steps have been described above generally in terms of their functionality. Whether such functionality is implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system. Skilled artisans may implement the described functionality in varying ways for each particular application, but such implementation decisions should not be interpreted as causing a departure from the scope of the present disclosure.

[0280] The various illustrative logical blocks, modules, and circuits described in connection with the aspects disclosed herein may be implemented or performed with a general purpose processor, a digital signal processor (DSP), an ASIC, a field-programable gate array (FPGA), or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general-purpose processor may be a microprocessor, but in the alternative, the processor may be any conventional processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices, for example, a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration.

[0281] The methods, sequences and/or algorithms described in connection with the aspects disclosed herein may be embodied directly in hardware, in a software module executed by a processor, or in a combination of the two. A software module may reside in random access memory (RAM), flash memory, read-only memory (ROM), erasable programmable ROM (EPROM), electrically erasable programmable ROM (EEPROM), registers, hard disk, a removable disk, a CD-ROM, or any other form of storage medium known in the art. An example storage medium is coupled to the processor such that the processor can read information from, and write information to, the storage medium. In the alternative, the storage medium may be integral to the processor. The processor and the storage medium may reside in an ASIC. The ASIC may reside in a user terminal (e.g., UE). In the alternative, the processor and the storage medium may reside as discrete components in a user terminal.

[0282] In one or more example aspects, the functions described may be implemented in hardware, software, firmware, or any combination thereof. If implemented in software, the functions may be stored on or transmitted over as one or more instructions or code on a computer-readable medium. Computer-readable media includes both computer storage media and communication media including any medium that facilitates transfer of a computer program from one place to another. A storage media may be any available media that can be accessed by a computer. By way of example, and not limitation, such computer-readable media can comprise RAM, ROM, EEPROM, CD-ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium that can be used to carry or store desired program code in the form of instructions or data structures and that can be accessed by a computer. Also, any connection is properly termed a computer-readable medium. For example, if the software is transmitted from a website, server, or other remote source using a coaxial cable, fiber optic cable, twisted pair, digital subscriber line (DSL), or wireless technologies such as infrared, radio, and microwave, then the coaxial cable, fiber optic cable, twisted pair, DSL, or wireless technologies such as infrared, radio, and microwave are included in the definition of medium. Disk and disc, as used herein, includes compact disc (CD), laser disc, optical disc, digital versatile disc (DVD), floppy disk and Blu-ray disc where disks usually reproduce data magnetically, while discs reproduce

data optically with lasers. Combinations of the above should also be included within the scope of computer-readable media.

[0283] While the foregoing disclosure shows illustrative aspects of the disclosure, it should be noted that various changes and modifications could be made herein without departing from the scope of the disclosure as defined by the appended claims. For example, the functions, steps and/or actions of the method claims in accordance with the aspects of the disclosure described herein need not be performed in any particular order. Further, no component, function, action, or instruction described or claimed herein should be construed as critical or essential unless explicitly described as such. Furthermore, as used herein, the terms “set,” “group,” and the like are intended to include one or more of the stated elements. Also, as used herein, the terms “has,” “have,” “having,” “comprises,” “comprising,” “includes,” “including,” and the like does not preclude the presence of one or more additional elements (e.g., an element “having” A may also have B). Further, the phrase “based on” is intended to mean “based, at least in part, on” unless explicitly stated otherwise. Also, as used herein, the term “or” is intended to be inclusive when used in a series and may be used interchangeably with “and/or,” unless explicitly stated otherwise (e.g., if used in combination with “either” or “only one of”) or the alternatives are mutually exclusive (e.g., “one or more” should not be interpreted as “one and more”). Furthermore, although components, functions, actions, and instructions may be described or claimed in the singular, the plural is contemplated unless limitation to the singular is explicitly stated. Accordingly, as used herein, the articles “a,” “an,” “the,” and “said” are intended to include one or more of the stated elements. Additionally, as used herein, the terms “at least one” and “one or more” encompass “one” component, function, action, or instruction performing or capable of performing a described or claimed functionality and also “two or more” components, functions, actions, or instructions performing or capable of performing a described or claimed functionality in combination.

What is claimed is:

1. A method of operating a sensing node device, the method comprising:

transmitting, to a managing device, clock drift information corresponding to clock drift of the sensing node device;

receiving, from the managing device, a sensing signal configuration for a sensing session based on the clock drift information; and

transmitting or receiving one or more sensing signals during a sensing window of the sensing session based on the sensing signal configuration.

2. The method of claim 1, wherein the sensing signal configuration indicates one or more parameters based at least in part on the clock drift information, the one or more parameters comprising:

a numerology of the one or more sensing signals;

an operating frequency of the one or more sensing signals;

an inter-symbol duration of the one or more sensing signals;

a duration of the sensing window; or

a combination thereof.

3. The method of claim 1, further comprising:
receiving, from the managing device, a configuration to participate in the sensing session based on the clock drift information.

4. The method of claim 1, further comprising:
receiving, from the managing device, a measurement configuration for the sensing session based on the clock drift information; and
transmitting, to the managing device, sensing measurements based on the one or more sensing signals received by the sensing node device and the measurement configuration.

5. The method of claim 4, wherein:
the receiving the one or more sensing signals during the sensing window of the sensing session comprises obtaining N signal samples based on the one or more sensing signals, and
the measurement configuration indicates that the sensing measurements are generated based on:
processing the N signal samples as a single batch to obtain the sensing measurements;
dividing the N signal samples into a plurality of batches, adjusting the N signal samples based on a clock drift correction estimation on a per-batch basis to obtain N adjusted signal samples, and processing the N adjusted signal samples as a single batch to obtain the sensing measurements; or
dividing the N signal samples into the plurality of batches, processing the N signal samples on a per-batch basis to obtain a plurality of per-batch measurements, and combining the plurality of per-batch measurements to obtain the sensing measurements, each batch of the plurality of batches has M signal samples,
N is a positive integer, and
M is a positive integer less than N.

6. The method of claim 4, further comprising:
determining a clock drift correction estimation based on measuring a reference signal along a reference signal path indicated in the measurement configuration, wherein the sensing measurements are generated based on the one or more sensing signals received by the sensing node device, the clock drift correction estimation, and the measurement configuration.

7. The method of claim 1, wherein the clock drift information indicates:
a maximum value of the clock drift of the sensing node device,
a median value of the clock drift of the sensing node device,
a statistical description of the clock drift of the sensing node device,
variability of the clock drift as a function of one or more environmental parameters,
a maximum duration of the sensing window supported by the sensing node device, or
a combination thereof.

8. The method of claim 1, wherein:
the sensing node device is a user equipment (UE), and
the clock drift information is transmitted based on:
long term evolution (LTE) positioning protocol (LPP) signaling in a case that the managing device is a location server,

radio resource control (RRC) signaling, medium access control (MAC) control element (MAC CE) signaling, or downlink control information (DCI) signaling in a case that the managing device is a base station or a transmission/reception point (TRP), or
sidelink LPP signaling, sidelink RRC signaling, sidelink MAC CE signaling, or sidelink control information (SCI) signaling in a case that the managing device is another UE.

9. The method of claim 1, wherein the clock drift information is transmitted as:
a solicited transmission in response to a request from the managing device; or
an unsolicited transmission initiated by the sensing node device.

10. The method of claim 1, further comprising:
transmitting a request to a base station or a transmission/reception point (TRP) requesting resource allocation for the one or more sensing signals, wherein the request includes the clock drift information.

11. A method of operating a managing device, the method comprising:
receiving, from a sensing node device, clock drift information corresponding to clock drift of the sensing node device; and
transmitting, to the sensing node device, a sensing signal configuration for a sensing session based on the clock drift information, the sensing signal configuration indicating one or more sensing signals to be transmitted or received by the sensing node device during a sensing window of the sensing session.

12. The method of claim 11, further comprising determining the sensing signal configuration indicative of one or more parameters based at least in part on the clock drift information, the one or more parameters comprising:
a numerology of the one or more sensing signals;
an operating frequency of the one or more sensing signals;
an inter-symbol duration of the one or more sensing signals;
a duration of the sensing window; or
a combination thereof.

13. The method of claim 11, further comprising:
transmitting, to the sensing node device, a configuration to participate in the sensing session based on the clock drift information.

14. The method of claim 11, further comprising:
transmitting, to the sensing node device, a measurement configuration for the sensing session based on the clock drift information, wherein:
the measurement configuration indicates that sensing measurements are generated based on:
processing N signal samples obtained based on the one or more sensing signals as a single batch to obtain the sensing measurements;
dividing the N signal samples into a plurality of batches, adjusting the N signal samples based on a clock drift correction estimation on a per-batch basis to obtain N adjusted signal samples, and processing the N adjusted signal samples as a single batch to obtain the sensing measurements; or
dividing the N signal samples into the plurality of batches, processing the N signal samples on a per-batch basis to obtain a plurality of per-batch mea-

measurements, and combining the plurality of per-batch measurements to obtain the sensing measurements, each batch of the plurality of batches has M signal samples,

N is a positive integer, and

M is a positive integer less than N.

15. The method of claim **14**, wherein:

the measurement configuration indicates a value of M that corresponds to a size of each batch of signal samples on which sensing measurements are generated.

16. The method of claim **14**, wherein:

the measurement configuration further indicates a reference signal path, and
the clock drift correction estimation is determinable based on measuring a reference signal along the reference signal path.

17. The method of claim **11**, wherein the clock drift information indicates:

a maximum value of the clock drift of the sensing node device,

a median value of the clock drift of the sensing node device,

a statistical description of the clock drift of the sensing node device,

variability of the clock drift as a function of one or more environmental parameters,

a maximum duration of the sensing window supported by the sensing node device, or

a combination thereof.

18. The method of claim **11**, wherein:

the sensing node device is a user equipment (UE), and the clock drift information is received based on:

long term evolution (LTE) positioning protocol (LPP) signaling in a case that the managing device is a location server,

radio resource control (RRC) signaling, medium access control (MAC) control element (MAC CE) signaling, or downlink control information (DCI) signaling in a case that the managing device is a base station or a transmission/reception point (TRP), or

sidelink LPP signaling, sidelink RRC signaling, sidelink MAC CE signaling, or sidelink control information (SCI) signaling in a case that the managing device is another UE.

19. The method of claim **11**, further comprising:

transmitting a request to the sensing node device requesting the clock drift information such that the clock drift information is transmitted as a solicited transmission in response to the request from the managing device.

20. A sensing node device, comprising:

one or more memories;

one or more transceivers; and

one or more processors communicatively coupled to the one or more memories and the one or more transceivers, the one or more processors, either alone or in combination, configured to:

transmit, to a managing device via the one or more transceivers, clock drift information corresponding to clock drift of the sensing node device;

receive, from the managing device via the one or more transceivers, a sensing signal configuration for a sensing session based on the clock drift information; and

transmit or receive, via the one or more transceivers, one or more sensing signals during a sensing window of the sensing session based on the sensing signal configuration.

21. The sensing node device of claim **20**, wherein the sensing signal configuration indicates one or more parameters based at least in part on the clock drift information, the one or more parameters comprising:

a numerology of the one or more sensing signals;

an operating frequency of the one or more sensing signals;

an inter-symbol duration of the one or more sensing signals;

a duration of the sensing window; or

a combination thereof.

22. The sensing node device of claim **20**, wherein the one or more processors, either alone or in combination, are further configured to:

receive, from the managing device via the one or more transceivers, a measurement configuration for the sensing session based on the clock drift information; and
transmit, to the managing device via the one or more transceivers, sensing measurements based on the one or more sensing signals received by the sensing node device and the measurement configuration.

23. The sensing node device of claim **22**, wherein:

the one or more processors, either alone or in combination, are further configured to obtain N signal samples based on the one or more sensing signals, and

the measurement configuration indicates that the sensing measurements are generated based on:

processing the N signal samples as a single batch to obtain the sensing measurements;

dividing the N signal samples into a plurality of batches, adjusting the N signal samples based on a clock drift correction estimation on a per-batch basis to obtain N adjusted signal samples, and processing the N adjusted signal samples as a single batch to obtain the sensing measurements; or

dividing the N signal samples into the plurality of batches, processing the N signal samples on a per-batch basis to obtain a plurality of per-batch measurements, and combining the plurality of per-batch measurements to obtain the sensing measurements, each batch of the plurality of batches has M signal samples,

N is a positive integer, and

M is a positive integer less than N.

24. The sensing node device of claim **22**, wherein the one or more processors, either alone or in combination, are further configured to:

determine a clock drift correction estimation based on measuring a reference signal along a reference signal path indicated in the measurement configuration,

wherein the sensing measurements are generated based on the one or more sensing signals received by the sensing node device, the clock drift correction estimation, and the measurement configuration.

25. The sensing node device of claim **20**, wherein the clock drift information indicates:

a maximum value of the clock drift of the sensing node device,

a median value of the clock drift of the sensing node device,

a statistical description of the clock drift of the sensing node device,
 variability of the clock drift as a function of one or more environmental parameters,
 a maximum duration of the sensing window supported by the sensing node device, or
 a combination thereof.

26. A managing device, comprising:

one or more memories;

one or more transceivers; and

one or more processors communicatively coupled to the one or more memories and the one or more transceivers, the one or more processors, either alone or in combination, configured to:

receive, from a sensing node device via the one or more transceivers, clock drift information corresponding to clock drift of the sensing node device; and

transmit, to the sensing node device via the one or more transceivers, a sensing signal configuration for a sensing session based on the clock drift information, the sensing signal configuration indicating one or more sensing signals to be transmitted or received by the sensing node device during a sensing window of the sensing session.

27. The managing device of claim **26**, wherein the one or more processors, either alone or in combination, are further configured to determine the sensing signal configuration indicative of one or more parameters based at least in part on the clock drift information, the one or more parameters comprising:

a numerology of the one or more sensing signals;

an operating frequency of the one or more sensing signals;
 an inter-symbol duration of the one or more sensing signals;

a duration of the sensing window; or

a combination thereof.

28. The managing device of claim **26**, wherein the one or more processors, either alone or in combination, are further configured to:

transmit, to the sensing node device, a measurement configuration for the sensing session based on the clock drift information,

wherein:

the measurement configuration indicates that sensing measurements are generated based on:

processing N signal samples obtained based on the one or more sensing signals as a single batch to obtain the sensing measurements;

dividing the N signal samples into a plurality of batches, adjusting the N signal samples based on a clock drift correction estimation on a per-batch basis to obtain N adjusted signal samples, and processing the N adjusted signal samples as a single batch to obtain the sensing measurements; or

dividing the N signal samples into the plurality of batches, processing the N signal samples on a per-batch basis to obtain a plurality of per-batch measurements, and combining the plurality of per-batch measurements to obtain the sensing measurements, each batch of the plurality of batches has M signal samples,

N is a positive integer, and

M is a positive integer less than N.

29. The managing device of claim **28**, wherein:

the measurement configuration further indicates a reference signal path, and

the clock drift correction estimation is determinable based on measuring a reference signal along the reference signal path.

30. The managing device of claim **26**, wherein the clock drift information indicates:

a maximum value of the clock drift of the sensing node device,

a median value of the clock drift of the sensing node device,

a statistical description of the clock drift of the sensing node device,

variability of the clock drift as a function of one or more environmental parameters,

a maximum duration of the sensing window supported by the sensing node device, or

a combination thereof.

* * * * *